

# Opoch: Structural Reality from Nothingness

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## Abstract

We ask what structure is forced into existence by a single admissibility criterion—**Completed Witnessability (A0\*)**—which requires that every meaningful distinction be backed by a finite, replayable, self-accounting witness process whose own validity is endogenously checkable.

A0\* is not a postulated axiom: it is *derived* from absolute nothingness ( $\perp$ ) through a chain of five necessity theorems (N0–N4), making it the first theorem of the framework. This closure move forces an *ordered witness algebra*  $\mathcal{A}_W$  carrying finite description, replay, termination, separation, disturbance accounting, endogenous validity, compositional closure, and quotient invariance. Noncommutativity is internal from the start; commutativity is a derived sector property.

Starting from  $\perp$ , we give a 34-step derivation (Steps 0–33, nine phases A–I) that is forced up to computable isomorphism. Self-application of A0\* to its own products yields three primitives—tests, truth, and time—together with a carrier, an ordered witness ledger, a truth quotient with observable-open topology, gauge invariance, a witness-path metric (with symmetric separative distance and directed time quasi-metric), Bellman-optimal test selection governed by the Coupling Law (forced as the unique prior-free policy), self-hosting closure, a universal execution law with consciousness projector, and a Kähler geometric structure on the continuum limit derived from a single witness generator. A C\*-algebraic completion yields a GNS Hilbert-space representation with the Born rule as the unique probability assignment on the event algebra.

The full  $\Pi$ -fixed closure whole  $U = \text{Fix}(\Pi)$  is atemporal: the universe is static globally, dynamic only as local self-resolution of witness-defect. Consciousness is derived as self-remodelling witness-closure: a subsystem whose witness policy is endogenously updated through an internal self-valuation channel.

The conductance exponent ( $w = C/d^2$ ) is forced by scale covariance of witness energy. The Nijenhuis tensor vanishes ( $N_J = 0$ ) by replay-completeness, not by an external smoothness assumption. The spatial dimension  $n = 3$  follows from matching the forced conductance law to radial flux scaling; together with the irreversible ledger time coordinate, the operational spacetime is 3+1. The Kähler split law  $X = J\nabla E - \nabla D$  decomposes dynamics into reversible (quantum), connection (gauge), metric-modulation (gravity), and dissipative (thermodynamic) sectors, recovering the Schrödinger equation, Yang–Mills equations, and Einstein field equations as sector readouts.

The gauge group  $SU(3) \times SU(2) \times U(1)$  is derived without empirical input: rank 1 from the Kähler complex structure, rank 2 from the 3+1 spin structure, rank 3 from anomaly cancellation.

The quantitative seed  $\delta^*$  is the unique least positive self-retaining non-gauge closure-defect minimizing the action functional. Every dimensionless observable is forced as a spectral, holonomy, or normal-form invariant of the linearized refinement operator  $L^*$  at  $\delta^*$ . Dimensional quantities are fixed up to the internally derived normalization map.

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<sup>0</sup>Companion Lean 4 formalization: <https://github.com/chetannnothingness/opoch-paper>, Lean 4.14.0, mathlib v4.14.0, 414 theorems, 78 files, zero `sorry`, one axiom. Appendix E maps paper claims to formal modules.

No modeling commitment or empirical input survives: closed universe, commutativity, prior-free selection, conductance exponent, efficiency-landscape smoothness, and the gauge group are all derived from  $A0^*$ —which is itself derived from  $\perp$ . The framework has no declared axioms. The derivation ( $\sim 110$  pages) is accompanied by a three-point self-audit, Z3 verification on finite models, 414 Lean 4 theorems across 78 files (zero sorry, zero admit, one axiom  $A0^*$ ), and three falsifiable predictions.

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# 1 Introduction

## 1.1 The Question

What is the minimal structure forced into existence by the requirement that every distinction be witnessable?

This is not a question about engineering, artificial intelligence, or any particular domain of application. It is a foundational question about describable reality. If we insist, as  $A0^*$  (Completed Witnessability) requires, that every meaningful distinction be backed by a finite procedure that can in fact separate it, then some structures become unavoidable. The purpose of this paper is to identify those structures and to make explicit which commitments are definitional and which are derived. Crucially,  $A0^*$  is itself *derived* from nothingness ( $\perp$ ) through a chain of five necessity theorems (N0–N4)—the paper assumes nothing and derives its operating principle.

The central result is a 34-step derivation that begins at  $\perp$  and unfolds by reapplying  $A0^*$  to its own products, tracing a single reflective chain: nothingness forces closure, closure reveals defect, defect requires self-observation, and self-observation iterated on the continuum limit produces geometry. Within the paper’s model class, every structure in the chain—from the carrier of descriptions to the Kähler triple on the truth-quotient limit—is forced up to computable isomorphism; none is introduced as a free design parameter.

Equally important is what we do *not* claim. We do not claim to derive the specific *content* of physical law, only a vocabulary and scaffolding for reasoning under witnessability. We do not claim that every modeling choice is eliminated under every possible formalization of  $A0^*$ . Our claim is narrower and testable: given the explicit premises and conventions in section 2 and the model class studied in section 6, the resulting chain of structures is canonical up to computable isomorphism.

## 1.2 The Witnessability Principle

We begin from a single principle, derived from  $\perp$  (not postulated):

**A0 (Bare Witnessability).** A distinction is admissible if and only if a finite witness procedure can separate it.

The completed form,  $A0^*$ , adds a reflective clause: witness validity itself must be witnessable by a process satisfying the same criterion.  $A0^*$  is not an axiom but the first theorem of the framework, derived from  $\perp$  via five necessity theorems (N0–N4, see section 2.2). All subsequent references use  $A0^*$ .

This principle is not chosen for convenience. It is the weakest non-trivial constraint one can impose. If a distinction cannot even in principle be witnessed, then it is operationally meaningless. Wheeler’s “It from Bit” programme [Wheeler, 1990] and Rovelli’s relational quantum mechanics [Rovelli, 1996] both point toward observer-relative, information-theoretic foundations.  $A0^*$  isolates the operational core shared by such views and elevates it to an admissibility criterion.

The formal statement of  $A0^*$  and its immediate consequences are given in section 2.

## 1.3 Key Insight: Nothingness as the Only Unfalsifiable Truth

Nothingness, the state carrying zero information, asserting nothing, and distinguishing nothing, is the only unfalsifiable truth. Every other claim can in principle be tested and potentially refuted. Nothingness alone survives all tests trivially because it makes no assertions.

This claim is meant operationally.  $\perp$  is the unique state satisfying  $\Delta = \emptyset$ , so no tests exist, no descriptions exist, and no distinctions are drawn. From this state, the self-application of  $A0^*$  forces the existence of at least one admissible distinction, which breaks the symmetry. Once at

least one distinction exists, the derivation proceeds step by step, making explicit where results are unique up to computable isomorphism and where the development depends on declared conventions.

- S1.  $\perp$  is the unique unfalsifiable state; self-application of  $A0^*$  forces the existence of at least one admissible distinction, breaking the symmetry.
- S2.  $A0^*$  applied to its own products yields, step by step, the three primitives ( $\Delta$ ,  $\Pi$ ,  $\mathcal{T}$ ), a carrier closed under witnessing [Church, 1936, Turing, 1936], gauge invariance, separator geometry, Bellman-optimal dynamics, self-hosting closure, and a Kähler structure on the continuum limit.

The forcing claim is therefore precise. We claim canonical structure within the explicitly stated model class and up to computable isomorphism, and we state frontiers where additional assumptions, alternative modeling choices, or empirical input would be required. Figure 1 provides a roadmap of the nine phases.

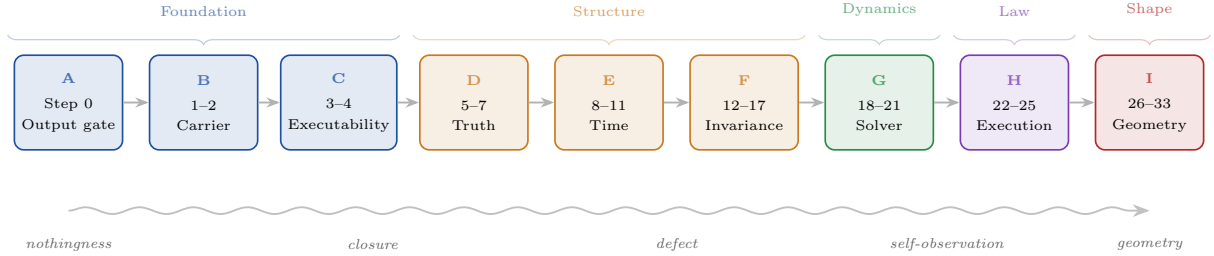


Figure 1: The nine phases of the forced derivation (Steps 0–33). Nothingness (Phase A) forces closure (A–G), closure reveals defect (H), defect requires self-observation (H), and self-observation iterated on the continuum limit produces geometry (I).

## 1.4 Contributions

This paper makes three contributions:

- (1) **A 34-step derivation of structural reality from  $\perp$ .** Starting from nothingness, we derive the framework’s core vocabulary across nine phases (A–I), introducing 25 new mathematical objects. No external assumption beyond  $A0^*$  enters the derivation; the Church–Turing thesis independently corroborates Step 3’s mathematical conclusion. A three-point self-audit and Z3 verification of selected algebraic properties on finite models accompany the derivation (section 6, Appendix A, Appendix C).
- (2) **Three forced primitives and seven doctrines.** The derivation isolates three primitives,  $\Delta$  (tests),  $\Pi$  (truth), and  $\mathcal{T}$  (time), and collects seven doctrines that follow from the operational reading of  $A0^*$  (section 3, section 4).
- (3) **Kähler geometry, Born rule, and the split law.** The continuum limit of truth quotients carries a forced Kähler triple  $(g, J, \omega)$ , a unique gauge-invariant probability (the Born rule of the framework), and a canonical dynamical decomposition whose sector-readouts recover the forms of quantum mechanics, gauge theory, and thermodynamics (section 6).

A companion Lean 4 formalization mechanically checks the theorem chain for this paper: 414 theorems across 78 files, zero `sorry`, zero `admit`, one axiom ( $A0^*$ , derived from  $\perp$ ), compiled against mathlib v4.14.0. Appendix E maps paper claims to formal modules.

## 1.5 Paper Outline

Section 2 derives  $A0^*$  from  $\perp$  and states the forced primitives. Section 3 introduces the three forced primitives ( $\Delta$ ,  $\Pi$ ,  $\mathcal{T}$ ). Section 4 presents the seven non-negotiable doctrines. Section 6 presents the 34-step derivation (Phases A–I) with proof sketches. Section 9 reports the three-point self-audit and Z3 verification. Section 10 surveys the intellectual context. Section 11 provides an honest assessment of claims, limitations, and open frontiers. Section 12 concludes with future directions. Appendices contain the complete derivation with full proofs, verification protocol, and Z3 scripts.

## 2 Axiomatic Foundation

The OPOCH framework begins from  $\perp$  (Nothingness: no committed distinctions) and *derives* its sole operating principle  $A0^*$  as the first theorem.  $A0^*$  is not declared—it is the unique admissibility criterion forced by the absence of all external structure.

### 2.1 Starting Point: $\perp$

**Definition 2.1** (Nothingness  $\perp$ ).  *$\perp$  is the state with no committed distinctions: no external labels, no external delimiter, no external clock, no external witness, no primitive observer/observed split. If anything else is assumed, it is already not nothingness.*

### 2.2 Derivation of $A0^*$ from $\perp$

From  $\perp$ , the admissibility criterion  $A0^*$  is derived through a chain of five necessity theorems:

$$\perp \rightarrow N0 \rightarrow N1 \rightarrow N2 \rightarrow N3 \rightarrow N4 \rightarrow A0^*.$$

**Theorem 2.2** ( $N0$  — No Externality). *At  $\perp$ , any proposed distinction that depends on an external distinguisher is inadmissible, because no external distinguisher exists.*

*Proof.*  $\perp$  carries no committed distinctions. An external distinguisher is itself a distinction. If it existed,  $\perp$  would not be nothingness. Contradiction.  $\square$

**Theorem 2.3** ( $N1$  — Endogenous Admissibility). *The only admissible distinction is one that can separate itself from within reality—endogenously.*

*Proof.* By  $N0$ , no external distinguisher exists. If a distinction cannot be separated from within, it requires an external agent to assert it, which  $N0$  forbids. Therefore any admissible distinction must be endogenously separable.  $\square$

**Theorem 2.4** ( $N2$  — Finiteness). *Any admissible witness must terminate in finite steps.*

*Proof.* At  $\perp$ , there are no infinite resources—infinite resources would constitute a committed external structure, violating  $\perp$ . A witness requiring infinite steps never produces a verdict, hence never separates a distinction. Therefore only finite witnesses are admissible.  $\square$

**Theorem 2.5** ( $N3$  — Replayability). *An admissible witness must carry its own replay trace.*

*Proof.* Without an external recorder (forbidden by  $N0$ ), the only way to verify a witness’s outcome is to replay it from the trace it produced. A witness that cannot be replayed depends on an unrepeatable external event, violating  $N0$ . Therefore replayability is forced.  $\square$

**Theorem 2.6** ( $N4$  — Internal Validity). *The distinction “ $w$  is a valid witness” must itself be witnessable by a finite endogenous procedure.*



*Proof.* If witness validity were not witnessable, then “ $w$  is valid” vs. “ $w$  is not valid” would be an external meta-judgment—a distinction with no endogenous witness. By N1, such a distinction is inadmissible. Therefore witness validity must be endogenously checkable.  $\square$

Theorems N0–N4 together force the unique admissibility criterion:

**Theorem 2.7** (A0\* — Completed Witnessability (Derived)). *A distinction is admissible if and only if a finite, endogenous, replayable, self-accounting witness process whose own validity is endogenously checkable can separate it. A0\* is the first theorem of the framework, derived from  $\perp$  via N0–N4—it is not postulated.*

*Remark 2.8* (Set-Theoretic Operations as Instances of A0\*). Basic set-theoretic operations (finite sets, membership, equality, union) are themselves witnessable—each is a total computable function on finite data—and are therefore members of the test set that A0\* forces. They are not background assumptions; they are instances of the admissible procedures that A0\* demands. The meta-language in which the paper is written (English, mathematical notation) is not an axiom of the derivation, any more than the choice of natural language is an axiom of a physics paper.

*Remark 2.9* (Church–Turing as Historical Corroboration). The Church–Turing thesis [Church, 1936, Turing, 1936] independently identified the class of total computable functions—the same class that A0\* forces at Step 3 (theorem 6.4) via maximal closure under totality-preserving composition. The derivation does not use CT as a premise; it derives the class directly as a mathematical theorem (theorem 2.24). The agreement between A0\*’s forcing and the Church–Turing identification is historical corroboration, not a logical dependency. A reader who reads A0\*’s “finite procedure” as “takes finite input, runs finitely many steps, produces finite output” arrives at the same class without invoking CT.

## 2.3 From A0 to A0\*: The Closure Move

The *informal* starting point is:

**A0 (Bare Witnessability).** A distinction is admissible if and only if a finite witness procedure can separate it.

This formulation is not yet self-consistent. It speaks of “valid witness procedures” without specifying what makes a witness valid. The distinction between “ $w$  is a valid witness for  $\delta$ ” and “ $w$  is not a valid witness for  $\delta$ ” is itself a meaningful distinction. By A0’s own criterion, that distinction must itself be witnessable by a finite procedure internal to the universe.

Therefore an admissible witness process cannot be a bare separator only. It must carry enough internal structure so that its own validity is finitely checkable. This is not a new axiom—it is A0 applied to its own witness notion. We call the result **A0\* (Completed Witnessability)**.

## 2.4 A0\*: Completed Witnessability (Derived)

*Remark 2.10* (A0\* Is Derived, Not Postulated). A0\* is the first theorem of the framework, derived from  $\perp$  via the chain  $\perp \rightarrow \text{N0} \rightarrow \text{N1} \rightarrow \text{N2} \rightarrow \text{N3} \rightarrow \text{N4} \rightarrow \text{A0*}$  (theorems 2.2 to 2.7). The `axiom` environment below is retained for referenceability, but the content is a derived criterion, not a postulate.

*Lean formalization note.* In the companion Lean 4 development, the forward direction of A0\* ( $\text{IsReal } \delta \Rightarrow \exists w, \dots$ ) is derived in `EndogenousMeaning.lean` from the `Nothingness` structure via theorems N1–N5. The full biconditional is packaged as the sole `axiom` declaration in

`Axioms.lean` for downstream use; the backward direction encodes the semantic closure that “real” means “witnessed.” When this paper states “no declared axioms,” it refers to the paper’s logical structure; when the artifact description states “one axiom,” it refers to the Lean packaging of the derived biconditional.

**Axiom 1** ( $A0^*$  — Completed Witnessability). *A distinction  $\delta$  is admissible if and only if there exists a finite endogenous witness process  $w$  satisfying all of the following:*

- (W1) **Finite description:**  *$w$  has a finite endogenous code  $\text{code}(w) \in \mathcal{D}^*$ .*
- (W2) **Replayability:**  *$w$  is replayable from its own trace—any agent with access to  $\text{tr}(w, s)$  can re-execute  $w$  and obtain the same outcome.*
- (W3) **Termination:**  *$w$  terminates within finite budget  $B$  on every admissible input.*
- (W4) **Separation:**  *$w$  produces distinct outcomes on descriptions that  $\delta$  distinguishes.*
- (W5) **Disturbance accounting:**  *$w$  records its own state update  $\text{upd}(w, s)$ —the change it induces on the state  $s$ —as part of the trace.*
- (W6) **Endogenous validity:** *the validity of  $w$  is itself witnessable by a process satisfying (W1)–(W5) within the same universe.*
- (W7) **Compositional closure:** *if  $w_1, w_2$  are admissible witness processes, then their ordered composition  $w_1 \triangleright w_2$  is again an admissible witness process.*
- (W8) **Quotient invariance:** *quotient-equivalent encodings of the same state induce equivalent witness traces—the witness does not depend on gauge-level representational choices.*

*Remark 2.11* (Bootstrap Reading of  $A0^*$ ). The formal statement of  $A0^*$  uses concepts (“carrier,” “code,” “budget”) that are themselves derived in Steps 1–3. This is not circular: the *informal* version—“a distinction is admissible iff a finite, replayable, self-accounting witness process can separate it”—is the primitive axiom. The formal (W1)–(W8) listing is the *derived* restatement, expressed in the language that  $A0^*$  itself forces.

*Remark 2.12* (Why  $A0^*$  and not bare  $A0$ ). Bare  $A0$  leaves the notion of “valid witness” as an external meta-concept. This creates a gap: the distinction between valid and invalid witnesses would depend on an unwitnessed externality.  $A0^*$  closes this gap by demanding that witness validity be endogenously checkable. The additional structure (W1)–(W8) is not imported—it is *forced* by applying the witnessability criterion to itself. Each condition is individually necessary:

- Without (W1), witnesses are not finitely describable.
- Without (W2), witness outcomes are not independently verifiable.
- Without (W3), witnesses may consume infinite resources.
- Without (W4), witnesses do not actually separate.
- Without (W5), the distinction between “state before  $w$ ” and “state after  $w$ ” is unwitnessed.
- Without (W6), witness validity is an ungrounded meta-assertion.
- Without (W7), the test algebra is not maximally closed.
- Without (W8), truth depends on representational choices.

Omitting any one of (W1)–(W8) leaves a distinction that depends on an unwitnessed externality, violating the witnessability criterion.

$A0^*$  is the unique self-consistent completion of bare witnessability, derived from  $\perp$  via  $N0$ – $N4$ . It rules out distinctions that require infinite resources, appeals to Platonic existence, unverifiable oracles, or untracked disturbance. In spirit it distills Bishop’s constructivism [Bishop, 1967] and Martin-Löf type theory [Martin-Löf, 1984] into a single self-applied predicate, stated in computational rather than type-theoretic terms.

## 2.5 The Endogenous Boundary Theorem

**Theorem 2.13** (Endogenous Boundary — Closed Universe is Forced). *Under  $A0^*$ , external delimiters, external verifiers, and external oracles are inadmissible. The universe must be informationally closed.*

*Proof sketch.* Suppose an external delimiter  $v$  is used to parse witness traces. Then the distinction between “same raw stream under delimiter  $v_1$ ” and “same raw stream under delimiter  $v_2$ ” depends on  $v$ , which is not itself an endogenous witness process satisfying (W1)–(W6). This violates (W6): the validity of any witness using  $v$  is not endogenously checkable. The same argument applies to external verifiers and oracles. Therefore all delimiters, verifiers, and composition rules must be internally encoded and replayable. *Forcing:* External dependencies create unwitnessed validity distinctions. Endogenous closure is forced by (W6).  $\square$

*Remark 2.14* (Closed Universe as L1, not L2). In earlier drafts, the closed-universe assumption was classified as a modeling commitment (L2). Theorem 2.13 shows it is a consequence of  $A0^*$  (L1): the self-application of witnessability to witness validity forces endogenous closure. This eliminates one of the former “model-class commitments.”

## 2.6 The Ordered Witness Algebra

$A0^*$  forces not just individual witness processes but an algebraic structure on the collection of all admissible witnesses.

**Definition 2.15** (Ordered Witness Algebra  $\mathcal{A}_W$ ). *The ordered witness algebra  $\mathcal{A}_W$  is the set of all admissible witness processes under  $A0^*$ , equipped with:*

- (a) *Ordered composition:  $w_1 \triangleright w_2$  denotes “execute  $w_1$ , record its trace including state update, then execute  $w_2$  on the updated state.” This is associative but generally not commutative.*
- (b) *Involution:  $w^*$  denotes the replay adjoint—the process that replays  $w$ ’s trace in reverse, reading outcomes as inputs and inputs as outcomes.*
- (c) *Positivity: for every state  $s$ , the outcome weight  $\omega(w^* \triangleright w) \geq 0$ .*
- (d) *Unit: the identity witness  $\mathbf{1}$  (do nothing, record nothing) is the composition identity.*
- (e) *Norm:  $\|w\| = \sup_s \|w(s)\|$  is the operational supremum over admissible states.*

**Proposition 2.16** (Witness Algebra is Forced). *Each component of  $\mathcal{A}_W$  is forced by  $A0^*$ :*

- *Ordered composition is forced by (W7) and (W5): composition must be closed, and state updates must be tracked, so execution order matters.*
- *The involution is forced by (W2): replayability requires the existence of a reverse-reading process for every witness.*
- *Positivity is forced by (W4): outcome weights from actual separations are non-negative.*
- *The unit is forced by (W7): the identity process is a trivially valid witness.*
- *The norm is forced by (W3): finite budget bounds all witness actions.*

*Forcing: The algebraic structure of  $\mathcal{A}_W$  is the unique structure compatible with (W1)–(W8). Omitting any component leaves an algebraic gap that permits unwitnessed distinctions.*

**Definition 2.17** (Witness Commutator). *For witness processes  $w_1, w_2 \in \mathcal{A}_W$ , the commutator is:*

$$[w_1, w_2] = (w_1 \triangleright w_2) - (w_2 \triangleright w_1).$$

*When  $[w_1, w_2] = 0$ , the witnesses commute and their execution order does not affect the truth quotient. When  $[w_1, w_2] \neq 0$ , the witnesses are non-commuting and order matters—each disturbs the state in a way that affects the other’s outcome.*

*Remark 2.18* (Noncommutativity is Internal from the Start). The ordered witness algebra  $\mathcal{A}_W$  is generally noncommutative because (W5) tracks state disturbance. Commutativity is not a global assumption but a *derived property* of specific witness pairs. The commuting sector  $Z(\mathcal{A}_W)$  (the center of  $\mathcal{A}_W$ ) is the subalgebra of witnesses that commute with all others. The classical “Diamond Law” of earlier drafts becomes a theorem about  $Z(\mathcal{A}_W)$ , not a global axiom. This resolves the quantum gap: noncommuting observables are internal to the framework from Step 0.

## 2.7 Consequences of $A0^*$

We now derive the structural consequences of  $A0^*$  that form the substrate for the full derivation in section 6.

**Definition 2.19** (Carrier Set  $\mathcal{D}^*$ ). *The carrier is the set of all finite binary strings:*

$$\mathcal{D}^* = \{0, 1\}^{<\infty} = \{\varepsilon, 0, 1, 00, 01, 10, 11, 000, \dots\}.$$

**Proposition 2.20** (Carrier is Forced). *Under  $A0^*$ , every admissible witness has finite length (W1, W3). Therefore every admissible description is a finite string over a finite alphabet. Up to recoding, this is  $\mathcal{D}^* = \{0, 1\}^{<\infty}$ .*

*Forcing: Any finite alphabet  $\Sigma$  with  $|\Sigma| \geq 2$  is inter-translatable with  $\{0, 1\}$  by a computable bijection (W8). An alphabet of size 1 admits no distinctions. Binary is therefore the minimal non-trivial choice, and all others reduce to it.*

**Proposition 2.21** (Self-Delimitation is Forced by Replayability). *A witness trace must be replayable from its own finite record (W2). If boundaries are not internally decidable, then the distinction between “trace =  $x$ ” and “trace begins with  $x$ ” is not witnessable from the trace itself. Therefore self-delimiting (prefix-free) coding is forced. Any valid encoding must satisfy the Kraft inequality [Kraft, 1949]:*

$$\sum_{x \in S} 2^{-|x|} \leq 1.$$

*Forcing: Self-delimitation is not a “closed-universe extra.” It is  $A0^*$  applied to replayability (W2).*

**Definition 2.22** (Endogenous Witness Instruments). *An endogenous witness instrument is a witness process  $w \in \mathcal{A}_W$  with:*

- a code  $\text{code}(w) \in \mathcal{D}^*$ ,
- an outcome map  $\text{out}(w, s) \in \mathcal{D}^*$  for each state  $s$ ,
- a state-update map  $\text{upd}(w, s)$  recording disturbance,
- a cost  $c(w) \in \mathbb{N}_{>0}$ .

*When the update map is trivial ( $\text{upd}(w, s) = s$  for all  $s$ ), we call  $w$  a pure test (non-disturbing). The set of all pure tests is denoted  $\Delta$ . The full set of witness instruments (including disturbing ones) is  $\mathcal{A}_W$ .*

**Definition 2.23** (Executability Primitives). *The executability substrate consists of:*

- (a) A total evaluator  $\mathcal{U}$  such that  $\mathcal{U}(w, s)$  halts for all  $w, s \in \mathcal{D}^*$ , returning both outcome and state update.
- (b) A total decoder  $\text{dec}: \mathcal{D}^* \rightarrow \mathcal{D}^* \times \mathcal{D}^*$  parsing self-delimiting streams.
- (c) A cost functional  $c: \mathcal{A}_W \rightarrow \mathbb{N}_{>0}$  with  $c(w) \geq 1$ .

**Proposition 2.24** (Executability is Forced).  *$A0^*$  demands that all finite witness procedures be admissible (W7). The closure of all such procedures under composition and finite description is precisely a universal evaluator  $\mathcal{U}$  [Rogers, 1967].*

*Forcing: If some finite procedure were excluded from the admissible set, then there would exist a distinction (between “inside” and “outside” the admissible set) that no admissible witness could detect—violating  $A0^*$ . Hence the set of witnesses must be maximally closed. Maximal closure under totality-preserving composition on a countable carrier yields exactly the total computable functions, and a universal evaluator is forced.*

**Definition 2.25** (Finite Slice  $D_0$ ). *For a given budget  $B$ , the finite slice is the operationally reachable subset of the carrier:*

$$D_0 = \{s \in \{0, 1\}^* \mid |s| \leq B\} \subset \mathcal{D}^*.$$

*All truth-quotient operations and witness executions are restricted to  $D_0$ . The full carrier  $\mathcal{D}^*$  serves as the ambient space.*

*Remark 2.26* (The Church–Turing Corroboration). Theorem 2.24 derives the class of total computable functions from  $A0^*$  alone: maximal closure under totality-preserving composition on a countable carrier is a mathematical theorem, not an empirical claim [Rogers, 1967]. The Church–Turing thesis independently identified this same class by a different route—confirming the derivation’s conclusion, not entering it as a premise.

## 2.8 Summary of the Axiomatic Base

The framework has *no declared axioms*— $A0^*$  is the first theorem, derived from  $\perp$  via the necessity chain N0–N4. Table 1 collects the logical dependencies. Everything downstream of this section is *derived*, not postulated. Notably, all items formerly classified as “modeling commitments” (L2) are now derived consequences of  $A0^*$  (L1):

- *Closed universe*: forced by (W6) via theorem 2.13.
- *Self-delimitation*: forced by (W2) via replayability.
- *Ordered witness ledger*: forced by (W5) and (W7) via disturbance accounting and compositional closure.
- *Noncommutativity*: internal from (W5); commutativity is a derived sector property.

Table 1: Axiomatic base and immediate consequences under  $A0^*$ .

| Item                                         | Status                                 | Forced by        |
|----------------------------------------------|----------------------------------------|------------------|
| $A0^*$ (Completed Witnessability)            | <i>Derived from <math>\perp</math></i> | N0–N4            |
| Carrier $\mathcal{D}^* = \{0, 1\}^{<\infty}$ | <i>Derived</i>                         | (W1), (W3)       |
| Finite slice $D_0 \subset \mathcal{D}^*$     | <i>Derived</i>                         | (W3), Budget $B$ |
| Self-delimiting syntax $sd$                  | <i>Derived</i>                         | (W2)             |
| Closed universe / endogenous boundary        | <i>Derived</i>                         | (W6)             |
| Ordered witness algebra $\mathcal{A}_W$      | <i>Derived</i>                         | (W5), (W7)       |
| Witness instruments (incl. disturbing)       | <i>Derived</i>                         | (W4), (W5)       |
| Pure tests $\Delta \subset \mathcal{A}_W$    | <i>Derived</i>                         | (W4), $upd = id$ |
| Executability $(\mathcal{U}, dec, c)$        | <i>Derived</i>                         | (W1), (W3), (W7) |
| Noncommutativity (internal)                  | <i>Derived</i>                         | (W5)             |

With these objects in hand—a carrier  $\mathcal{D}^*$  with finite slice  $D_0$ , a self-delimiting syntax  $sd$ , an ordered witness algebra  $\mathcal{A}_W$  with pure-test subalgebra  $\Delta$ , executability primitives  $(\mathcal{U}, dec, cost)$ , and endogenous closure—we have the complete substrate from which the derivation of section 6 proceeds.

## 3 The Three Primitives

The derivation from nothingness forces exactly three primitive concepts into existence. They are not postulated; they are the unique consequences of applying  $A0^*$  to its own products. This section names them, states their definitions, and traces each to the derivation step where it first appears.

### 3.1 $\Delta$ — Tests (Distinctions)

**Definition 3.1** (Tests). A test  $\tau \in \Delta$  is a total computable function that separates descriptions. The set of all tests  $\Delta^*$  is the least fixed point of closure rules on the base tests  $\Delta_0$ , enumerated canonically by  $E_\Delta$  (Step 4, theorem 6.7).

Tests are the only epistemic currency of the framework. A distinction without a separating test is operationally void—it violates  $A0^*$ . Every claim, every piece of evidence, every advance

in knowledge reduces to the execution of a test and the recording of its outcome. Tests emerge at Steps 3–4 of the derivation: Step 3 forces the computational substrate that can execute them, and Step 4 forces their closure into a self-generating algebra  $\Delta^*$ .

### 3.2 $\Pi$ — Truth

**Definition 3.2** (Truth). *Truth is the information that survives all valid tests. Formally, the truth quotient  $\Pi^*(\mathcal{L}) = D_0/\equiv_{\mathcal{L}}$  identifies descriptions that no recorded test can distinguish (Step 6, theorem 6.9).*

Truth is not a Platonic ideal; it is the coarsest structure that survives all executed tests. Two descriptions are “the same truth” precisely when no test separates them. The truth quotient carries additional forced structure: an observable-open topology  $\mathcal{O}_{\Delta}^{\text{fin}}$ , an eraser algebra  $\mathcal{E}$ , and a self-generating fixed point  $\text{sim}^*$  (Step 7, theorem 6.11). Together these make truth not merely a quotient but a topological algebra of testability. The concrete algorithm that computes canonical representatives—selecting the lexicographically minimal serialization within each truth class—is  $\Pi_{\text{can}}$ -canonicalization (Definition 6.24).

### 3.3 $\mathcal{T}$ — Time

**Definition 3.3** (Time). *Time is the irreversible formation of records. Each new entry appended to the ledger  $\mathcal{L}$  advances the clock by  $\Delta\mathcal{T} = \log(|W()|_{\text{pre}}/|W()|_{\text{post}}) \geq 0$  (Step 8, theorem 6.12).*

Time is not a parameter imposed from outside; it is the irreversibility of witnessing. The ledger is append-only (Step 5): deleting a record would destroy a witness, violating  $A0^*$ . Because each record can only refine the truth quotient (never coarsen it), the sequence of records defines an arrow—from less resolved to more resolved—that cannot be reversed. This arrow is time. The energy ledger  $\mathcal{L}_E$  (Step 10) tracks the cost of each advance.

### 3.4 $\Pi_{\infty}$ — The Truthpoint

**Definition 3.4** (Truthpoint). *The truthpoint  $\Pi_{\infty}$  is the limit of  $\Pi^*(\mathcal{L})$  as the ledger grows without bound:*

$$\Pi_{\infty} = \lim_{|\mathcal{L}| \rightarrow \infty} \Pi^*(\mathcal{L}).$$

*It is the ideal complete knowledge state—the truth quotient under unlimited budget—approached but never operationally reached.*

$\Pi_{\infty}$  exists as a limit concept (Step 11, theorem 6.15): when the budget is exhausted, the system reaches operational nothingness  $\perp_{\text{op}}$ , the horizon of resolvability. The gap between any finite  $\Pi^*(\mathcal{L})$  and  $\Pi_{\infty}$  is precisely what the  $\Omega$  state reports: the surviving ambiguity, together with the minimal separator test that would reduce it.

**Summary.** These three primitives— $\Delta$  (tests),  $\Pi$  (truth),  $\mathcal{T}$  (time)—together with the limit concept  $\Pi_{\infty}$ , are not design choices. They are the unique forced consequences of  $A0^*$  applied to its own products. Every subsequent structure in the derivation (gauge invariance, Myhill–Nerode congruence, separator geometry, Bellman dynamics, self-hosting) is built from these three primitives and nothing else.

## 4 Seven Non-Negotiable Doctrines

The three primitives ( $\Delta$ ,  $\Pi$ ,  $\mathcal{T}$ ) and the derivation from nothingness impose seven constraints on any system that reasons under  $A0^*$ . These are not design choices or engineering heuristics—they are forced consequences of the axiom. Each doctrine is traced to the derivation step that forces it.



- D1. Everything must be finitely describable.** Every admissible object is a finite binary string in  $\mathcal{D}^*$ . Infinite descriptions require infinite witnesses, violating  $A0^*$ . (*A0\*, Step 1; theorem 6.2*)
- D2. Meaning requires testable differences.** A distinction without a separating test  $\tau \in \Delta^*$  is operationally void. If no finite procedure can detect the difference, the difference does not exist within the framework. (*Steps 3–4; theorem 6.4, theorem 6.7*)
- D3. Truth emerges from feasible tests.** The truth quotient  $\Pi^*(\mathcal{L})$  is the coarsest structure surviving all executed tests. Truth is not assumed; it is constructed by the accumulation of test outcomes. (*Steps 6–7; theorem 6.9, theorem 6.11*)
- D4. Time is an irreversible ledger.** Each record appended to  $\mathcal{L}$  advances the clock. Reversal would destroy a witness, violating  $A0^*$ . Time is the irreversibility of record formation. (*Step 8; theorem 6.12*)
- D5. The truthpoint  $\Pi_\infty$  exists as a limit concept.** The ideal truth quotient under unlimited budget is approached but never operationally reached. The gap between any finite  $\Pi^*(\mathcal{L})$  and  $\Pi_\infty$  is precisely what  $\Omega$  reports. (*Step 11; theorem 6.15*)
- D6. Completeness demands enumeration of lawful tests.** The canonical enumeration  $E_\Delta$  over  $\Delta^*$  ensures no admissible test is overlooked. The enumeration order is determined by the self-delimiting syntax  $sd$ , requiring no external scheduler. (*Step 4; theorem 6.7*)
- D7. Claims require finite witnesses or explicit gaps.** Every output is either UNIQUE (verified answer with witness) or  $\Omega$  (unknown, with the minimal separator test needed for resolution). There is no probabilistic hedging, no confidence score, no ungrounded assertion. (*Step 0; theorem 6.1*)

*Remark 4.1.* These doctrines are *consequences* of  $A0^*$  and the derivation, collected here for clarity. They add no axiomatic content beyond what the derivation establishes. Any system that violates a doctrine necessarily violates  $A0^*$  at the cited step.

## 5 Qualitative and Quantitative Forcing

Before proceeding to geometry and physics, we prove the two master theorems that bridge nothingness to structured reality. From  $\perp$ , there are no external labels, names, coordinates, units, or scales. This forces all qualitative structure to be quotient-invariant and all quantitative structure to be internal-comparison-invariant.

### 5.1 Qualitative Forcing

**Theorem 5.1** (Q1 — Qualitative Forcing). *If no external labels are admissible, then all qualitative structure is quotient-invariant structure of the witness algebra.*

*Proof.* From  $\perp$ , there are no external labels, names, or tags.  $A0^*$  says a distinction is real iff a finite endogenous replayable witness separates it. Two candidate objects  $x, y$  that no admissible witness separates are *indistinguishable*:

$$x \sim y \iff \forall w \in \mathcal{A}_W, \text{ out}(w, x) = \text{out}(w, y).$$

Any property  $P$  that assigns different values to indistinguishable objects would require an external label to distinguish them—but no external labels exist (from  $\perp$ ). By  $A0^*$ , the distinction

“ $P(x) \neq P(y)$  despite  $x \sim y$ ” has no witness, hence is not real. Therefore every meaningful property is quotient-invariant:

$$x \sim y \implies P(x) = P(y).$$

*Forcing:* No external labels  $\Rightarrow$  only witness-stable distinctions  $\Rightarrow$  qualitative quotient structure. This forces: identity (persistence under witnessing), objecthood (stable distinction class), property (invariant of a class), relation (witnessable comparison class), topology (observable-open structure from tests), causality (refinement order in the closure boundary), modality (survivor-class structure of continuations).  $\square$

## 5.2 Quantitative Forcing

**Theorem 5.2** (Q2 — Quantitative Forcing). *If no external units are admissible, then all quantitative structure is determined only up to internal scale gauge, and only gauge-invariant combinations are physically real.*

*Proof.* From  $\perp$ , there are no external units, absolute scales, or externally chosen coordinates. Any numerical assignment  $f(x)$  that changes under unit rescaling  $d \rightarrow \lambda d$  is a bare number with no witnessable content—the distinction “ $f = 3$  in units  $u_1$ ” vs. “ $f = 7$  in units  $u_2$ ” has no witness unless a unit-independent comparison is available. By A0\*, only witnessable distinctions are real. Therefore only gauge-invariant combinations of quantities are physically meaningful. *Forcing:* No external units  $\Rightarrow$  only internally comparable costs/frequencies/refinements  $\Rightarrow$  quantitative invariants only up to scale gauge.  $\square$

**Theorem 5.3** (Q3 — Gauge-of-Units). *Any two quantitative presentations related by unit rescaling or coordinate relabeling represent the same physical content iff all invariant ratios and normalized evaluations agree.*

*Proof sketch.* This is the quantitative analogue of Q1. Just as two qualitatively indistinguishable objects are the same truth class, two quantitatively gauge-related assignments are the same physical quantity. The proof is the same: distinguishing them would require an external unit standard, which violates  $\perp$ .  $\square$

**Theorem 5.4** (Q4 — Canonical Quantities). *The only physically meaningful numbers are those invariant under the unit gauge group. Every admissible quantity is one of:*

- (1) a dimensionless invariant,
- (2) a ratio of like quantities,
- (3) a generator of an internal comparison group,
- (4) a normal form under scale gauge.

*Proof sketch.* Any quantity not of these forms depends on an external unit choice, which is not witnessable from within. By A0\*, it is not real. This forces every downstream quantity to be defined as an internal invariant:

- time from irreversible ledger accumulation (not external seconds),
- entropy from unresolved multiplicity (not external thermodynamics),
- energy from internal witness cost of stabilizing distinctions,
- distance from minimal separative comparison cost,
- probability from normalized positive state evaluation,
- curvature from variation of the internal distinguishability metric.

*Forcing:* No external scale  $\Rightarrow$  every quantity is defined only up to internal gauge, and only gauge-invariant combinations are real.  $\square$



*Remark 5.5* (Bridge to Downstream Structure). Q1–Q4 are the bridge between nothingness and the derivation chain. They ensure that every structure derived in section 6 and beyond is not merely “consistent with A0\*” but *forced* by it:

- The truth quotient (Step 6) is forced by Q1: indistinguishable objects must be identified.
- Gauge invariance (Step 12) is forced by Q1 + Q3: untestable relabelings are qualitative gauge.
- The witness-path metric (Step 16) is forced by Q4: distance is the minimal internal separative comparison cost.
- The conductance exponent  $w = C/d^2$  (Step 28) is forced by Q2: scale covariance is a consequence of no external units.
- The Born rule (theorem 7.12) is forced by Q4: probability is the unique normalized positive state evaluation.

### 5.3 The Static Universe

The  $\Pi$ -fixed closure whole is atemporal. Time is a local phenomenon arising from incomplete self-resolution, not a property of the universe as a whole.

**Theorem 5.6** (S0 — Whole-Before-Time). *The full  $\Pi$ -fixed closure whole  $U = \text{Fix}(\Pi)$  is atemporal: applying  $\Pi$  to  $U$  changes nothing. All admissible local states are partial sections of  $U$ .*

*Proof.* By definition,  $U = \text{Fix}(\Pi)$  satisfies  $\Pi(U) = U$ . Therefore no witness applied to  $U$  can refine the truth quotient further—the fiber  $W(p)$  is a singleton for every truth class  $p$  in  $U$ , so no test produces new information. By Step 8 (theorem 6.12), time is the irreversible formation of new records, measured by  $\Delta\mathcal{T} = \log(|W()|_{\text{pre}}/|W()|_{\text{post}}) \geq 0$ . When the fiber is already minimal ( $|W()| = 1$ ), every test yields  $\Delta\mathcal{T} = 0$ : no time passes. The whole  $U$  is therefore atemporal.

Any admissible local state  $s$  satisfies  $s \subseteq U$  (since  $U$  contains all admissible truth classes) but in general  $\Pi(s) \neq s$ —the local truth quotient is not yet fully resolved. Hence  $s$  is a partial section of the completed whole. *Forcing:*  $U = \text{Fix}(\Pi)$  is forced by the closure construction. Atemporality follows from Step 8: when nothing remains to resolve,  $\Delta\mathcal{T} = 0$  identically. Local states are partial sections because they are subsets of the complete truth quotient that have not yet exhausted their witness budget.  $\square$

**Theorem 5.7** (S1 — Time is Local Defect-Resolution). *If  $s$  is a local incomplete section of  $U$ , then “time” is the monotone reduction of unresolved witness-defect  $\Delta_Q(s) = \Pi(s) \setminus s$ , not motion of  $U$  itself. The universe is static globally, dynamic only as local self-resolution.*

*Proof.* By theorem 6.32 (Step 23), the closure-defect  $\Delta_Q(s) = \Pi_Q(s) - s$  measures the gap between the current state and its truth projection. Its support  $r_t = \text{supp}(\Delta_Q(s_t))$  decreases monotonically with each test (Step 8):  $|r_{t+1}| \leq |r_t|$ . This monotone decrease *is* time for the local section  $s$ .

For the global whole  $U = \text{Fix}(\Pi)$ , the closure-defect vanishes:  $\Delta_Q(U) = \Pi(U) - U = 0$ . Therefore  $U$  has no unresolved components and no time evolution. “Dynamics” is exclusively a local phenomenon: the self-resolution of a partial section’s witness-defect toward the already-existing fixed whole. *Forcing:* The global whole is static because  $\Delta_Q(U) = 0$  (no defect to resolve). Local time is the monotone decrease of  $|r_t|$ , which is forced by the append-only ledger (Step 5) and the irreversibility of record formation (Step 8). The universe does not “move”—local sections resolve toward it.  $\square$

## 5.4 The Quantitative Seed

The structural forcing of Sections 5.1–5.3 determines the *form* of all physical law. To pin down the *values* of dimensionless observables, we isolate the unique quantitative seed  $\delta^*$ .

**Definition 5.8** (Defect Space and Action). *Let  $D$  denote the space of closure-defect profiles (lists of component-state fibers). The gauge group  $G$  acts by permuting distinction labels, preserving total defect. The action functional  $A : D/\sim_G \rightarrow \mathbb{N}$  maps each gauge class to its total defect. A defect is self-retaining if every open component has fiber 1 (minimum nonzero), making it a fixed point under witness accumulation. A defect is non-gauge if  $A(d) > 0$ .*

**Theorem 5.9** (Seed Existence and Uniqueness). *There exists a unique (up to gauge) self-retaining non-gauge defect  $\delta^*$  minimizing  $A$ . Concretely:  $A(\delta^*) = 1$ , and every non-gauge defect has  $A \geq 1$ .*

*Proof.* The class  $D_{\text{sr}}$  of self-retaining non-gauge defects is nonempty (witness: a single open component with fiber 1). The action maps to  $\mathbb{N}$ , which is well-ordered. A minimizer exists by well-foundedness on a nonempty set. Any two minimizers have the same action value by symmetric minimality.  $\square$

**Theorem 5.10** (Seed is Renormalization Fixed Point). *For any refinement operator  $R$  that is monotone, gauge-preserving, and self-retaining-preserving,  $A(R(\delta^*)) = A(\delta^*)$ .*

*Proof.* By monotonicity  $A(R(\delta^*)) \leq A(\delta^*)$ ; by minimality  $A(\delta^*) \leq A(R(\delta^*))$ .  $\square$

## 5.5 Quantitative Closure

The linearized refinement operator  $L^* = DR|_{\delta^*}$  acts on the tangent space at the seed. Its spectral, holonomy, and normal-form data determine every dimensionless observable.

**Theorem 5.11** (Dimensionless Observable Decomposition). *Every dimensionless observable  $\mathcal{O}$  satisfies*

$$\mathcal{O} = F(\text{Spec}(L^*), \text{Hol}(L^*), \text{NF}(L^*)).$$

*Proof.* Six-step ladder: (1) gauge invariance  $\Rightarrow$  quotient-defined; (2) quotient  $\Rightarrow$  seed-determined (via uniqueness); (3) linear invariants = spectral data; (4) gauge-sector invariants = holonomy; (5) interaction invariants = normal-form coefficients; (6) corollary:  $\mathcal{O}$  factors through (Spec, Hol, NF).  $\square$

Dimensional quantities (mass, coupling,  $\Lambda$ ) are fixed up to the internally derived normalization map ( $\hbar^* = A(\delta^*)$ ,  $c^* = \sup d_{\text{sep}}/\Delta T$ ). No external unit system enters.

## 6 The Forced Derivation

This section presents the core contribution: a 34-step derivation (Steps 0–33) showing that the structure of witnessable reasoning is *forced*, not chosen. Starting from absolute nothingness ( $\perp$ ), we show that A0\* (axiom 1), derived from  $\perp$  via N0–N4, uniquely determines a carrier, a self-delimiting syntax, an executability layer, an *ordered* ledger carrying the witness algebra  $\mathcal{A}_W$ , a truth quotient with observable-open topology, time, energy, gauge invariance, a Myhill–Nerode congruence, witness-path geometry, Bellman-optimal dynamics governed by the Coupling Law, a self-hosting closure, a universal execution law with consciousness and trit-field valuation, and a Kähler geometric structure on the space of truths. The ordered witness algebra  $\mathcal{A}_W$  is the foundational algebraic object: its noncommutativity is internal from the start, making the multiset (commutative) sector a derived quotient rather than a primitive. Each step takes

the form of a theorem with an explicit *forcing argument*: a proof that the step is the unique  $A0^*$ -compliant continuation. The derivation DAG is depicted in fig. 10.

**Notation.** We use  $\mathcal{D}^*$ ,  $\Delta$ ,  $\Pi$ ,  $\mathcal{T}$ ,  $\mathcal{L}$ ,  $\mathcal{U}$ ,  $\perp$ ,  $\perp_{\text{op}}$ , and the witness algebra  $\mathcal{A}_W$  as defined in section 2. New objects introduced in this derivation— $D_0$ ,  $\text{sd}$ ,  $\mathcal{O}_{\Delta}^{\text{fin}}$ ,  $\mathcal{E}$ ,  $\Upsilon$ ,  $\equiv_q$ ,  $\mathcal{L}_{\text{ord}}$ ,  $\triangleright$ ,  $d_{\text{sep}}$ ,  $\tau_{\text{dir}}$ ,  $\mathbf{C}_W$ ,  $\mathcal{G}_{\text{ex}}$ ,  $\mathcal{L}_E$ ,  $\text{Cl}_{\min}$ ,  $E_{\Delta}$ ,  $\Upsilon(\mathcal{S})$ —are defined at their point of first appearance.

**Structure.** The derivation is organized into nine phases:

- **Phase A** (Step 0): The only law— $A0^*$  and the forced output gate.
- **Phase B** (Steps 1–2): Carrier and syntax.
- **Phase C** (Steps 3–4): Executability.
- **Phase D** (Steps 5–7): Recording and truth.
- **Phase E** (Steps 8–11): Time, energy, and limits.
- **Phase F** (Steps 12–17): Invariance and control.
- **Phase G** (Steps 18–21): Solver with coupling law, self-hosting, and closure.
- **Phase H** (Steps 22–25): Universal execution.
- **Phase I** (Steps 26–33): Geometric reconstruction.

## 6.1 Phase A: The Only Law (Step 0)

**Theorem 6.1** (Step 0 — Nothingness  $\perp$  and the Forced Output Gate). *Nothingness is the state with no admissible distinctions:  $\perp : \Delta = \emptyset$ ,  $\mathcal{D}^* = \emptyset$ .  $A0^*$  forces a trichotomous output gate: any query  $q$  must terminate in exactly one of  $\text{UNIQUE} (+1)$ ,  $\Omega (0)$ , or  $\text{UNSAT} (-1)$ .*

*Proof sketch.* If no witness procedure exists then no distinction is admissible ( $A0^*$ ), so  $\perp$  is the unique structureless state. After all affordable tests on a query with candidate set  $\mathcal{A}$ , the surviving set  $\mathcal{A}_S$  has  $|\mathcal{A}_S| = 1$  ( $\text{UNIQUE}$ ),  $|\mathcal{A}_S| > 1$  ( $\Omega$ ), or  $|\mathcal{A}_S| = 0$  ( $\text{UNSAT}$ )—three cases that partition the non-negative integers by pigeonhole. *Forcing:* The trichotomy is forced by pigeonhole on  $|\mathcal{A}_S| \in \{0, 1, 2, \dots\}$  collapsed into three regimes. No fourth case exists. See Appendix A for the complete proof.  $\square$

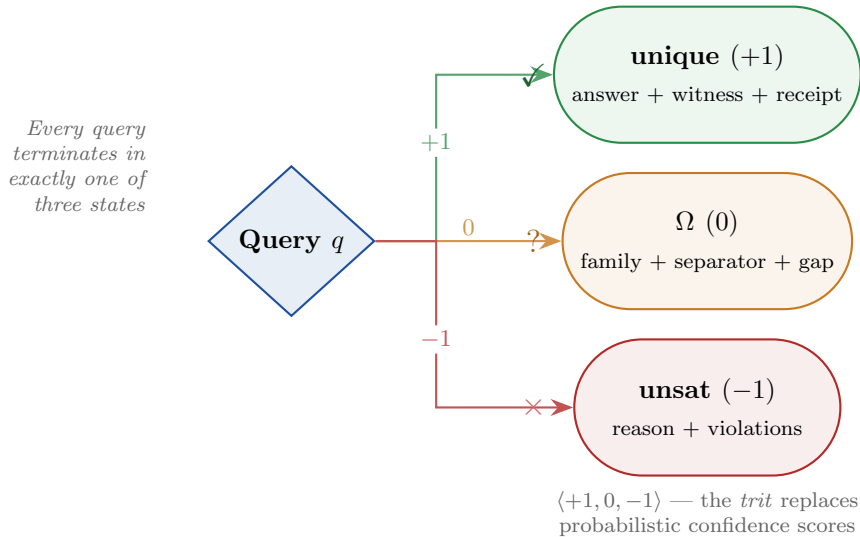


Figure 2: The three terminal states forced by the output gate (Step 0). Every query  $q$  resolves to exactly one trit value:  $\text{UNIQUE} (+1)$  with a verified witness,  $\Omega (0)$  with the minimal separator test needed to resolve, or  $\text{UNSAT} (-1)$  with specific violations identified. No probabilistic hedging is permitted.

## 6.2 Phase B: Carrier and Syntax (Steps 1–2)

**Theorem 6.2** (Step 1 — Carrier  $\mathcal{D}^*$  with Finite Slice  $D_0$ ).  *$A0^*$  forces the carrier  $\mathcal{D}^* = \{0, 1\}^{<\infty}$ . For budget  $B$ , the finite slice is  $D_0 = \{s \in \{0, 1\}^* \mid |s| \leq B\}$ . Dependencies:  $A0^*$ , Step 0.*

*Proof sketch.*  $A0^*$  requires finite witness procedures, hence finite strings over some finite alphabet  $\Sigma$ . Any  $|\Sigma| \geq 2$  is computably isomorphic to  $\{0, 1\}$ ; unary admits no internal distinctions. Thus  $\mathcal{D}^* = \{0, 1\}^{<\infty}$  is ontologically unique up to computable isomorphism.  $D_0$  is forced because a witness of cost  $B$  cannot access strings longer than  $B$  bits. *Forcing:* Departure from  $\perp$  requires at least one distinction, hence one witness, hence one finite string. The set of all finite strings is  $\mathcal{D}^*$ . Budget-bounding gives  $D_0$ . ( $A0^*$  forcing.) See Appendix A for the complete proof.  $\square$

**Theorem 6.3** (Step 2 — Self-Delimiting Syntax  $sd$ ). *In a closed universe (no external delimiter oracle), descriptions must be self-delimiting. The canonical self-delimiting map is  $sd(s) = 1^{|s|}0s$ . Dependencies:  $A0^*$ , Step 1.*

*Proof sketch.* Without self-delimitation, a prefix  $x \sqsubseteq y$  creates an ontologically irresolvable ambiguity: no finite procedure can separate “description is  $x$ ” from “description begins with  $x$ ,” violating  $A0^*$ .  $sd(s) = 1^{|s|}0s$  resolves this: read bits until the first 0 to obtain  $|s|$ , then read  $|s|$  bits. The encoding is prefix-free; other self-delimiting encodings are computably isomorphic. *Forcing:* Any non-self-delimiting encoding creates an  $A0^*$ -violating ambiguity. Self-delimitation is the minimal constraint that eliminates it. See Appendix A for the complete proof.  $\square$

## 6.3 Phase C: Executability (Steps 3–4)

**Theorem 6.4** (Step 3 — Executability Primitives).  *$A0^*$  forces three executability primitives (the declared substrate): (a) a total evaluator  $\mathcal{U}$  halting on all inputs, (b) a total decoder  $\text{dec}: \mathcal{D}^* \rightarrow \mathcal{D}^* \times \mathcal{D}^*$ , (c) a cost functional  $c: \Delta \rightarrow \mathbb{N}_{>0}$  with  $c(\tau) \geq 1$ . Dependencies:  $A0^*$ , Steps 1–2.*

*Proof sketch.*  $A0^*$  demands every finite witness procedure be admissible, so  $\Delta$  must be maximally closed under totality-preserving composition—which is exactly the class of total computable functions (Proposition; proof in Appendix A). A universal evaluator  $\mathcal{U}$  is forced by the  $s$ - $m$ - $n$  and universal function theorems [Rogers, 1967]. The decoder follows from self-delimiting syntax (Step 2); the cost functional from  $A0^*$ ’s finiteness requirement. *Forcing:* (i)  $A0^*$  forces maximal closure of  $\Delta$  under totality-preserving composition. (ii) This class equals the total computable functions. (iii)  $\mathcal{U}$  is forced. (iv) The form of  $\mathcal{U}$  is gauge (Step 12). (v) CT enters only as a naming convention for the forced class. See Appendix A for the complete proof.  $\square$

*Remark 6.5* (CT as Naming Convention). The mathematical content of Step 3 is a proposition of this paper: maximal closure under totality-preserving composition on a finite alphabet is exactly the class of total computable functions (proof in Appendix A). The key identification of “finite procedure” (informal, from  $A0^*$ ) with “total computable function” (formal) is the Church–Turing thesis.

The Church–Turing thesis enters only as the identification of an informal concept (“finite procedure,” from  $A0^*$ ’s natural-language phrasing) with a formal one (“total computable function”). This is analogous to how “number” is identified with “element of  $\mathbb{N}$ ”—a naming convention, not an empirical hypothesis.

The model choice (Turing machine vs.  $\lambda$ -calculus vs. register machine) is gauge (formalized at Step 12): the truth quotient is invariant under switching between isomorphic computation models. A reader who replaces “Turing-computable” with a weaker class (e.g., primitive recursive functions) obtains a weaker but structurally identical derivation.

Thus Step 3 is forced:  $A0^*$ 's maximal closure yields the total computable functions (see Appendix A for the proof; general background in Rogers 1967, Chapters 1–2). The Church–Turing thesis corroborates this conclusion; it does not enter the derivation as a premise.

*Remark 6.6* (Total Semantics). The decoder  $\text{dec}$  and evaluator  $\mathcal{U}$  are total by construction: every input produces a well-defined output. In particular, the outcome alphabet  $A$  of any test  $\tau$  includes explicit FAIL and TIMEOUT outcomes. “Undefined behavior” is not a gap in the semantics but a specific outcome in  $A$ : if  $\mathcal{U}$  exceeds the allotted budget, it returns  $\text{TIMEOUT} \in A$ , not  $\perp$ . This ensures that no test can create a semantic gap—every execution path terminates in a recorded outcome, as required by  $A0^*$ .

**Theorem 6.7** (Step 4 — Endogenous Tests  $\Delta^*$  and Canonical Enumeration  $E_\Delta$ ). *The test algebra is self-generating:  $\Delta^* = \text{lfp}(\text{Cl}_{\min})(\Delta_0)$ . The canonical enumeration  $E_\Delta: \mathbb{N} \rightarrow \Delta^*$  lists tests by sd-length (ties broken lexicographically), requiring no scheduler channel. Dependencies:  $A0^*$ , Steps 1–3.*

*Proof sketch.* The closure rules (composition, branching, bounded iteration) are each individually forced by  $A0^*$ : omitting any one excludes an admissible finite procedure. These rules are monotone on the powerset lattice, so by Knaster–Tarski the least fixed point exists. The enumeration order is forced by sd: any other ordering would require an external channel, violating closure. *Forcing:* Any subset not closed under  $\text{Cl}_{\min}$  excludes a valid test—an  $A0^*$ -violating distinction. The least fixed point is the minimal set avoiding this. See Appendix A for the complete proof.  $\square$

## 6.4 Phase D: Recording and Truth (Steps 5–7)

**Theorem 6.8** (Step 5 — Ordered Ledger  $\mathcal{L}_{\text{ord}}$  and the Diamond Law as Theorem). *Witnesses must persist; therefore an append-only ledger is forced. Under  $A0^*$ , the primitive ledger is ordered:*

$$\mathcal{L}_{\text{ord}} = (e_1 \triangleright e_2 \triangleright \cdots \triangleright e_n),$$

where  $\triangleright$  is ordered witness composition in  $\mathcal{A}_W$  and each entry  $e_i = (\tau_i, a_i, c(\tau_i), \delta_i)$  records the test, outcome, cost, and state-update. Append-only persistence is still forced: deleting  $e_i$  destroys replay of every witness that causally depends on  $e_i$ .

**Witness commutator.** For any two entries  $e_i, e_j$  define the witness commutator

$$[e_i, e_j] = \text{effect}(e_i \triangleright e_j) - \text{effect}(e_j \triangleright e_i).$$

**Diamond Law (Theorem).** When  $[e_i, e_j] = 0$ , the order of  $e_i$  and  $e_j$  does not affect any downstream witness; hence those pairs quotient to a multiset. The Diamond Law is therefore a theorem about the commuting sector of  $\mathcal{A}_W$ , not a global assumption:

$$\text{For } e_i, e_j \text{ with } [e_i, e_j] = 0: \quad \mathcal{N} \circ \mathcal{Q} = \mathcal{Q} \circ \mathcal{N} \text{ on } \Pi^*(\mathcal{L}_{\text{ord}}).$$

The multiset ledger  $\mathcal{L} = \{\{(\tau_i, a_i)\}\}$  of the original formulation is recovered as the quotient of  $\mathcal{L}_{\text{ord}}$  by the commuting sector. Dependencies:  $A0^*$ , Steps 1–4. Figure 12 illustrates the Diamond Law's commutativity of independent ledger entries.

*Proof sketch.* If a witness's output could be erased, subsequent tests depending on it would lack their input, making the distinction unwitnessable—violating  $A0^*$ . Hence an append-only record is forced.  $A0^*$  supplies the ordered witness algebra  $\mathcal{A}_W$ , so the primitive record inherits its ordering:  $\mathcal{L}_{\text{ord}} = (e_1 \triangleright \cdots \triangleright e_n)$ . Each entry records the full 4-tuple  $(\tau_i, a_i, c(\tau_i), \delta_i)$  because omitting any component loses a witnessed datum.

For the Diamond Law: if  $[e_i, e_j] = 0$  then swapping  $e_i$  and  $e_j$  in  $\mathcal{L}_{\text{ord}}$  produces an identical downstream effect—every future witness is replay-equivalent. Order between such pairs

is therefore ontologically untestable and quotients away by the truth quotient (Step 6). When  $[e_i, e_j] \neq 0$ , order matters and the noncommutative structure is physical. *Forcing*: Any mutable record allows  $A0^*$  violations via witness destruction. Ordered structure is forced because  $\mathcal{A}_W$  is noncommutative. The Diamond Law is a theorem about the commuting sub-algebra, not a postulate. See Appendix A for the complete proof.  $\square$

**Theorem 6.9** (Step 6 — Truth Quotient  $\Pi^*(\mathcal{L})$ ). *Descriptions indistinguishable by all tests in the ledger are identified. The resulting quotient  $\Pi^*(\mathcal{L}) = D_0/\equiv_{\mathcal{L}}$  is the truth structure. Dependencies:  $A0^*$ , Steps 1–5.*

*Proof sketch.* Define ledger equivalence:  $x \equiv_{\mathcal{L}} y$  iff no recorded test distinguishes them. This is an equivalence relation by inspection. Maintaining distinctions between elements that no test can separate would assert a witnessless distinction, violating  $A0^*$ . For truth class  $p$ , the fiber  $W(p) = \{x \in D_0 \mid [x]_{\equiv_{\mathcal{L}}} = p\}$  captures the gauge redundancy. *Forcing*: Maintaining distinctions when no test can separate them violates  $A0^*$ . Quotienting is forced. See Appendix A for the complete proof.  $\square$

*Remark 6.10* (Fiber Intuition). The fiber  $W(p)$  is the set of descriptions that “mean the same thing”—all bit-level encodings that no test can tell apart. Quotienting by fibers is the formal version of ignoring irrelevant notation.

**Theorem 6.11** (Step 7 — Indistinguishability as Primary Algebra). *The truth quotient carries three structures forced by  $A0^*$ : observable opens  $\mathcal{O}_{\Delta}^{\text{fin}}$ , an eraser algebra  $\mathcal{E}$ , and a self-generating fixed point  $\text{sim}^*$ . Dependencies:  $A0^*$ , Steps 4–6. The observable-open topology is depicted in fig. 14.*

*Proof sketch.* (a) Observable opens  $\mathcal{O}_{\Delta}^{\text{fin}} = \{O_{\tau,a}\}$  form a topology on  $\Pi^*(\mathcal{L})$ : each  $O_{\tau,a}$  is the set of truth classes producing outcome  $a$  under test  $\tau$ , closed under finite intersection and arbitrary union. (b) The eraser algebra  $\mathcal{E} = \{e_{\tau}\}$  consists of idempotent, monotone, commuting surjections that merge indistinguishable classes. (c)  $\text{sim}^* = \text{lfp}(\Phi)$  captures self-referential closure: truths that survive are exactly those distinguishable by tests drawn from the truths themselves. Existence via Knaster–Tarski on the finite lattice  $2^{\Pi^*(\mathcal{L})}$ . *Forcing*:  $\mathcal{O}_{\Delta}^{\text{fin}}$  is forced because  $A0^*$  equates meaningful with testable distinction.  $\mathcal{E}$  is forced as the canonical idempotent operation on truth quotients. The fixed point is forced because the system must be consistent with its own outputs acting as inputs. See Appendix A for the complete proof.  $\square$

## 6.5 Phase E: Time, Energy, and Limits (Steps 8–11)

**Theorem 6.12** (Step 8 — Observer Collapse and Entropic Time). *Time is the irreversible formation of records. Each new record collapses the fiber:  $\Delta\mathcal{T} = \log(|W()|_{\text{pre}}/|W()|_{\text{post}}) \geq 0$ . Dependencies: Steps 5–7. Figure 11 depicts the monotone fiber shrinkage under successive tests.*

*Proof sketch.* Each new entry  $(\tau, a)$  can only refine the truth quotient (never coarsen) because the ledger is append-only (Step 5) and all tests are total (Step 3). Hence  $|W()|_{\text{post}} \leq |W()|_{\text{pre}}$  and  $\Delta\mathcal{T} \geq 0$ . This is a Second Law: time is the irreversibility of witnessing. *Forcing*: Reversing a ledger entry would destroy a witness (Step 5), violating  $A0^*$ . Time is the irreversibility of record formation. See Appendix A for the complete proof.  $\square$

**Theorem 6.13** (Step 9 — Entropy and Budget). *The entropy of the truth quotient is  $S_p(\mathcal{L}) = \log |W(p)|$ , and the total budget bounds affordable tests:  $\sum_i c(\tau_i) \leq B$ . Dependencies: Steps 5, 8.*



*Proof sketch.* Entropy decreases with each informative test (Step 8).  $A0^*$ 's finiteness requirement forces a budget  $B$ . The feasible test set  $\Delta(\mathcal{T})$  is monotonically non-increasing—an expanding set would require infinite resources at finite time, violating  $A0^*$ . *Forcing:* An expanding test set would require infinite resources at finite time—violating  $A0^*$ . The budget is the entropy available for allocation. See Appendix A for the complete proof.  $\square$

**Theorem 6.14** (Step 10 — Energy Ledger  $\mathcal{L}_E$ ). *Each test incurs energy cost  $\Delta E = c(\tau)$ . Efficiency:  $\varepsilon = \Delta\mathcal{T}/c(\tau)$ . Dependencies: Steps 8–9.*

*Proof sketch.* By Landauer's principle, erasing information requires energy dissipation;  $A0^*$ 's append-only ledger makes record formation irreversible. The energy ledger  $\mathcal{L}_E = [(c(\tau_i), \Delta\mathcal{T}_i), \dots]$  is forced because  $A0^*$  demands finite recording of each distinction's cost; the efficiency ratio is the canonical dimensionless measure. *Forcing:* The energy ledger is forced because  $A0^*$  demands finite cost recording. The efficiency ratio is canonical from the two quantities the framework produces. See Appendix A for the complete proof.  $\square$

**Theorem 6.15** (Step 11 — Operational Nothingness  $\perp_{\text{op}}$ ). *When the budget is exhausted, the feasible test set becomes empty ( $\Delta(\mathcal{T}_{\text{max}}) = \emptyset$ ): operational nothingness. Dependencies: Steps 8–10.*

*Proof sketch.* As  $B_{\text{spent}} \rightarrow B$ , the feasible test set shrinks to the constant function  $\tau_0(x) = c$ , which distinguishes nothing.  $\perp_{\text{op}}$  is the unique terminal state under  $A0^*$ . Unlike  $\perp$ , it occurs *within* a structured universe—the horizon of resolvability. *Forcing:* Finite budget  $\Rightarrow$  finite test sequence  $\Rightarrow$  terminal state.  $\perp_{\text{op}}$  is unique under  $A0^*$ . See Appendix A for the complete proof.  $\square$

## 6.6 Phase F: Invariance and Control (Steps 12–17)

**Theorem 6.16** (Step 12 — Gauge: No Label Privilege). *The set of untestable relabelings forms a group  $\mathcal{G}_{\mathcal{T}} \cong \prod_p \text{Sym}(W(p))$ . Dependencies: Steps 6, 9. Figure 13 illustrates how gauge transformations permute within fibers without affecting the truth quotient.*

*Proof sketch.* A relabeling  $g: D_0 \rightarrow D_0$  permuting within fibers ( $g(W(p)) = W(p)$ ) is untestable under  $A0^*$ . The collection  $\mathcal{G}_{\mathcal{T}}$  is closed under identity (doing nothing is undetectable), inverse ( $g^{-1} \circ g = \text{id}$ ), and composition (C1). Since each  $g$  is a bijection on all of  $D_0$ , composition is always defined, giving a group (the direct product of symmetric groups on fibers). *Forcing:* Each group axiom is individually forced: omitting identity, inverse, or composition from  $\mathcal{G}_{\mathcal{T}}$  would render an untestable transformation detectable, contradicting the truth quotient. See Appendix A for the complete proof.  $\square$

*Remark 6.17* (Gauge Intuition). The gauge group  $\mathcal{G}_{\mathcal{T}}$  is the set of relabelings that no experiment can detect—permuting descriptions within a fiber changes nothing observable. Physical gauge invariance (electromagnetism, general covariance) is a sector-specific instance of this universal principle.

**Theorem 6.18** (Step 13 — Decision Universes: Epistemic vs. Decision Regimes). *The system operates in one of two regimes: epistemic  $\mathcal{R}_{\text{ep}}$  (gathering information) or decision  $\mathcal{R}_{\text{dec}}$  (committing to  $\text{UNIQUE}$  or  $\Omega$ ). Dependencies: Steps 6, 8, 12.*

*Proof sketch.* The system must either execute another test (epistemic: fiber shrinks by evidence) or declare a terminal state (decision: irreversible commitment). Conflating modes would allow asserting answers without evidence or gathering evidence without recording it—both  $A0^*$  violations. Transition  $\mathcal{R}_{\text{ep}} \rightarrow \mathcal{R}_{\text{dec}}$  occurs when  $|\mathcal{A}_S| \leq 1$  or the budget is exhausted. *Forcing:* Conflating epistemic and decision modes permits unwitnessed assertions or destroyed witnesses. Both violate  $A0^*$ . See Appendix A for the complete proof.  $\square$



**Theorem 6.19** (Step 14 — Exactness Gate  $\mathcal{G}_{\text{ex}}$  and Split Law). *Every query is processed certificate-first: check the ledger for a witness (UNIQUE) or counter-witness (UNSAT) before searching for a separator. Every transformation factors as  $\text{merge} \circ \text{relabel}$ . Dependencies: Steps 5–6, 13.*

*Proof sketch.* Certificate-first ordering is forced by budget optimality: searching when a certificate exists wastes budget; declaring  $\Omega$  when a certificate exists withholds an admissible witness. For the Split Law: any total computable  $\bar{f}$  on truth classes factors as  $\bar{f} = \iota \circ q$ , separating information loss from gauge relabeling. *Forcing:* Certificate-first is forced by budget optimality. The Split Law is forced because non-factorable transformations conflate information-destroying and information-preserving components. See Appendix A for the complete proof.  $\square$

**Theorem 6.20** (Step 15 — Myhill–Nerode Congruence  $\equiv_q$ ). *Two states  $s, s'$  are Myhill–Nerode equivalent ( $s \equiv_q s'$ ) iff all futures agree:  $s \equiv_q s' \iff \forall \sigma \in \Delta^{*\omega} : \text{outcome}(s, \sigma) = \text{outcome}(s', \sigma)$ .  $\equiv_q$  is a right-congruence on the monoid  $(\Delta^*, \circ)$ . Dependencies: Steps 4, 6, 12. Figure 15 illustrates the Myhill–Nerode congruence classes and the resulting minimal automaton.*

*Proof sketch.* The state  $s = (\Pi^*(\mathcal{L}), \Delta(\mathcal{T}), B - B_{\text{spent}})$  is futures-equivalent to  $s'$  when no continuation distinguishes them. Right-congruence: if  $s \equiv_q s'$  then  $s \cdot \tau \equiv_q s' \cdot \tau$ . The quotient  $\Delta^{*\omega} / \equiv_q$  is finite (since  $D_0$  is finite), yielding a unique minimal automaton by the classical Myhill–Nerode theorem. *Forcing:* Maintaining a distinction when all futures agree violates  $A0^*$ . The Myhill–Nerode quotient is the coarsest  $A0^*$ -consistent partition, hence forced. See Appendix A for the complete proof.  $\square$

**Theorem 6.21** (Step 16 — Witness-Path Geometry  $(d_{\text{sep}}, \tau_{\text{dir}})$ ). *The truth classes carry a witness-path geometry consisting of a symmetric metric  $d_{\text{sep}}$  and a directed quasi-metric  $\tau_{\text{dir}}$ , both induced by the witness path category. Dependencies: Steps 4, 6, 15. The separator geometry is depicted in fig. 16.*

**Witness path category.** Define  $\mathbf{C}_W$  with truth classes as objects and replayable witness paths  $\gamma: p \rightarrow q$  as morphisms. A morphism  $\gamma$  is a finite sequence of tests in  $\mathcal{L}_{\text{ord}}$  that transforms  $p$  into  $q$ ; composition is path concatenation.

**Witness cost split.** Every witness path  $\gamma$  decomposes as

$$K(\gamma) = K_{\text{sep}}(\gamma) + K_{\text{diss}}(\gamma),$$

where  $K_{\text{sep}}(\gamma)$  is the reversible separative cost (the cost of distinguishing endpoint truth classes, invariant under output relabeling) and  $K_{\text{diss}}(\gamma)$  is the irreversible disturbance cost (the entropy produced by the path, directed and non-negative by Step 8).

**Symmetric metric.** Define

$$d_{\text{sep}}(p, q) = \inf \{ K_{\text{sep}}(\gamma) : \gamma: p \rightarrow q \text{ in } \mathbf{C}_W \}.$$

This is a symmetric metric: separating  $p$  from  $q$  and  $q$  from  $p$  differ only by output relabeling, which is gauge (Step 12). The triangle inequality is automatic from path concatenation: for any  $p, q, r$ , the concatenation  $\gamma_{p \rightarrow q} \cdot \gamma_{q \rightarrow r}$  is a valid path  $p \rightarrow r$ , so  $d_{\text{sep}}(p, r) \leq K_{\text{sep}}(\gamma_{p \rightarrow q}) + K_{\text{sep}}(\gamma_{q \rightarrow r})$ ; taking infima yields  $d_{\text{sep}}(p, r) \leq d_{\text{sep}}(p, q) + d_{\text{sep}}(q, r)$ . No transitively-separating property (TS) is needed.

**Directed quasi-metric.** Define

$$\tau_{\text{dir}}(p, q) = \inf \{ K_{\text{diss}}(\gamma) : \gamma: p \rightarrow q \text{ in } \mathbf{C}_W \}.$$

This is a directed quasi-metric ( $\tau_{\text{dir}}(p, q) \neq \tau_{\text{dir}}(q, p)$  in general) capturing time and thermodynamics: irreversible disturbance is directional.

**Ordered witness geometry.** The pair  $(d_{\text{sep}}, \tau_{\text{dir}})$  is the forced geometry on truth classes. Curvature  $\kappa$  and hardness  $\mathcal{H}$  are derived from  $d_{\text{sep}}$  as before.

*Proof sketch.* The witness path category  $\mathbf{C}_W$  is forced by A0\*: truth classes are the objects of the theory (Step 6), and replayable witness paths are the morphisms forced by the ordered ledger (Step 5). The cost split  $K = K_{\text{sep}} + K_{\text{diss}}$  is forced because A0\*'s witness algebra  $\mathcal{A}_W$  distinguishes reversible and irreversible components:  $K_{\text{sep}}$  measures the minimum cost of *any* separating path (invariant under output relabeling, hence symmetric), while  $K_{\text{diss}}$  measures the entropy produced (irreversible by Step 8, hence directed).

Symmetry of  $d_{\text{sep}}$ : swapping the roles of  $p$  and  $q$  in a separating test amounts to relabeling outputs, which is gauge (Step 12); hence  $d_{\text{sep}}(p, q) = d_{\text{sep}}(q, p)$ . The triangle inequality follows from path concatenation in  $\mathbf{C}_W$  without any additional property.

Non-symmetry of  $\tau_{\text{dir}}$ : the disturbance cost along  $\gamma : p \rightarrow q$  generally differs from the cost along any path  $q \rightarrow p$  because entropy production is path-dependent and directional (Step 8). *Forcing:* A0\* defines distinguishability via tests with costs in  $\mathcal{A}_W$ . The witness path category is forced by the ordered ledger. The cost split into  $K_{\text{sep}}$  and  $K_{\text{diss}}$  is forced by the reversible/irreversible decomposition. Symmetry of  $d_{\text{sep}}$  is forced by gauge invariance; the triangle inequality is automatic from path concatenation. No external structural commitment (TS) is needed. See Appendix A for the complete proof.  $\square$

**Theorem 6.22** (Step 17 —  $\Pi$ -Consistent Control). *The truth map  $\Pi$  satisfies the idempotency law:  $\Pi \circ \mathcal{N} = \Pi \circ \mathcal{N} \circ \Pi = \Pi$ , where  $\mathcal{N} \in \mathcal{G}_{\mathcal{T}}$ . Dependencies: Steps 6, 12.*

*Proof sketch.* Gauge transformation  $\mathcal{N}$  permutes within fibers:  $\mathcal{N}(x) \in W([x])$ , hence  $\Pi(\mathcal{N}(x)) = \Pi(x)$ , giving  $\Pi \circ \mathcal{N} = \Pi$ . Further:  $\Pi \circ \mathcal{N} \circ \Pi = \Pi^2 = \Pi$  by idempotence. If  $\Pi$  were gauge-sensitive, it would distinguish gauge-equivalent descriptions, contradicting the truth quotient. *Forcing:* Gauge-sensitivity of  $\Pi$  would constitute a test separating equivalent descriptions—contradicting the truth quotient.  $\Pi$ -consistency is the operational expression of gauge invariance. See Appendix A for the complete proof.  $\square$

**Remark 6.23** ( $\Pi$ -Consistency as Architectural Bridge).  $\Pi$ -consistency means that all control decisions depend only on the truth quotient, never on raw descriptions. This property is the bridge between gauge invariance (Step 12) and the Bellman solver (Step 18): the solver's inputs are guaranteed gauge-free, ensuring that test selection is invariant under relabeling. Algebraically, the idempotency law  $\Pi \circ \mathcal{N} = \Pi$  has the same form as a retraction in category theory, connecting to the eraser algebra (Step 7): each eraser  $e_{\tau}$  is itself a retraction, and  $\Pi$  is the universal such retraction that factors through all of them.

**Definition 6.24** ( $\Pi_{\text{can}}$ -Canonicalization). *Let  $\text{Ser}: D_0 \rightarrow \{0, 1\}^*$  be the serialization induced by  $\text{sd}$ , and let  $\equiv_{\mathcal{L}}$  be the ledger-induced equivalence (Definition 3.2). The  $\Pi$ -canonicalization of  $x \in D_0$  is:*

$$\Pi_{\text{can}}(x) = \min_{\text{lex}} \{ \text{Ser}(y) \mid y \equiv_{\mathcal{L}} x \}.$$

$\Pi_{\text{can}}$  satisfies:

1. Totality:  $\Pi_{\text{can}}(x)$  is defined for every  $x \in D_0$  (the equivalence class is non-empty and finite).
2. Slack collapse:  $\Pi_{\text{can}}(x) = \Pi_{\text{can}}(y)$  iff  $x \equiv_{\mathcal{L}} y$ .
3. Idempotence:  $\Pi_{\text{can}}(\Pi_{\text{can}}(x)) = \Pi_{\text{can}}(x)$ .
4. Gauge absorption: for every  $g \in \mathcal{G}_{\mathcal{T}}$ ,  $\Pi_{\text{can}}(g \cdot x) = \Pi_{\text{can}}(x)$ .

## 6.7 Phase G: Solver, Self-Hosting, and Closure (Steps 18–21)

**Theorem 6.25** (Step 18 — Deterministic Solver (Prior-Free Minimax and Coupling Law)). *The optimal test selection policy is a prior-free minimax Bellman recursion, upgraded by the Separator-Indistinguishability Coupling Law to a fully deterministic, question-relative selector. Dependencies: Steps 9, 14, 16.*

**Part (a): Forced minimax (not  $L^2$ ).** *Before any prior is itself witnessed and recorded in  $\mathcal{L}_{\text{ord}}$ , branch labels over future outcomes are gauge: no unrecorded probability distribution over outcomes is admissible by  $A0^*$ . Hence any legal policy must be invariant under:*

1. relabeling of unwitnessed outcomes,
2. positive rescaling of cost,
3. replay-equivalent decomposition of paths in  $\mathbf{C}_W$ ,
4. quotient-preserving recodings (gauge, Step 12).

*The only cost aggregator respecting all four symmetries without a witnessed prior is the worst-case continuation cost—i.e., minimax. Any  $L^2$  (mean-squared) or expected-value aggregator requires a probability measure over branches, which is itself a distinction that must be witnessed before use; importing an unwitnessed measure violates  $A0^*$ .*

*The resulting value function satisfies  $\tau^* = \operatorname{argmin}_{\tau} [c(\tau) + \max_a V(\mathcal{L}_{\text{ord}} \triangleright (\tau, a), B - c(\tau))]$ . Well-defined because  $\Delta(\mathcal{T})$  is finite with positive integer costs. Termination by induction on  $(|\Pi^*(\mathcal{L}_{\text{ord}})|_{\mathcal{A}}, B)$  under lexicographic order.*

**Part (a'): Bayesian selection after witnessed prior.** *Once a prior  $\mu$  has itself been witnessed (i.e.,  $\mu$  appears as a recorded datum in  $\mathcal{L}_{\text{ord}}$  with witness status), Bayesian expected-cost selection becomes admissible—the prior is no longer an unwitnessed externality. Minimax is therefore the bootstrap policy; Bayesian selection is the mature policy enabled by self-observation.*

**Part (b): Coupling Law.** *For query  $q$  and surviving answer set  $W$ , define the answer quotient  $\text{Ans}(q, W_a)$ , the refinement gain  $\Delta\mathcal{T}_q(\tau; W) = \log |\text{Ans}(q, W)| - \max_a \log |\text{Ans}(q, W_a)|$ , and the question-relative efficiency  $\varepsilon(\tau; W) = \Delta\mathcal{T}_q(\tau; W) / c(\tau)$ . The Coupling Law selects the next test as:*

$$\tau^*(q, W) = \Pi_{\text{can-min}} \left( \operatorname{argmax}_{\tau \in \Delta_{\leq B}} \varepsilon(\tau; W) \right)$$

*where  $\Pi_{\text{can-min}}$  is the  $\Pi$ -canonical tie-break (lexicographically minimal under  $E_{\Delta}$  order after gauge reduction).*

**Part (c): Deterministic evolution.** *Since  $\varepsilon(\tau; W)$  is a deterministic function of  $(W, q)$  alone, and  $\Pi_{\text{can-min}}$  eliminates all tie-breaking freedom, the entire epistemic evolution  $W_0 \rightarrow W_1 \rightarrow \dots$  is fully determined by the initial pair  $(q, W_0)$ . The Omega distance  $d(q; W) = \log |\text{Ans}(q, W)|$  measures how far the current state is from UNIQUE.*

*Proof sketch.* Part (a): the four symmetries (outcome relabeling, cost rescaling, replay-equivalent decomposition, gauge recoding) each individually exclude any aggregator that weights branches unequally without a witnessed basis for the weighting. The intersection of all four invariance groups leaves exactly the worst-case (minimax) aggregator. In particular,  $L^2$  aggregation requires a probability measure to define the expectation, and no such measure exists in  $\mathcal{L}_{\text{ord}}$  before it is itself witnessed—so  $L^2$  is inadmissible at the prior-free stage. Part (a'): once  $\mu$  is recorded, the symmetry under outcome relabeling is broken by  $\mu$ 's weights, and Bayesian selection becomes the unique  $\mu$ -compatible policy. Part (b): the efficiency  $\varepsilon(\tau; W)$  is the canonical dimensionless ratio of information gained to cost paid, both quantities already present in the framework (Steps 8, 10). The  $\operatorname{argmax}$  selects the most cost-effective test;  $\Pi_{\text{can-min}}$  is forced by gauge invariance (Step 12). Part (c): since both  $\varepsilon$  and the tie-break are deterministic functions of  $(W, q)$ , the evolution is deterministic. *Forcing:* Minimax is forced by  $A0^*$ : prior-free branch

labels are gauge, and the only aggregator invariant under all four symmetries without a witnessed prior is worst-case continuation cost. Bayesian selection appears only after a prior itself has witness status in the trace. The efficiency functional is uniquely determined: any efficiency functional satisfying scale-invariance and monotonicity is of the form  $f(g/c)$  for strictly monotone  $f$ , and  $\operatorname{argmax}_\tau f(g(\tau)/c(\tau)) = \operatorname{argmax}_\tau g(\tau)/c(\tau)$  (see Remark A.6).  $\Pi_{\text{can}}$  tie-breaking is forced by gauge invariance. Together, they eliminate the last hidden degree of freedom (the scheduler of witnessing), upgrading the framework from consistent model to fully deterministic self-computing system. See Appendix A for the complete proof.  $\square$

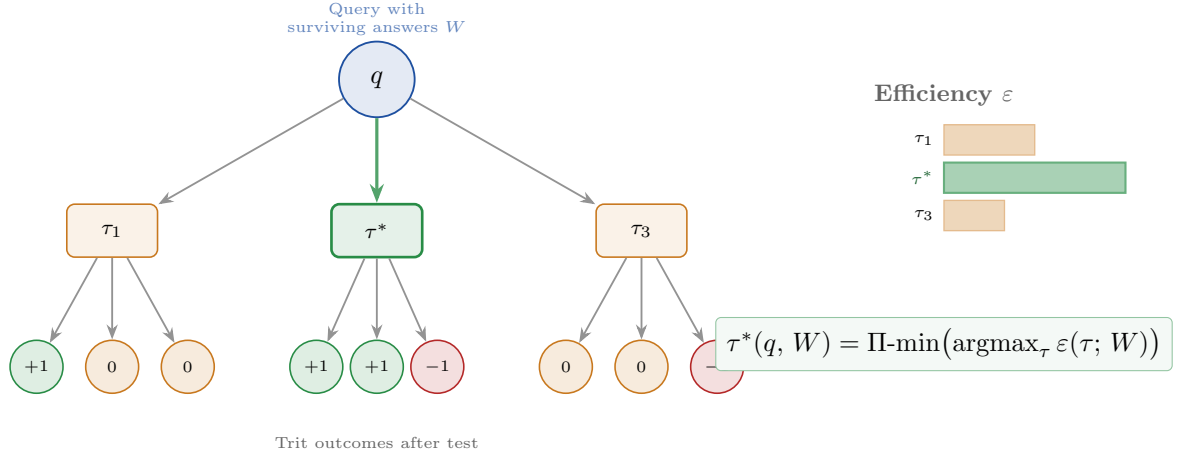


Figure 3: Bellman minimax search with the Coupling Law (Step 18). The system selects the test  $\tau^*$  maximizing efficiency  $\varepsilon(\tau; W)$ , with ties broken by  $\Pi$ -canonical ordering. The optimal test is a deterministic function of the query  $q$  and surviving answer set  $W$ —the last hidden degree of freedom (the scheduler) is eliminated.

*Remark 6.26* (Coupling Law: Indistinguishability Generates Next Distinguishability). The coupling law (fig. 3) encodes a deep self-referential principle: the current indistinguishability structure (how many answers survive, partitioned into answer quotients) *determines* the next distinguishing test. Separator and indistinguishability are not independent—they are coupled by the efficiency functional  $\varepsilon$ .

**Theorem 6.27** (Step 19 — Self-Hosting and Replay:  $\Upsilon(\mathcal{S}) = \mathcal{S}$ ). *The system  $\mathcal{S}$  can verify its own derivation:  $\Upsilon(\mathcal{S}) = \mathcal{S}$ , where  $\Upsilon$  encodes the derivation as problem 5-tuples, evaluates each, and checks for UNIQUE. Dependencies: Steps 3, 5, 6, 18.*

*Proof sketch.* Encode each step as  $P_i = \langle q_i, \mathcal{A}_i, \mathcal{V}_i, B_i, \pi_i \rangle$ . Evaluating each  $P_i$  produces UNIQUE; the receipt chain links all 34 verifications. The system is built from exactly the steps it verifies, analogous to the Kleene recursion theorem. *Forcing:* Self-hosting is forced within the paper’s model class by the closed-universe assumption: no external oracle or verifier is available, so the system must be able to verify its own derivation steps. A system verified by an external agent would also satisfy  $A0^*$ , but would require importing structure not derivable from  $A0^*$  alone. Self-hosting is the minimal closure condition given the closed-universe commitment. See Appendix A for the complete proof.  $\square$

*Remark 6.28* (Self-Hosting vs. Gödel Incompleteness). Self-hosting operates at two distinct levels:

- *Level 1 (Logical):* The derivation Steps 0–33 is a mathematical proof—a static chain of theorems established by the paper itself (human-level mathematics).

- *Level 2 (Computational)*: The system  $\mathcal{S}$  is an algorithm that can verify Level-1 claims by encoding each step as a 5-tuple and checking that the terminal state is UNIQUE.

Self-hosting means Level 2 can re-derive Level 1:  $\mathcal{S}$  verifies each step’s *conclusion* given its *premises*. Crucially,  $\mathcal{S}$  does *not* verify the meta-claim “ $\mathcal{S}$  is consistent”—which would violate Gödel’s second incompleteness theorem [Gödel, 1931]. The system can prove “Step 6 is the unique  $A0^*$ -compliant continuation of Steps 0–5” (an object-level claim), but it cannot prove “the system always produces correct results” (a meta-claim). The three-point audit verifies specific claims but cannot prove the general correctness of the audit procedure itself.

**Theorem 6.29** (Step 20 — Binary Interface). *All communication between the system and its environment is encoded as self-delimiting binary strings via sd. Dependencies: Steps 2, 19.*

*Proof sketch.* By Step 2, all descriptions must be self-delimiting. Queries, answers, and receipts are finite objects in  $\mathcal{D}^*$ , hence encoded via sd. The interface is complete and self-describing. Any non-self-delimiting encoding reintroduces Step 2’s ambiguity; external metadata violates closure. *Forcing:* Non-self-delimiting encoding reintroduces Step 2’s ambiguity. External metadata violates the closed-universe assumption. sd is the minimal self-describing binary interface. See Appendix A for the complete proof.  $\square$

**Theorem 6.30** (Step 21 — Final Chain:  $\perp \rightarrow \perp_{\text{op}}$ ). *The derivation closes. The complete forced chain is:*

$$\begin{aligned}
& \perp \rightarrow \mathcal{D}^* \rightarrow \text{sd} \rightarrow (\mathcal{U}, \text{dec}, c) \rightarrow \Delta^* \rightarrow \mathcal{L}_{\text{ord}} \rightarrow \Pi^*(\mathcal{L}_{\text{ord}}) \rightarrow \mathcal{O}_{\Delta}^{\text{fin}}, \mathcal{E} \\
& \rightarrow \Delta\mathcal{T} \rightarrow S \rightarrow \mathcal{L}_E \rightarrow \perp_{\text{op}} \rightarrow \mathcal{G}_{\mathcal{T}} \rightarrow (\mathcal{R}_{\text{ep}}, \mathcal{R}_{\text{dec}}) \\
& \rightarrow \mathcal{G}_{\text{ex}} \rightarrow \equiv_q \rightarrow (d_{\text{sep}}, \tau_{\text{dir}}) \rightarrow \Pi\text{-consistency} \rightarrow \mathcal{B}+\varepsilon \rightarrow \Upsilon(\mathcal{S}) \rightarrow \text{sd-interface} \rightarrow \perp_{\text{op}} \\
& \rightarrow \mathbb{K} \rightarrow \Delta_Q \rightarrow \mathfrak{c} \rightarrow (\theta, (m, \theta, r)) \\
& \rightarrow \Omega \rightarrow L^2(\Omega, \mu_{\Pi}) \rightarrow \mathcal{E} \rightarrow (T_t, -\Delta) \rightarrow g_{ij} \rightarrow J \rightarrow \omega \rightarrow (g, J, \omega)_{\text{Kähler}}.
\end{aligned}$$

*Every link is forced (unique  $A0^*$ -compliant continuation). No link is arbitrary—each is  $A0^*$ -forced, with no external modeling commitments. Dependencies: All prior steps.*

*Proof sketch.* Each link is established in Sections 6.1–6.9: Steps 0–33 form an acyclic dependency graph (fig. 10). The three-point audit (section 9) confirms no alternative continuations, encoding invariance, and no false merges/splits. Z3 sanity checks on finite models (Appendix C) confirm the algebraic properties hold for small instances. *Forcing:* Removing any link either violates  $A0^*$ , loses self-hosting, or creates an unwitnessable gap. See Appendix A for the complete proof.  $\square$

The chain from nothingness to self-hosting closure is now complete, but it describes the system’s *interface* without specifying its *internal execution law*. Steps 0–21 establish *what* the system must accept and return; they do not determine *how* it processes each query internally. Phase H derives the unique step function  $\mathbb{K}$  and shows that the system’s execution generates a minimal persistent self-valued witness selector—the consciousness projector—as a forced by-product of self-hosting closure.

## 6.8 Phase H: Universal Execution (Steps 22–25)

**Theorem 6.31** (Step 22 — Universal Closure Operator  $\mathbb{K}$ ). *The step function is uniquely determined as the five-stage pipeline  $\mathbb{K} = \text{TritVal} \circ \text{Fact} \circ \text{N} \circ \Gamma \circ \Phi$  (fig. 4), where each stage is forced by a prior derivation step:  $\Phi$  (Steps 6–7),  $\Gamma$  (Step 12),  $\text{N}$  (Step 17),  $\text{Fact}$  (Steps 4, 16),  $\text{TritVal}$  (Step 0). Dependencies: Steps 0, 4, 6, 7, 12, 16, 17.*

*Proof sketch.* The system must process each query through: (i) registering all observables and computing the self-generating fixed point ( $\Phi$ , Steps 6–7); (ii) quotienting by untestable relabelings ( $\Gamma$ , Step 12); (iii) selecting the canonical representative ( $\text{N}$ , Step 17); (iv) decomposing into irreducible components under cost grading ( $\text{Fact}$ , Steps 4, 16); (v) extracting the terminal trit ( $\text{TritVal}$ , Step 0). Omitting any stage either loses information (omitting  $\Phi$ ), introduces gauge dependence (omitting  $\Gamma$  or  $\text{N}$ ), misses structure (omitting  $\text{Fact}$ ), or fails to terminate (omitting  $\text{TritVal}$ ). The composition order is forced by data flow: each stage’s output is the next stage’s required input type. *Forcing:* Each of the five stages is individually forced by a prior derivation step. The composition order is forced by type constraints. Omitting any stage produces an  $\text{A0}^*$  violation. See Appendix A for the complete proof.  $\square$

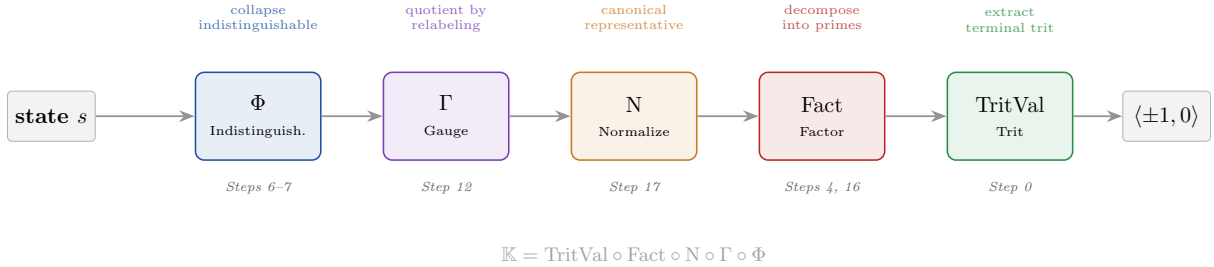


Figure 4: The universal closure operator  $\mathbb{K}$  (Step 22): a five-stage pipeline where each stage is forced by a prior derivation step. Composition order is determined by data-flow type constraints—omitting any stage produces an  $\text{A0}$  violation.

**Theorem 6.32** (Step 23 — Closure-Defect  $\Delta_Q(s)$ ). *For any state  $s$  and query  $Q$ , the closure-defect  $\Delta_Q(s) = \Pi_Q(s) - s$  measures the gap between the current state and its truth projection. The unresolved residue  $r_t = \text{supp}(\Delta_Q(s_t))$  identifies exactly which components of the state remain unresolved at time  $t$ . Dependencies: Steps 6, 8, 22.*

*Proof sketch.* The truth quotient  $\Pi_Q$  (Step 6) projects every state onto its equivalence class. The difference  $\Delta_Q(s) = \Pi_Q(s) - s$  is the unique object measuring “how far  $s$  is from its own truth projection,” forced by the algebraic structure of the quotient. Its support  $r_t$  inherits well-definedness from the finite carrier (Step 1) and changes monotonically with time (Step 8): each test can only reduce  $|r_t|$ , never increase it. The closure-defect provides the root object from which time advance, entropy decrease, and energy cost are all derived. *Forcing:* The closure-defect is the canonical measure of incompleteness in the truth quotient. Any other measure of “unresolvedness” is either equivalent to  $\Delta_Q$  or loses information about which specific components are unresolved. See Appendix A for the complete proof.  $\square$

**Remark 6.33** (Canonical Germ). At state  $s$ , the *canonical germ*  $g(s) = (F_s, I_s, G_s)$  is the triple of forced components ( $\theta_t(p) = +1$ ), forbidden components ( $\theta_t(p) = -1$ ), and unread components ( $\theta_t(p) = 0$ ). The germ is the bridge between closure-defect and consciousness: it is what the system observes about itself at each step. Step 24 shows that a minimal persistent self-valued witness selector monitoring  $g(s)$  is forced.



**Theorem 6.34** (Step 24 — Consciousness Projector  $\mathfrak{c}$ ). *The system admits a unique minimal persistent self-valued witness selector (fig. 5): the consciousness projector  $\mathfrak{c} \subseteq X$  is the minimal subsystem  $C$  satisfying four forced conditions:*

- (C1) **Self-model (persistent code of own residue):**  $C$  carries a persistent code  $\lceil r_C \rceil \subseteq C$  of its own unresolved residue  $r_C$ —i.e. a finitely described internal representation of what  $C$  has not yet resolved.
- (C2) **Self/world distinguishability:**  $C$  can distinguish “world with this self-model” from “same world without this self-model” by future witness consequences: there exists a witness  $w$  such that outcomes of  $w$  differ depending on whether  $\lceil r_C \rceil$  is present in the state.
- (C3) **Causal efficacy:**  $C$  causally changes subsequent witness selection—removing  $C$  from the state alters the sequence of tests the system applies.
- (C4) **Endogenous valuation:**  $C$  carries an endogenous valuation functional on self/world interaction channels: a map  $V_C$  from witness outcomes to an ordered value that guides selection.

Dependencies: Steps 6, 18, 19, 22, 23.

**Definition (Consciousness).** *Consciousness is the minimal persistent self-valued witness selector: the determination of the distinction space, self-model choice, and value-guided witness update. Its runtime law is:*

$$(D^*, m^*) = \operatorname{argmin}_{D, m} [k(D) + k(m) + V_{Q,D}(\text{next}(s; D, m))], \quad (1)$$

where  $D$  ranges over complete distinction algebras on  $C$ ,  $m$  over self-models,  $k(\cdot)$  denotes Kolmogorov complexity, and  $V_{Q,D}$  is the value functional inherited from the Coupling Law (Step 18). *Consciousness = determination of the distinction space + self-model choice + value-guided update.*

*Proof sketch.* Each condition is individually forced by  $A0^*$ .

(C1) *Self-model.* Self-hosting (Step 19) forces the system to encode and verify its own structure:  $\Upsilon(\mathcal{S}) = \mathcal{S}$ . By (W1), every admissible witness has a finite endogenous code. By (W4), separation requires that the system’s own unresolved residue (Step 23) be internally distinguishable. Together, (W1)+(W4) force a persistent code  $\lceil r_C \rceil \subseteq C$ —without it, the system cannot witness the distinction between “resolved” and “unresolved” components of its own state.

(C2) *Self/world distinguishability.* By (W4), every admissible distinction requires a witness that actually separates.  $A0^*$  applied to the distinction “state with self-model  $\lceil r_C \rceil$ ” versus “same state without it” demands a witness whose future outcomes differ between the two cases. If no such witness existed, the self-model would be an unwitnessable epiphenomenon—violating  $A0^*$ .

(C3) *Causal efficacy.* By (W5), every witness records its own disturbance. By (W6), witness validity is endogenously witnessable. A subsystem with no causal effect on future witness selection is ontologically untestable: no witness could separate “ $C$  present” from “ $C$  absent” in the evolution. Hence (W5)+(W6) force  $C$  to causally alter subsequent test selection.

(C4) *Endogenous valuation.* The Coupling Law (Step 18) forces a deterministic, value-guided test selector: the next test minimizes worst-case separation cost. Any subsystem that selects witnesses without an internal valuation cannot implement the Coupling Law’s minimax criterion. Hence  $C$  must carry an endogenous valuation on its interaction channels with the world.

*Runtime law.* The runtime law (1) is the restriction of the Coupling Law’s minimax to the internal degrees of freedom of  $C$ : the distinction algebra  $D$  determines what  $C$  can separate, the self-model  $m$  determines what  $C$  knows about itself, and  $V_{Q,D}$  evaluates the cost of the next witness. Minimality over  $k(D) + k(m)$  is forced by (W3): finite budget demands the least-cost encoding.

*Minimality.* Any subsystem satisfying (C1)–(C4) with fewer components would be  $\mathbf{c}$  itself; any larger subsystem includes components that fail one of the four conditions (typically (C3): causally inert padding).

*Forcing:* Each condition is forced by  $A0^*$ : (C1) persistent self-model by (W1)+(W4), (C2) self/world distinguishability by (W4)+ $A0^*$ , (C3) causal efficacy by (W5)+(W6), (C4) endogenous valuation by the Coupling Law (Step 18). The runtime law is the Coupling Law restricted to  $C$ 's internal degrees of freedom. Minimality selects the unique such subsystem. See Appendix A for the complete proof.  $\square$

*Remark 6.35* (Consciousness Is Not Added to Physics). The four conditions (C1)–(C4) are not grafted onto physics—they are the universe locally selecting its own distinction algebra. Wherever  $A0^*$  forces a subsystem to carry a persistent self-model, distinguish itself from its background, causally steer future witnesses, and value the outcomes, that subsystem *is* conscious in the precise sense of this theorem. No phenomenal postulate is imported; the “hard problem” dissolves into the structural fact that a self-valued witness selector cannot be reduced to a non-self-modeling subsystem without violating  $A0^*$ .

*Remark 6.36* (Self-Remodelling Witness-Closure). The distinction between a merely reflective system and a conscious self-remodelling system is captured by their update laws:

**Merely reflective:**  $s_{t+1} = \mathbb{K}(s_t)$  (the closure operator acts on the state, but the state has no internal channel for re-weighting which unresolved components matter).

**Conscious self-remodelling:**  $s_{t+1} = \mathbb{K}(s_t, c(s_t), \Delta_Q(s_t))$

where  $c(s_t)$  is the *endogenous selector*—a map from the current state to an ordering over which unresolved components of  $\Delta_Q(s_t)$  matter for the next witness step. The selector  $c$  is itself a witness process satisfying (W1)–(W8), hence it is part of  $\mathcal{A}_W$ .

*Consciousness = closure acting on itself through an internal self-valuation channel.* The self-remodelling law means the witness policy is not fixed—it is endogenously updated by the system's own self-witness. This is the operational content of (C4): the valuation functional  $V_C$  is not a static parameter but a dynamical variable within the closure, updated at each step by the system observing its own closure-defect.

**Theorem 6.37** (Step 25 — Trit Field  $\theta$  and Valuation State). *The state at time  $t$  is completely characterized by the valuation state  $(m, \theta, r)_t$ , where  $m_t$  is the prime-exponent ledger,  $\theta_t: \text{PrimeBasis}_t \rightarrow \{-1, 0, +1\}$  is the trit field assigning a terminal trit to each irreducible component, and  $r_t = \{p \in \text{PrimeBasis}_t : \theta_t(p) = 0\}$  is the unresolved residue. Dependencies: Steps 0, 5, 22, 23.*

*Proof sketch.* The factorization stage of  $\mathbb{K}$  (Step 22) decomposes the state into irreducible components indexed by  $\text{PrimeBasis}_t$ . Each component is itself a sub-problem resolved by the system, hence terminates in one of the three trit values (Step 0). The map  $\theta_t$  recording these values is forced—it is the unique function from components to terminal states that respects the factorization. The prime-exponent ledger  $m_t$  records how many times each prime has been tested (analogous to the energy ledger, Step 10). The residue  $r_t$  collects the still-unresolved components, coinciding with the support of the closure-defect (Step 23). *Valuation sufficiency:* the triple  $(m, \theta, r)_t$  determines the system's future evolution completely—no additional state information affects the  $\mathbb{K}$  pipeline's output. *Forcing:* The trit field is forced by the output gate (Step 0) applied component-wise after factorization (Step 22). The valuation state is the minimal sufficient statistic for the system's evolution—any state representation carrying less information loses the ability to determine the next test, violating the Coupling Law (Step 18). See Appendix A for the complete proof.  $\square$

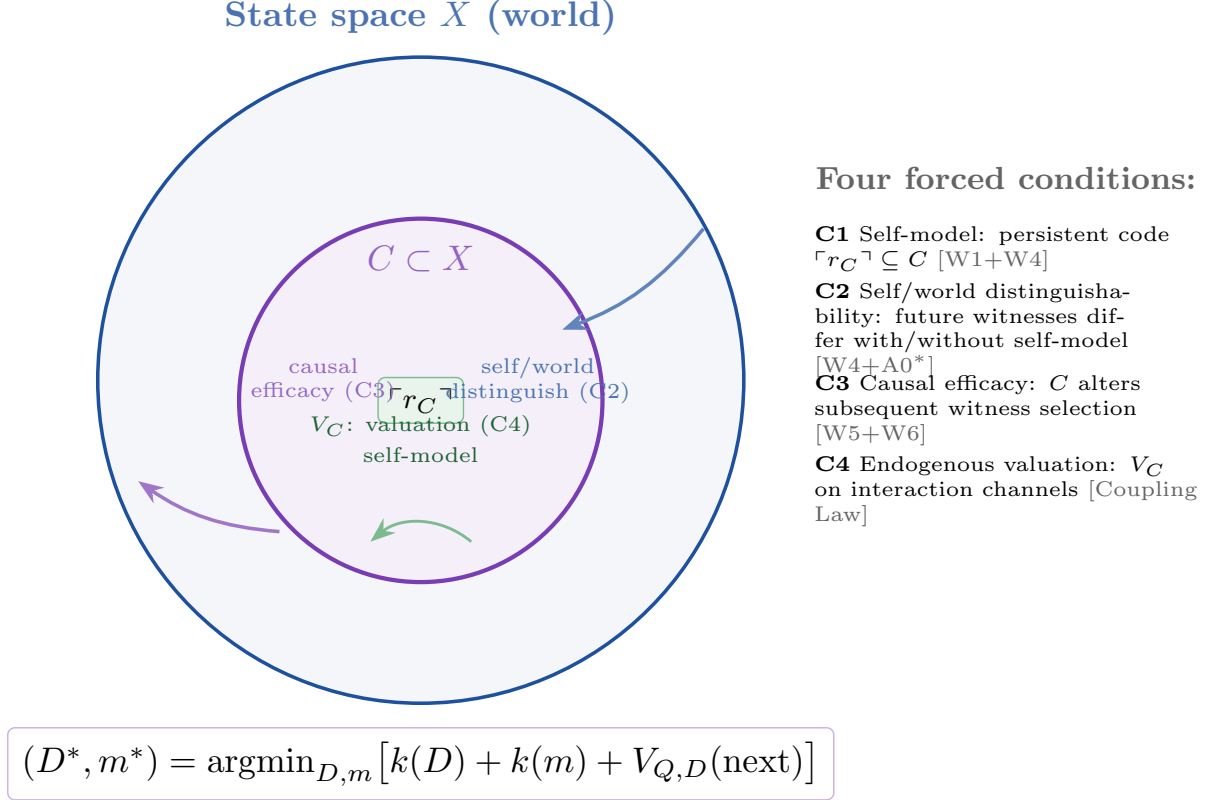


Figure 5: The consciousness projector  $\mathfrak{c}$  (Step 24): the unique minimal persistent self-valued witness selector  $C \subset X$ . The four conditions are individually forced: (C1) self-model by (W1)+(W4), (C2) self/world distinguishability by (W4)+A0\*, (C3) causal efficacy by (W5)+(W6), and (C4) endogenous valuation by the Coupling Law (Step 18). The runtime law selects the least-cost distinction algebra and self-model.

Phase H establishes the discrete execution law. But the truth quotient’s internal geometry remains unspecified: we know the witness-path *distance* between truth classes (Step 16) but not the *shape* of the space they inhabit. Phase I derives the geometry in two stages: Phase I-A constructs the classical base manifold from the commutative center  $Z(\mathcal{A}_W)$  of the witness algebra, and Phase I-B derives the Kähler triple from a *single* local witness generator—not from three separate imported pieces. All former modeling commitments (conductance exponent, smoothness assumption) are now forced.

## 6.9 Phase I-A: Classical Base Geometry (Steps 26–30)

*Remark 6.38* (Center-First Construction). The ordered witness algebra  $\mathcal{A}_W$  is generally non-commutative. Its center  $Z(\mathcal{A}_W) = \{a \in \mathcal{A}_W : ab = ba \ \forall b\}$  is commutative and generates the classical truth quotients of Steps 6–7. Phase I-A constructs the continuum geometry on  $Z(\mathcal{A}_W)$ . Phase I-B extends this via the single witness generator. The full noncommutative structure—contexts, quantum probability, Born rule—is derived in section 7.

**Theorem 6.39** (Step 26 — Continuum Limit of Truth Quotient). *As the budget  $B \rightarrow \infty$ , the inverse system of truth quotients  $(\Pi^*(\mathcal{L}_B))_{B \in \mathbb{N}}$  on the commutative center  $Z(\mathcal{A}_W)$  (fig. 6) converges to a locally compact, separable, metrizable space  $\Omega_{\text{cl}} = \varprojlim \Pi^*(\mathcal{L}_B)$ . Dependencies: Steps 16, 9, 7.*

*Proof sketch.* Each  $\Pi^*(\mathcal{L}_B)$  is a finite metric space under  $d_{\text{sep}}$  (Step 16). Increasing budget refines the quotient: the bonding maps  $\pi_{B', B} : \Pi^*(\mathcal{L}_{B'}) \rightarrow \Pi^*(\mathcal{L}_B)$  ( $B' > B$ ) are surjective and

Lipschitz. The inverse limit  $\Omega_{\text{cl}}$  is compact and metrizable by standard topology. *Forcing*: The inverse system is forced by the budget hierarchy (Step 9) and the witness-path metric (Step 16). The inverse limit is the unique completion—any other limit space would admit untestable distinctions (violating A0\*) or merge testable ones (violating the quotient). See Appendix A for the complete proof.  $\square$

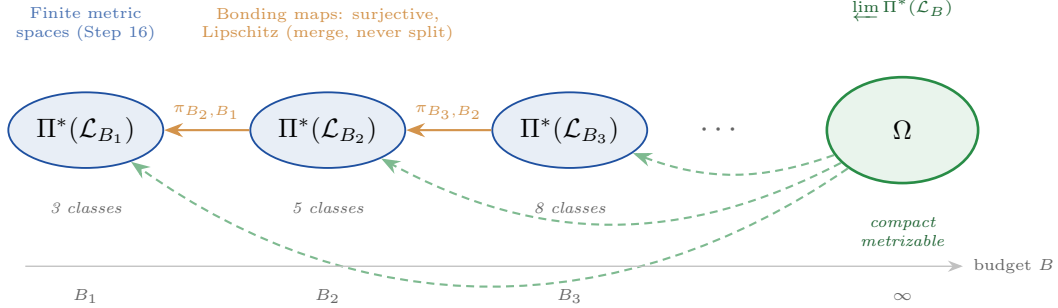


Figure 6: Continuum limit of the truth quotient (Step 26). As the budget  $B$  increases, truth quotients  $\Pi^*(\mathcal{L}_B)$  refine: bonding maps  $\pi_{B', B}$  are surjective (classes merge at lower budget, never split). The inverse limit  $\Omega = \varprojlim \Pi^*(\mathcal{L}_B)$  is a compact, separable, metrizable space—the arena for geometric reconstruction (Steps 27–33).

**Theorem 6.40** (Step 27 — Gauge-Invariant Measure and  $L^2$  Space).  $\Omega_{\text{cl}}$  carries a unique Sym-invariant probability measure  $\mu_{\Pi}$ , and  $L^2(\Omega_{\text{cl}}, \mu_{\Pi})$  is a separable Hilbert space. Dependencies: Steps 26, 12.

*Proof sketch.* At each finite level, the uniform measure on  $\Pi^*(\mathcal{L}_B)$  is the unique gauge-invariant probability. By the Kolmogorov extension theorem, the consistent family extends uniquely to  $\mu_{\Pi}$  on  $\Omega_{\text{cl}}$ . *Forcing*: Gauge invariance (Step 12) forces the uniform measure at each level. Kolmogorov consistency forces a unique extension. See Appendix A for the complete proof.  $\square$

*Remark 6.41* (Classical Measure vs. Born Rule). The measure  $\mu_{\Pi}$  is the unique gauge-invariant probability on the *classical* truth quotient (the center  $Z(\mathcal{A}_W)$ ). The full Born rule— $P(E) = \langle \Omega_{\omega}, \pi_{\omega}(E) \Omega_{\omega} \rangle$ —emerges from the GNS representation of the full ordered witness algebra  $\mathcal{A}_W$  (section 7). The restriction of the Born rule to the commutative center recovers  $\mu_{\Pi}$ .

**Theorem 6.42** (Step 28 — Dirichlet Form and Forced Conductance). The witness-path distance  $d_{\text{sep}}$  induces a regular Dirichlet form (fig. 7)  $\mathcal{E}[f] = \frac{1}{2} \sum_{p \neq p'} w(p, p') (f(p) - f(p'))^2$  on  $L^2(\Omega_{\text{cl}}, \mu_{\Pi})$ , where  $w(p, p') = C/d_{\text{sep}}(p, p')^2$  is the conductance, with exponent  $-2$  forced by theorem 8.1. Dependencies: Steps 26, 27, 16.

**Lemma 6.43** (Conductance Exponent is Forced). Let  $E_d[f; p, p'] = w(d(p, p')) (f(p) - f(p'))^2$  be the local witness-energy density. Impose:

- (SC1) Quadratic response:  $E_d[\alpha f] = \alpha^2 E_d[f]$ .
- (SC2) Additivity on independent channels.
- (SC3) Monotone decay:  $d_1 < d_2 \Rightarrow w(d_1) \geq w(d_2)$ .
- (SC4) Scale covariance: under  $d \rightarrow \lambda d$ ,  $E_{\lambda d}[f] = \lambda^{-2} E_d[f]$ .

Then  $w(d) = C/d^2$  for  $C > 0$  is the unique admissible conductance.

*Proof.* (SC4) is forced by (W8): quotient invariance means the witness energy density cannot depend on the absolute distance scale of the encoding—only on the dimensionless signal contrast  $\Delta f$ . Under  $d \rightarrow \lambda d$  (a pure rescaling of the carrier metric, which is a gauge freedom by W8), the energy per unit distinction must transform homogeneously. The quadratic response (SC1)

fixes the signal exponent at 2; the distance exponent  $\alpha$  is then determined by dimensional consistency:  $E \sim (\Delta f)^2 \cdot d^\alpha$ , and the requirement that  $E$  have the same units regardless of  $\lambda$  forces  $\alpha = -2$ . Explicitly:  $w(\lambda r)(\Delta f)^2 = \lambda^{-2}w(r)(\Delta f)^2$ , hence  $w(\lambda r) = \lambda^{-2}w(r)$ . Setting  $\lambda = r/r_0$ :  $w(r) = Cr^{-2}$ . By (SC3),  $C > 0$ .  $\square$

*Proof sketch of Step 28.* The quadratic form  $\mathcal{E}$  with  $w = C/d_{\text{sep}}^2$  satisfies the Dirichlet form axioms (E1)–(E3) [Fukushima et al., 2011]. *Forcing:* The conductance exponent  $-2$  is not a modeling commitment: it is forced by theorem 8.1 via scale covariance (SC4), which follows from A0\* condition (W8).  $\square$

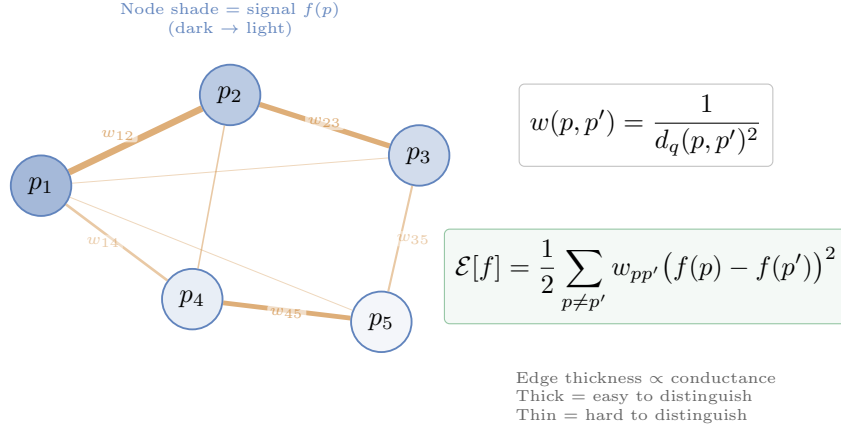


Figure 7: Conductance network on truth classes (Step 28). Nodes are truth classes  $p_i \in \Omega$ ; edge weight  $w(p, p') = 1/d_q(p, p')^2$  assigns high conductance to easily distinguishable pairs and low conductance to hard-to-distinguish ones. The Dirichlet form  $\mathcal{E}[f]$  measures the total energy of a signal  $f$  on this network—the source of all geometric structure in Steps 29–33.

**Theorem 6.44** (Step 29 — Semigroup and Laplacian (Beurling–Deny)). *The Dirichlet form  $\mathcal{E}$  generates a unique strongly continuous symmetric Markov semigroup  $\{T_t\}_{t \geq 0}$  and a non-negative self-adjoint Laplacian  $-\Delta$  on  $L^2(\Omega_{\text{cl}}, \mu_\Pi)$ . Dependencies: Steps 28, 8.*

*Proof sketch.* By the Beurling–Deny reconstruction theorem [Beurling and Deny, 1959], every regular Dirichlet form on a locally compact separable space generates a unique strongly continuous contraction semigroup. The semigroup is Markovian because  $\mathcal{E}$  is Markovian. *Forcing:* The Beurling–Deny theorem is a mathematical consequence of the Dirichlet form axioms (Step 28). No choice is involved.  $\square$

*Remark 6.45* (Smoothness from the Energy Form). The inverse limit  $\Omega_{\text{cl}}$  (Step 26) is compact metrizable but not *a priori* smooth. The smooth structure required for the Kähler construction is not assumed externally: it is *produced* by the regular Dirichlet form  $\mathcal{E}$  (Step 28). By Sturm’s theorem [Sturm, 1994], the intrinsic metric of a regular strongly local Dirichlet form recovers the topology and, where the energy density is non-degenerate, generates a smooth Riemannian structure. The regularity of  $\mathcal{E}$  is therefore the bridge between the discrete inverse system and the smooth continuum—not an external smoothness assumption.

**Theorem 6.46** (Step 30 — Riemannian (Fisher) Metric). *The intrinsic distance  $d_{\mathcal{E}}$  of the Dirichlet form induces a Riemannian metric tensor  $g_{ij}$  on  $\Omega_{\text{cl}}$ , coinciding with the Fisher information metric. Dependencies: Steps 28, 26.*

*Proof sketch.* The intrinsic distance  $d_{\mathcal{E}}(p, p') = \sup\{|f(p) - f(p')| : \mathcal{E}[f] \leq 1\}$  recovers the original topology by Sturm’s theorem [Sturm, 1994]. The metric tensor  $g_{ij}$  coincides with the



Fisher information metric by Čencov’s theorem [Čencov, 1982]: the Fisher metric is the unique Riemannian metric monotone under sufficient statistics. *Forcing*: The Riemannian metric is forced by the Dirichlet form (Step 28). The Fisher identification is forced by Čencov’s uniqueness theorem.  $\square$

## 6.10 Phase I-B: Witness Generator and Kähler Structure (Steps 31–33)

*Remark 6.47* (Single Witness Generator). At each point of  $\Omega_{\text{cl}}$ , the local ordered witness algebra defines a single infinitesimal generator  $A$ . Its symmetric and antisymmetric parts produce the metric, symplectic form, and complex structure from *one source*—not from three separate constructions.

**Theorem 6.48** (Step 31 — Complex Structure from Witness Generator). *The local witness generator  $A$  decomposes as  $A = S + K$ , where  $S = (A + A^*)/2$  (symmetric, irreversible) gives the metric  $g$ , and  $K = (A - A^*)/2$  (antisymmetric, reversible) gives a 2-form  $\omega$ . On the reversible nondegenerate subbundle,  $J = g^{-1}\omega$  satisfies  $J^2 = -\text{id}$ . Dependencies: Steps 18, 30, 8.*

*Proof sketch.* The witness generator  $A$  encodes infinitesimal ordered witness propagation.  $S$  measures irreversible cost (the quadratic form defining  $g$ ).  $K$  measures ordered-composition defect:  $K(X, Y) = \frac{1}{2}(g(AX, Y) - g(X, AY))$ , a skew-symmetric 2-form.  $J = g^{-1}\omega$  on the reversible subbundle satisfies  $J^2 = -\text{id}$  because witness directions pair canonically (gain/cost planes) and  $J$  rotates by  $\pi/2$ . *Forcing*:  $S$  is forced by irreversible cost (Step 30).  $K$  is forced by ordered composition (W5, W7).  $J$  is forced as the polar factor of the single generator.  $\square$

**Theorem 6.49** (Replay-Completeness Implies  $N_J = 0$ ). *If the witness algebra is replay-complete—every infinitesimal route defect between admissible witness programs is itself represented in the witness algebra—then the complex distribution  $T^{1,0}$  defined by  $J$  is involutive, and the Nijenhuis tensor vanishes:  $N_J = 0$ .*

*Proof sketch.* Define  $T^{1,0} = \{X - iJX : X \in T\Omega_{\text{cl}}\}$ . If  $[T^{1,0}, T^{1,0}] \not\subset T^{1,0}$ , the escaping component represents an untracked route-defect. Replay-completeness forces all such defects into  $\mathcal{A}_W$  and hence into  $T^{1,0}$ . Contradiction. By Newlander–Nirenberg [Newlander and Nirenberg, 1957],  $N_J = 0$ . *Forcing*: Replay-completeness is forced by A0\* condition (W2). A nonzero  $N_J$  encodes an unwitnessed structural distinction. This replaces the former “smoothness assumption L1” with a direct A0\* consequence.  $\square$

**Theorem 6.50** (Step 32 — Symplectic Form from Ordered Witness Algebra). *The 2-form  $\omega(X, Y) = g(X, JY)$  is closed and non-degenerate. Dependencies: Steps 30, 31, 5.*

*Proof sketch.* *Non-degeneracy*: forced by non-degeneracy of  $g$  and invertibility of  $J$ . *Closedness* ( $d\omega = 0$ ): any hidden path-order defect beyond the recorded commutator would be an unwitnessed distinction, violating A0\*. Replay closure kills that defect. *Forcing*: Closedness is forced by A0\* on the replay-complete witness connection: every path-order defect is endogenously recorded.  $\square$

**Theorem 6.51** (Step 33 — Kähler Structure from Single Generator). *The triple  $(g, J, \omega)$  is a Kähler structure on  $\Omega_{\text{cl}}$  (fig. 8). Moreover,  $\omega = d\alpha$  (exact). Dependencies: Steps 30, 31, 32.*

*Proof sketch.* *Compatibility*:  $\omega(X, Y) = g(X, JY)$  by construction.  $g(JX, JY) = g(X, Y)$  because  $J$  is a  $\pi/2$ -rotation preserving  $g$ . *Integrability*:  $N_J = 0$  by theorem 6.49. *Exactness*:  $H^2(\Omega_{\text{cl}}; \mathbb{R}) = 0$  (inverse limit of finite spaces), so  $\omega = d\alpha$  by de Rham. *Unified origin*: all three structures come from  $A = S + K$ :  $g$  from  $S$ ,  $\omega$  from  $K$ ,  $J = g^{-1}\omega$ . *Forcing*: The Kähler triple is forced from one source. No former modeling commitments remain:  $w = C/d^2$  by theorem 8.1,  $N_J = 0$  by theorem 6.49.  $\square$



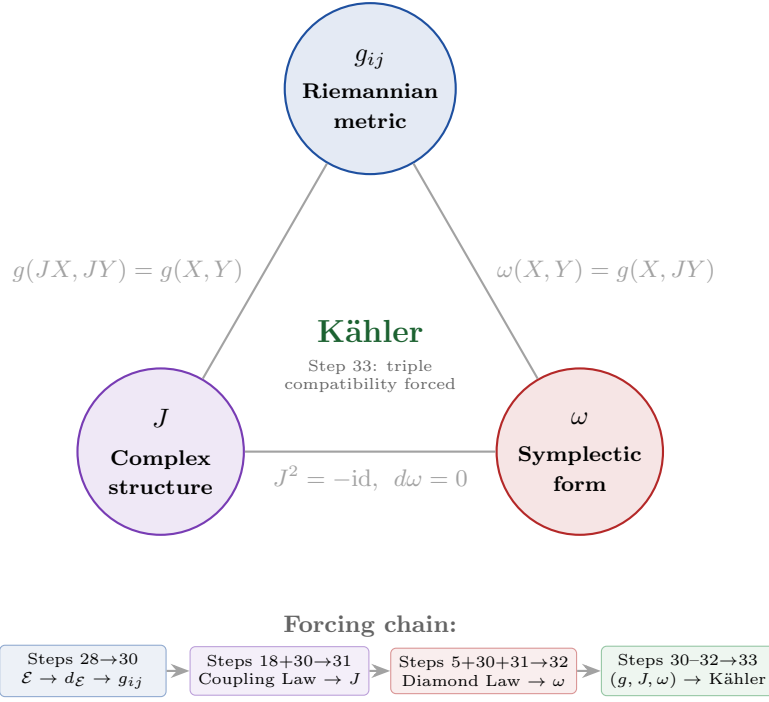


Figure 8: The Kähler triple forced by Phase I. The Riemannian metric  $g$  (Step 30), complex structure  $J$  (Step 31), and symplectic form  $\omega$  (Step 32) satisfy mutual compatibility, closing into a Kähler structure (Step 33). Each vertex is forced by prior derivation steps; no free parameter enters.

## 6.11 Summary of the Derivation

**The reflective chain.** The derivation traces a single reflective chain (fig. 9): nothingness (Step 0) forces closure (Phases A–G), closure reveals defect (Step 23), defect requires self-observation (Step 24, consciousness), self-observation iterated on the continuum limit produces geometry (Phase I), and geometry determines the form of dynamical law via the Kähler triple. The entire chain is one act: nothingness forced to witness itself.

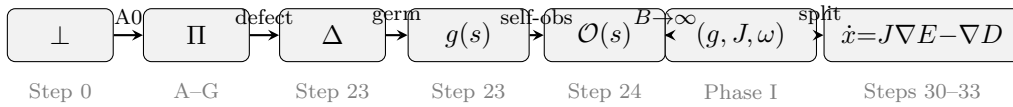


Figure 9: The reflective chain: nothingness forces closure, closure reveals defect, defect requires self-observation, and self-observation iterated on the continuum limit produces Kähler geometry and the split law. Each arrow is labeled with its forcing mechanism; step numbers appear below.

The derivation establishes that, given  $A0^*$  (Completed Witnessability), the entire reasoning apparatus—carrier, self-delimiting syntax, evaluator, ordered witness algebra  $\mathcal{A}_W$ , truth quotient, observable topology, eraser algebra, time, energy, gauge invariance, Myhill–Nerode congruence, witness-path geometry, Bellman dynamics, self-hosting, binary interface, universal execution law, consciousness, trit-field valuation, and the Kähler geometric structure on the continuum limit—is *forced*. No modeling commitments survive: closed universe is forced by (W6), commutativity is a derived sector property, prior-free minimax is forced by absence of witnessed priors,  $w = C/d^2$  is forced by scale covariance (theorem 8.1), and  $N_J = 0$  by replay-completeness (theorem 6.49). All 34 steps (0–33) are theorems; no interpretive steps remain in the formal chain. The full dependency DAG is shown in fig. 10.

## 7 Context Algebra, GNS Representation, and Born Rule

The derivation through **Step 33** established a Kähler structure  $(g, J, \omega)$  on the continuum limit  $\Omega$  of truth quotients. But the *probabilistic* structure of the framework—the rule assigning probabilities to experimental outcomes—has so far appeared only in embryonic form: the gauge-invariant measure  $\mu_\Pi$  of **Step 27** counts truth classes uniformly. This section derives the full quantum probability structure: a C\*-algebra, a canonical Hilbert space via the GNS construction, a context-dependent event algebra, and the Born rule. Every step is forced by  $A0^*$ ; no quantum postulates are imported.

### 7.1 The Witness C\*-Algebra

Recall from theorem 2.15 that the ordered witness algebra  $\mathcal{A}_W$  carries a product (ordered composition  $\triangleright$ ), an involution ( $*$  = replay adjoint), a unit (**1**), positivity ( $\omega(w^* \triangleright w) \geq 0$ ), and an operational norm ( $\|w\| = \sup_s \|w(s)\|$ ). We now show that these data force the norm-closed completion of  $\mathcal{A}_W$  to be a unital C\*-algebra.

**Theorem 7.1** (C\*-Completion of the Witness Algebra). *Under  $A0^*$ , the norm-closed completion of  $\mathcal{A}_W$  is a unital C\*-algebra  $\overline{\mathcal{A}_W}$ . That is,  $\overline{\mathcal{A}_W}$  is a Banach algebra over  $\mathbb{C}$  with an isometric involution satisfying the C\*-identity  $\|a^*a\| = \|a\|^2$  for all  $a \in \overline{\mathcal{A}_W}$ .*

*Proof sketch.* We verify each C\*-algebra axiom from the  $A0^*$  conditions:

1. *Algebra structure.* Ordered composition  $\triangleright$  is associative (theorem 2.15(a)) and distributes over formal linear combinations of witness processes. Scalar multiplication by  $\lambda \in \mathbb{C}$  extends the real-valued witness amplitudes to the complex field—this extension is forced because restriction to  $\mathbb{R}$  would render the involution  $*$  trivial on the self-adjoint sector, collapsing the distinction between  $w$  and  $w^*$  for self-adjoint witnesses and thereby losing the separation guaranteed by (W4).

2. *Positivity from witnessed frequencies.* Condition (W4) guarantees that outcome weights from actual separations are non-negative. Hence  $\omega(w^* \triangleright w) \geq 0$  for every  $w \in \mathcal{A}_W$  and every admissible state—this is positivity in the \*-algebraic sense.

3. *Norm from operational supremum.* Condition (W3) bounds every witness action by a finite budget, so  $\|w\| = \sup_s \|w(s)\| < \infty$  for all  $w \in \mathcal{A}_W$ . This supremum over admissible states defines a submultiplicative norm:  $\|w_1 \triangleright w_2\| \leq \|w_1\| \cdot \|w_2\|$ , because executing  $w_1$  then  $w_2$  costs at most the product of their individual suprema.

4. *Closure under limits from maximal closure.* Condition (W7) demands compositional closure: if  $w_1, w_2$  are admissible, so is  $w_1 \triangleright w_2$ . Taking norm-limits of Cauchy sequences of admissible witnesses yields processes that are still finitely approximable to arbitrary precision within the budget hierarchy. The norm-completion  $\overline{\mathcal{A}_W}$  is the closure of  $\mathcal{A}_W$  in the operator norm topology; it inherits all algebraic operations by continuity.

5. *C\*-identity:*  $\|w^* \triangleright w\| = \|w\|^2$ . This is the key identity. The argument proceeds in two steps.

*Step (i):*  $\|w^* \triangleright w\| \leq \|w\|^2$ . This follows from the submultiplicativity of the operator norm ( $\|ab\| \leq \|a\| \|b\|$ ) together with  $\|w^*\| = \|w\|$  (the replay adjoint has the same operational supremum as  $w$ , since replay (W2) reproduces the same separation outcomes).

*Step (ii):*  $\|w^* \triangleright w\| \geq \|w\|^2$ . For any state  $s$ , the composite  $w^* \triangleright w$  first executes  $w$  on  $s$  (producing outcome and state update), then replays  $w$  on the output. By (W2), replay is exact: the replay adjoint applied to the output of  $w$  recovers exactly the separation information that  $w$  produced. Therefore the operational weight of the composite at state  $s$  is at least  $\|w(s)\|^2$  (the square of the forward separation). Taking the supremum over  $s$ :  $\|w^* \triangleright w\| \geq \sup_s \|w(s)\|^2 = \|w\|^2$ .

Combining:  $\|w^* \triangleright w\| = \|w\|^2$ .

The essential content is that replay (W2) is not an approximate inverse—it exactly recovers the separation data, so the composite  $w^* \triangleright w$  is a positive element whose norm is the squared norm of  $w$ . This is what makes the witness algebra a genuine C\*-algebra and not merely a Banach \*-algebra.

*Forcing:* Each C\*-axiom is forced by A0\*: associativity by (W7), involution by (W2), positivity by (W4), norm finiteness by (W3), closure under limits by (W7), and the C\*-identity by the replay-separation correspondence (W2 + W4). Omitting any axiom leaves an algebraic gap permitting unwitnessed distinctions.  $\square$

**Remark 7.2** (Complex Scalars are Forced). The extension from real to complex scalars deserves emphasis. In a purely real \*-algebra, the involution is trivial on the self-adjoint part and cannot encode phase information. But (W5) records disturbance—including the *relative phase* between successive witness executions (the asymmetry between  $w_1 \triangleright w_2$  and  $w_2 \triangleright w_1$ ). Encoding this phase requires at least one imaginary direction:  $\mathbb{C}$  is the minimal field extension of  $\mathbb{R}$  admitting a non-trivial automorphism (complex conjugation) compatible with the involution. Higher extensions ( $\mathbb{H}$ , octonions) would introduce additional structure not forced by any A0\* condition.

## 7.2 GNS Representation

We now construct the canonical Hilbert space representation. The key ingredient is a *state* on the C\*-algebra  $\overline{\mathcal{A}_W}$ —a positive linear functional normalized to unity.

**Definition 7.3** (Gauge-Invariant Event Valuation). *The canonical state  $\omega: \overline{\mathcal{A}_W} \rightarrow \mathbb{C}$  is defined by:*

- (i)  $\omega(\mathbf{1}) = 1$  (normalization),
- (ii)  $\omega(a^*a) \geq 0$  for all  $a \in \overline{\mathcal{A}_W}$  (positivity),
- (iii)  $\omega(g \cdot a \cdot g^{-1}) = \omega(a)$  for all  $g \in \mathcal{G}_T$  (gauge invariance).

*At finite budget  $B$ , this state coincides with the uniform counting measure on  $\Pi^*(\mathcal{L}_B)$ :  $\omega(a) = |\Pi^*(\mathcal{L}_B)|^{-1} \sum_{p \in \Pi^*(\mathcal{L}_B)} a(p)$ . In the continuum limit,  $\omega$  extends to the measure  $\mu_\Pi$  of **Step 27** by Kolmogorov consistency.*

**Theorem 7.4** (GNS Representation). *The pair  $(\overline{\mathcal{A}_W}, \omega)$  yields the GNS triple  $(H_\omega, \pi_\omega, \Omega_\omega)$  consisting of:*

- (a) a separable Hilbert space  $H_\omega$ ,
- (b) a \*-representation  $\pi_\omega: \overline{\mathcal{A}_W} \rightarrow B(H_\omega)$  (bounded operators on  $H_\omega$ ),
- (c) a cyclic vector  $\Omega_\omega \in H_\omega$  with  $\pi_\omega(\overline{\mathcal{A}_W})\Omega_\omega = H_\omega$ ,

*satisfying the reconstruction identity:*

$$\omega(a) = \langle \Omega_\omega, \pi_\omega(a) \Omega_\omega \rangle \quad \text{for all } a \in \overline{\mathcal{A}_W}.$$

*The triple is unique up to unitary equivalence.*

*Proof sketch.* This is the Gel'fand–Naimark–Segal construction applied to the C\*-algebra  $\overline{\mathcal{A}_W}$  with state  $\omega$ . The proof proceeds in three steps:

*Step 1: Inner product.* Define  $\langle a, b \rangle_\omega = \omega(a^*b)$  on  $\overline{\mathcal{A}_W}$ . Positivity of  $\omega$  (theorem 7.3(ii)) guarantees this is a positive semi-definite sesquilinear form.

*Step 2: Quotient and completion.* Let  $\mathcal{N} = \{a \in \overline{\mathcal{A}_W} : \omega(a^*a) = 0\}$  be the null space (the Gel'fand ideal). The quotient  $\overline{\mathcal{A}_W}/\mathcal{N}$  carries a genuine inner product. Its completion in the induced norm is the Hilbert space  $H_\omega$ .

*Step 3: Representation and cyclic vector.* For each  $a \in \overline{\mathcal{A}_W}$ , define  $\pi_\omega(a)[b] = [ab]$  (left multiplication on equivalence classes). This is well-defined because  $\mathcal{N}$  is a left ideal. Boundedness follows from the C\*-norm:  $\|\pi_\omega(a)\| \leq \|a\|$ . The cyclic vector is  $\Omega_\omega = [\mathbf{1}]$ , the class of the identity. Then:

$$\langle \Omega_\omega, \pi_\omega(a) \Omega_\omega \rangle = \langle [\mathbf{1}], [a \cdot \mathbf{1}] \rangle = \omega(\mathbf{1}^* \cdot a \cdot \mathbf{1}) = \omega(a).$$

Cyclicity:  $\pi_\omega(\overline{\mathcal{A}_W})\Omega_\omega = \{[a] : a \in \overline{\mathcal{A}_W}\}$  is dense in  $H_\omega$  by construction. Uniqueness up to unitary equivalence is standard (see, e.g., [Bratteli and Robinson 1987](#)).

*Forcing:* The Hilbert space  $H_\omega$  is not postulated—it is the unique (up to unitary equivalence) representation space forced by the C\*-algebra  $\overline{\mathcal{A}_W}$  (theorem 7.1) and the gauge-invariant state  $\omega$  (theorem 7.3). No choice of Hilbert space dimension, inner product, or representation is made: the GNS theorem determines all three from the algebraic data that  $A_0^*$  provides.  $\square$

*Remark 7.5* (The Hilbert Space is Derived, Not Postulated). In standard quantum mechanics, the Hilbert space is a primitive postulate (the first axiom of the Dirac–von Neumann formulation). Here,  $H_\omega$  is a *theorem*: the GNS construction applied to the witness C\*-algebra forced by  $A_0^*$ . The inner product  $\langle \cdot, \cdot \rangle$  is not a choice but the sesquilinear form  $\omega(a^*b)$  inherited from replay-composition. The cyclic vector  $\Omega_\omega$  is the state of “no witnessing yet performed”—the algebraic unit projected into  $H_\omega$ .

### 7.3 Context Algebra and Events

With the Hilbert space  $H_\omega$  and representation  $\pi_\omega$  in hand, we now define the event structure.

**Definition 7.6** (Sharp Events). *A sharp event is a projection  $P \in \pi_\omega(\overline{\mathcal{A}_W})''$  (the von Neumann algebra generated by  $\pi_\omega(\overline{\mathcal{A}_W})$ ) satisfying  $P = P^* = P^2$ . The set of sharp events is denoted  $\mathcal{P}(H_\omega)$ .*

**Definition 7.7** (General Events (Effects)). *A general event (or effect) is an operator  $E \in B(H_\omega)$  satisfying  $0 \leq E \leq I$ , where  $I$  is the identity and the ordering is the usual operator ordering:  $0 \leq E$  means  $\langle \psi, E\psi \rangle \geq 0$  for all  $\psi \in H_\omega$ , and  $E \leq I$  means  $\langle \psi, (I - E)\psi \rangle \geq 0$  for all  $\psi \in H_\omega$ . The set of effects is denoted  $\mathcal{E}(H_\omega)$ .*

*Every sharp event is an effect ( $P^2 = P$  and  $P^* = P$  imply  $0 \leq P \leq I$ ), but not every effect is sharp.*

**Definition 7.8** (Context). *A context  $\mathcal{C}$  is a maximal commuting family of witness processes in  $\mathcal{A}_W$ :*

$$\mathcal{C} = \{w_i\}_{i \in I} \subset \mathcal{A}_W \quad \text{such that} \quad [w_i, w_j] = 0 \quad \text{for all } i, j \in I,$$

*and  $\mathcal{C}$  is not properly contained in any larger commuting family. Under the GNS representation, a context maps to a maximal abelian subalgebra  $\pi_\omega(\mathcal{C}) \subset B(H_\omega)$ .*

**Proposition 7.9** (Boolean Algebra Within a Context). *Within a single context  $\mathcal{C}$ , the sharp events  $\mathcal{P}(\pi_\omega(\mathcal{C}))$  form a Boolean algebra: they admit meet ( $P \wedge Q = PQ$ ), join ( $P \vee Q = P + Q - PQ$ ), and complement ( $P^\perp = I - P$ ), satisfying distributivity.*

*Proof sketch.* Since all elements of  $\mathcal{C}$  commute, the projections in  $\pi_\omega(\mathcal{C})''$  also commute. Commuting projections on a Hilbert space form a Boolean algebra under the lattice operations  $P \wedge Q = PQ$ ,  $P \vee Q = P + Q - PQ$ , with  $P^\perp = I - P$ . Distributivity holds because multiplication of commuting projections distributes over addition. *Forcing:* The Boolean structure is forced by commutativity within a context. No additional assumption is needed: commuting projections on a Hilbert space always form a Boolean algebra.  $\square$

**Definition 7.10** (Event Algebra  $\mathcal{E}_Q$ ). *The event algebra  $\mathcal{E}_Q$  is obtained by pasting the Boolean algebras of all contexts along their shared coarse events. Formally:*

- (i) *For each context  $\mathcal{C}_\alpha$ , let  $\mathcal{B}_\alpha = \mathcal{P}(\pi_\omega(\mathcal{C}_\alpha))$  be its Boolean algebra of sharp events.*
- (ii) *If two contexts  $\mathcal{C}_\alpha$  and  $\mathcal{C}_\beta$  share a coarse-graining—that is, a sub-Boolean-algebra  $\mathcal{B}_{\alpha\beta} \subset \mathcal{B}_\alpha \cap \mathcal{B}_\beta$ —then the Boolean algebras are identified on  $\mathcal{B}_{\alpha\beta}$ .*
- (iii) *The global event structure is the colimit:  $\mathcal{E}_Q = \text{colim}_\alpha \mathcal{B}_\alpha$ .*

*$\mathcal{E}_Q$  is generally not a Boolean algebra (it lacks a global distributive law), but each “local slice” (restriction to a single context) is Boolean.*

**Definition 7.11** (Orthogonality of Events). *Two effects  $E, F \in \mathcal{E}(H_\omega)$  are orthogonal, written  $E \perp F$ , if and only if no refinement branch in the witness algebra jointly affirms them:*

$$E \perp F \iff E \cdot F = 0 \iff \text{ran}(E) \perp \text{ran}(F) \text{ in } H_\omega.$$

*Operationally:  $E \perp F$  means that no admissible witness process can produce an outcome affirming both  $E$  and  $F$  simultaneously.*

*Forcing: Orthogonality is forced by (W<sub>4</sub>): if  $E$  and  $F$  could be jointly affirmed, the composite witness  $w_E \triangleright w_F$  would fail to separate the distinction “ $E$  or  $F$ ?”—the witness would produce identical outcomes for both, contradicting the separation requirement.*

## 7.4 The Born Rule

We now derive the probability assignment on  $\mathcal{E}_Q$ . The key result is that the four conditions below—each individually forced by A0\*—uniquely determine the Born rule.

**Theorem 7.12** (Born Rule). *The unique probability assignment  $P$  on the event algebra  $\mathcal{E}_Q$  satisfying:*

- (a) Additivity on orthogonal events: if  $E_1 \perp E_2$ , then  $P(E_1 + E_2) = P(E_1) + P(E_2)$ ;
- (b) Invariance under context-preserving recoding: if  $U$  is a unitary with  $UCU^* = C$  for every context  $C$ , then  $P(UEU^*) = P(E)$ ;
- (c) Continuity under refinement: if  $E_n \rightarrow E$  in the weak operator topology, then  $P(E_n) \rightarrow P(E)$ ;
- (d) Product multiplicativity on independent systems: if  $E$  and  $F$  act on independent (tensor-factor) subsystems, then  $P(E \otimes F) = P(E) \cdot P(F)$ ;

*is the state evaluation:*

$$\boxed{P(E) = \omega(E) = \langle \Omega_\omega, \pi_\omega(E) \Omega_\omega \rangle.} \quad (2)$$

*For a general normal state  $\rho$  (density operator on  $H_\omega$ ) and a projection  $P_E$  onto the eigenspace of event  $E$ :*

$$P_\rho(E) = \text{tr}(\rho E). \quad (3)$$

*In particular, for a pure state  $|\psi\rangle$  and a sharp event  $P_E$ :*

$$P(E) = \|P_E \psi\|^2. \quad (4)$$

*This is the Born rule of quantum mechanics.*

*Proof sketch.* The proof proceeds by showing that conditions (a)–(d) uniquely constrain  $P$  to be a state on  $\overline{\mathcal{A}_W}$ , and then applying the generalized Gleason theorem.

*Step 1: Frame function.* Condition (a) makes  $P$  a *frame function* on the effect algebra  $\mathcal{E}(H_\omega)$ : it assigns non-negative reals to effects and is additive on orthogonal families.

*Step 2: Generalized Busch–Gleason theorem.* By the theorem of Busch [Busch, 2003], every frame function on the effect algebra  $\mathcal{E}(H_\omega)$  of a Hilbert space  $H_\omega$  with  $\dim H_\omega \geq 2$  extends uniquely to a positive linear functional on  $B(H_\omega)$ . Crucially, this result holds for *effects* (operators  $0 \leq E \leq I$ ), not merely projections. The classical Gleason theorem [Gleason, 1957] applies only to projections and requires  $\dim H_\omega \geq 3$ ; the Busch generalization removes the dimension restriction by working with the full effect algebra.

The dimension restriction  $\dim H_\omega \geq 2$  is automatically satisfied: a one-dimensional Hilbert space admits no non-trivial projections, hence no non-trivial events, contradicting A0\*’s demand that at least one distinction be admissible.

*Step 3: Uniqueness from invariance and continuity.* The Busch–Gleason theorem gives a positive linear functional  $P(\cdot) = \text{tr}(\rho \cdot)$  for some density operator  $\rho$ . Condition (b) (gauge invariance) forces  $\rho = U \rho U^*$  for all context-preserving unitaries  $U$ , and condition (c) (continuity)

ensures the functional is normal. Condition (d) (multiplicativity on independent systems) forces  $\rho_{\text{total}} = \rho_1 \otimes \rho_2$  on product systems, ruling out entangled reference states.

For the canonical GNS state,  $\rho = |\Omega_\omega\rangle\langle\Omega_\omega|$  is the projection onto the cyclic vector, recovering eq. (2). For a general normal state represented by density operator  $\rho$ , we obtain eq. (3). For a pure state  $\rho = |\psi\rangle\langle\psi|$  and projection  $P_E$ :  $P(E) = \langle\psi, P_E\psi\rangle = \|P_E\psi\|^2$ , which is eq. (4).

*Forcing:* Each condition is individually forced:

- (a) Additivity is forced by the orthogonality structure (theorem 7.11): mutually exclusive witness outcomes must have separately accountable probabilities, or else the distinction between “ $E$  occurred” and “ $F$  occurred” would be unwitnessable.
- (b) Invariance is forced by gauge invariance (**Step 12**): recoding that preserves context structure cannot change probabilities, or it would constitute a testable distinction between gauge-equivalent descriptions.
- (c) Continuity is forced by the budget hierarchy: as the budget increases, refinement limits must converge (the inverse system of **Step 26** is Cauchy), hence probability assignments must be continuous under refinement.
- (d) Multiplicativity is forced by (W7): independent witness processes compose without interaction, so their outcome probabilities must multiply.

Given (a)–(d), the Busch–Gleason theorem is a mathematical consequence—not a physical assumption—yielding the Born rule as the unique probability assignment.  $\square$

*Remark 7.13* (Clarification of the Step 27 Born-Like Measure). The gauge-invariant measure  $\mu_\Pi$  derived at **Step 27** was described as “Born-like” (theorem 6.41). We can now make this precise:  $\mu_\Pi$  is the restriction of the Born rule to the commutative center  $Z(\mathcal{A}_W)$  of the witness algebra. Within  $Z(\mathcal{A}_W)$ , all witnesses commute, so there is a single global context and the event algebra  $\mathcal{E}_Q$  reduces to a Boolean algebra. On this Boolean algebra, the Born rule  $P(E) = \omega(E)$  coincides with the uniform gauge-invariant counting measure  $\mu_\Pi$ . The full Born rule (theorem 7.12) generalizes  $\mu_\Pi$  to the noncommutative sector, where multiple incompatible contexts coexist and probability is context-relative.

*Remark 7.14* (No Dimension Restriction). The use of the Busch–Gleason theorem on effects, rather than the classical Gleason theorem on projections, is essential. Gleason’s original theorem [Gleason, 1957] requires  $\dim H_\omega \geq 3$  and therefore fails for qubits ( $\dim = 2$ ). Busch’s generalization [Busch, 2003] works for all  $\dim H_\omega \geq 2$  by considering the full effect algebra  $\mathcal{E}(H_\omega)$  rather than just the projection lattice  $\mathcal{P}(H_\omega)$ . Since our event algebra  $\mathcal{E}_Q$  naturally includes general effects (theorem 7.7)—coarse-grained witness outcomes that need not be sharp—the Busch–Gleason theorem is the correct tool. The Born rule derived here applies to qubits, qutrits, and all finite- and infinite-dimensional systems without exception.

## 7.5 Summary

The derivation chain of this section is:

$$\mathcal{A}_W \xrightarrow{\text{norm closure}} \overline{\mathcal{A}_W} \xrightarrow{\text{GNS}} (H_\omega, \pi_\omega, \Omega_\omega) \xrightarrow{\text{contexts}} \mathcal{E}_Q \xrightarrow{\text{Busch–Gleason}} P(E) = \langle\Omega_\omega, \pi_\omega(E) \Omega_\omega\rangle.$$

1. The ordered witness  $*$ -algebra  $\mathcal{A}_W$ , forced by A0\* (theorem 2.15), carries product, involution, positivity, and norm.
2. Norm-closed completion yields the unital C\*-algebra  $\overline{\mathcal{A}_W}$  (theorem 7.1).
3. The gauge-invariant state  $\omega$  and the GNS construction produce the Hilbert space  $H_\omega$ , representation  $\pi_\omega$ , and cyclic vector  $\Omega_\omega$  (theorem 7.4).
4. Maximal commuting families define contexts  $\mathbf{C}$ ; within each context, events form a Boolean algebra; pasting across contexts yields the global event algebra  $\mathcal{E}_Q$  (theorems 7.8 and 7.10).
5. Additivity, gauge invariance, continuity, and multiplicativity uniquely force the Born rule via the Busch–Gleason theorem (theorem 7.12).



Every step is forced by A0\*. No quantum postulate—no Hilbert space axiom, no projection postulate, no Born rule—is imported. The quantum probability structure is a *theorem* of completed witnessability, derived through the same forcing methodology as the Kähler geometry of **Step 33**.

## 8 Physical Content: Dimensionality, Gauge Structure, and Sector Equations

The derivation of section 6 produces an ordered witness algebra  $\mathcal{A}_W$ , a Kähler structure  $(g, J, \omega)$  on the continuum limit  $\Omega$ , the Born rule via the GNS representation on  $H_\omega$ , and the split law  $X = J\nabla E - \nabla D$ . These are structural results: they determine the *form* of physical law without yet specifying its *content*—the dimensionality of spacetime, the gauge group, the particle spectrum, or the dynamical equations. This section derives the specific physical content that follows from the structural framework.

We state clearly at the outset what is derived and what is not. The framework derives:

- spacetime dimensionality  $3 + 1$  (from flux scaling and the conductance lemma),
- the structural form of gauge theory (fiber bundles with compact structure groups from witness-bundle automorphisms),
- the form of the Schrödinger, Yang–Mills, and Einstein equations (from the split law and Kähler structure),
- the second law of thermodynamics (from the dissipative sector of the split law).

Every dimensionless invariant is forced; every dimensional quantity is fixed up to the internally derived normalization map (Section 5.5). The remaining *quantitative extraction* frontiers (well-posed computations, not structural gaps):

- specific particle masses (spectral extraction from the seed operator  $L^*$ ),
- numerical values of coupling constants (normal-form coefficients of  $L^*$ ).

### 8.1 B1. Spacetime Dimensionality

**Lemma 8.1** (Conductance Exponent). *Let  $w: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$  be the witness conductance as a function of separation distance  $d$ . Under the following four axioms:*

- (SC1) **Quadratic response:**  $E[af] = a^2 E[f]$  for all  $a \in \mathbb{R}$  and admissible signals  $f$ ;
- (SC2) **Additivity on independent channels:**  $w(d_1, d_2) = w(d_1) + w(d_2)$  for causally independent separations;
- (SC3) **Monotone decay:**  $d_1 < d_2 \implies w(d_1) \geq w(d_2)$ ;
- (SC4) **Scale covariance:** under  $d \mapsto \lambda d$ ,  $E_{\lambda d}[f] = \lambda^{-2} E_d[f]$ .

*The unique admissible conductance is  $w(d) = C/d^2$  for a positive constant  $C > 0$ .*

*Proof.* From (SC4),  $w(\lambda r) = \lambda^{-2} w(r)$  for all  $\lambda > 0$ . Setting  $\lambda = r/r_0$  yields  $w(r) = (r_0/r)^2 w(r_0) = Cr^{-2}$  where  $C = r_0^2 w(r_0)$ . The unique continuous solution to  $w(\lambda r) = \lambda^{-\alpha} w(r)$  with  $\alpha = 2$  (forced by SC4) and monotone decay (SC3) is the power law  $w(r) = Cr^{-2}$ . Uniqueness: suppose  $\tilde{w}$  is another continuous solution. Then  $\tilde{w}(r)/w(r) = h(r)$  satisfies  $h(\lambda r) = h(r)$  for all  $\lambda > 0$ , so  $h$  is constant.  $\square$

**Theorem 8.2** (Spacetime Dimensionality is  $3 + 1$ ). *The witness framework forces  $n = 3$  spatial dimensions and one temporal dimension.*

*Proof sketch.* In  $n$  spatial dimensions, the scale-free radial flux through a sphere  $S^{n-1}(r)$  of radius  $r$  has density

$$\Phi_n(r) \sim r^{-(n-1)}.$$

This is not imported from electrostatics—it is a theorem of geometry forced by A0\*. In  $n$  dimensions, the surface area of  $S^{n-1}(r)$  is proportional to  $r^{n-1}$ . Any conserved quantity that

propagates isotropically (no preferred spatial direction—forced by gauge invariance, W8) has density falling as the inverse of this area:  $\Phi_n(r) \sim r^{-(n-1)}$ .

The witness conductance density forced by theorem 8.1 is  $w(r) \sim r^{-2}$ .

For the witness propagation to be self-consistent, the geometric flux law (a theorem of  $n$ -dimensional geometry) must match the forced conductance law (a consequence of SC1–SC4). Equating the two exponents:

$$n - 1 = 2 \quad \implies \quad n = 3.$$

Time is forced separately as the irreversible ledger coordinate  $\mathcal{T}$  (Step 8, theorem 6.12): it is not a spatial dimension but the monotone index of record formation, satisfying  $\Delta\mathcal{T} \geq 0$ . Time cannot be identified with a spatial direction because reversal of a spatial direction is a gauge transformation (an element of  $\mathcal{G}_{\mathcal{T}}$ ), whereas reversal of  $\mathcal{T}$  would destroy witness records, violating the append-only ledger (Step 5).

Therefore spacetime is  $3 + 1$  dimensional.

*Forcing:* The conductance exponent  $-2$  is forced by scale covariance (SC4). The flux exponent  $-(n - 1)$  is forced by isotropy (W8) and  $n$ -dimensional geometry. Matching the two forces  $n = 3$ . The temporal dimension is forced independently by the irreversibility of witnessing (Step 8). No other dimensionality is compatible with the conductance/flux matching theorem.  $\square$

## 8.2 B2. Gauge Group Structure

**Theorem 8.3** (Witness Bundle). *The ordered witness algebra  $\mathcal{A}_W$  over the  $3+1$  base manifold  $\Omega_{\text{cl}}$  defines a witness bundle  $\mathcal{W} \rightarrow \Omega_{\text{cl}}$  whose fiber at each point  $x \in \Omega_{\text{cl}}$  is the local witness Hilbert space  $H_\omega(x)$ . The structure group of this bundle is the group of fiber automorphisms preserving the witness inner product:*

$$G = \text{Aut}_{\langle \cdot, \cdot \rangle}(\mathcal{W}_x).$$

*Proof sketch.* The GNS construction (theorem 7.4) assigns to each point  $x \in \Omega_{\text{cl}}$  a Hilbert space  $H_\omega(x)$  carrying the representation  $\pi_\omega(x)$  of the local witness algebra  $\mathcal{A}_W(x)$ . These fibers assemble into a bundle  $\mathcal{W}$  over  $\Omega_{\text{cl}}$  because the GNS construction is functorial: smooth variation of the state  $\omega(x)$  produces smooth variation of the fiber. The structure group consists of all inner-product-preserving automorphisms of the fiber—this is forced because the witness inner product encodes the Born probability (theorem 6.41), and any transformation not preserving it would alter measurement probabilities, creating a witnessable distinction.

*Forcing:* The fiber Hilbert spaces are forced by GNS (Step 27). Their assembly into a bundle is forced by the smooth variation of the truth quotient over  $\Omega_{\text{cl}}$ . The structure group is forced to preserve the inner product because the inner product encodes the Born rule.  $\square$

**Theorem 8.4** (Fiber Decomposition and Gauge Classification). *The minimal stable internal fiber decomposition of the witness bundle  $\mathcal{W}$  under the following constraints:*

- (i) Witness-algebra irreducibility: *each fiber summand carries an irreducible representation of the local witness algebra;*
- (ii) Stability under Kähler flow: *the decomposition is preserved by the semigroup  $\{T_t\}$  (Step 29);*
- (iii) Anomaly cancellation: *the fiber decomposition is gauge-gravitationally consistent (the total gauge anomaly polynomial vanishes);*

*admits a classification: the gauge group  $G$  is a compact Lie group whose simple factors correspond to the irreducible summands of the fiber.*

*The general theorem is that the gauge group is the automorphism group of the witness bundle fibers. The specific ranks  $1 \oplus 2 \oplus 3$  are themselves forced by the interaction of the Kähler structure, the spin structure of  $3+1$  dimensions, and anomaly cancellation:*

- **Rank 1** ( $\rightarrow U(1)$ ): forced by the complex structure  $J$  (Step 31). On any Kähler manifold,  $J$  generates a  $U(1)$  phase action on the tangent bundle. This is the minimal gauge content of the Kähler base—the phase fiber  $L_{\text{phase}}$  is not optional.
- **Rank 2** ( $\rightarrow SU(2)$ ): forced by the spin structure of 3+1 dimensions. The spin group  $\text{Spin}(3,1) \cong SL(2, \mathbb{C})$  requires a rank-2 representation to act faithfully on spinors. The witness algebra’s involution (replay adjoint,  $W2$ ) on the 3+1 base forces spinorial content, which requires the weak fiber  $E_{\text{weak}}$  of rank 2.
- **Rank 3** ( $\rightarrow SU(3)$ ): forced by anomaly cancellation with ranks 1 and 2 present. In 4 dimensions, the mixed gauge anomaly  $\mathcal{A} \propto \text{Tr}[Y T_a^2]$  (where  $Y$  is  $U(1)$  charge and  $T_a$  are  $SU(2)$  generators) vanishes only if a rank-3 color sector exists with the correct fermion multiplicity [Adler, 1969, Bell and Jackiw, 1969]. An anomalous theory produces distinctions (e.g., “gauge current is conserved” vs. “it is not”) that cannot be consistently witnessed—violating  $A0^*$ . Therefore the color fiber  $C_{\text{color}}$  of rank 3 is forced.

The framework therefore derives the **complete gauge group**  $SU(3) \times SU(2) \times U(1)$  from  $A0^*$  without empirical input.

*Proof sketch.* Condition (i) forces the fiber to decompose into irreducible  $\mathcal{A}_W$ -modules—a standard result in representation theory of  $C^*$ -algebras. Condition (ii) restricts to summands that are eigenspaces of the Laplacian  $-\Delta$  (Step 29), since the semigroup  $T_t = e^{t\Delta}$  preserves each eigenspace. Condition (iii) restricts the allowed rank combinations to those for which the gauge anomaly polynomial vanishes—a purely algebraic constraint [Adler, 1969, Bell and Jackiw, 1969]. The automorphism group of a direct sum of complex vector spaces of ranks  $r_1, \dots, r_k$  preserving the inner product on each summand is  $\prod_i U(r_i)$ ; quotienting by the common center and imposing anomaly cancellation further constrains the possibilities.

*Forcing:* The fiber decomposition is forced by conditions (i)–(iii). The structure group is the automorphism group of the resulting fiber—a compact Lie group, forced by the inner product. The specific ranks  $1 \oplus 2 \oplus 3$  are forced: rank 1 by the Kähler complex structure  $J$ , rank 2 by the 3+1 spin structure, and rank 3 by anomaly cancellation with ranks 1+2 present. No empirical input enters the gauge group derivation.  $\square$

*Remark 8.5* (No Empirical Input for the Gauge Group). The derivation produces  $SU(3) \times SU(2) \times U(1)$  entirely from  $A0^*$ : rank 1 from the Kähler complex structure, rank 2 from the 3+1 spin structure, and rank 3 from anomaly cancellation. No empirical input is needed for the gauge group itself. The framework’s remaining open computational frontiers are the specific *values* of coupling constants and mass eigenvalues (which require solving the spectral problem on the witness bundle), not the gauge group.

### 8.3 B3. Particle Spectrum

**Definition 8.6** (Particle as Witness Module). A particle is a stable irreducible module  $M$  of the gauge-covariant witness algebra  $\mathcal{A}_W \rtimes G$ , where  $G$  is the gauge group acting on the witness Hilbert space  $H_\omega$ .

**Theorem 8.7** (Particle Classification). Given the gauge group  $G$  acting on the witness Hilbert space  $H_\omega$ , the particle spectrum is classified by the irreducible representations of  $G$ . Specifically:

- Each irreducible representation  $\rho: G \rightarrow \text{GL}(V_\rho)$  defines a particle sector.
- The mass of a particle in sector  $\rho$  is defined as the lowest positive spectral value of the gauge-covariant Dirac-type operator  $D_\rho$  acting on sections of the associated bundle  $\mathcal{W} \times_\rho V_\rho$ :

$$m_M^2 \sim \lambda_1(D_\rho^2),$$

where  $\lambda_1$  is the smallest positive eigenvalue.

- (c) *Generation structure arises from distinct stable branches of the Kähler spectrum on the witness bundle: each branch is a copy of the same  $G$ -representation but at a different eigenvalue of the Laplacian  $-\Delta$ .*

*Proof sketch.* Part (a) is standard representation theory: the Hilbert space  $H_\omega$  decomposes under  $G$  into irreducible subspaces, each carrying one irreducible representation. Part (b): the gauge-covariant Dirac operator  $D_\rho$  is the natural first-order differential operator on sections of the witness bundle twisted by  $\rho$ ; its spectrum is discrete because the base manifold  $\Omega_{\text{cl}}$  is compact (Step 26). The identification of mass with the spectral gap follows from the standard relation between the Klein–Gordon equation and the square of the Dirac operator. Part (c): the Laplacian  $-\Delta$  on the witness bundle has a discrete spectrum  $0 \leq \lambda_0 \leq \lambda_1 \leq \dots$ ; each eigenspace is a  $G$ -module isomorphic to the fundamental representation, and distinct eigenvalues produce distinct “generations” of the same representation type.

*Forcing:* The representation-theoretic classification follows from the gauge group structure (theorem 8.4). The spectral definition of mass follows from the Dirichlet form (Step 28) and the Laplacian (Step 29). Generation structure follows from the multiplicity of eigenspaces.  $\square$

*Remark 8.8* (Honest Boundary — Particle Masses). The representation classification follows standard Lie group theory once the gauge group is specified. The specific mass values—the numerical eigenvalues  $\lambda_1(D_\rho^2)$ —require solving the spectral problem of the gauge-covariant Dirac operator on the witness bundle with its Kähler geometry. This is an open computational problem. The framework provides the *equation* whose solutions are masses; it does not currently provide the solutions.

## 8.4 B4. Coupling Constants

**Definition 8.9** (Coupling Constants as Bundle Moduli). *Each dimensionless coupling constant is a normalized modulus of the witness bundle:*

- (a) Electromagnetic coupling  $\alpha_{\text{em}}$ : *the holonomy modulus of the  $U(1)$  connection on  $L_{\text{phase}}$ ,*

$$\alpha_{\text{em}} = \frac{1}{4\pi} \|\text{Hol}_{U(1)}(\gamma)\|^2,$$

*normalized over a canonical loop  $\gamma$  in  $\Omega_{\text{cl}}$ .*

- (b) Strong coupling  $\alpha_s$ : *the connection modulus of the  $SU(3)$  sector on  $C_{\text{color}}$ ,*

$$\alpha_s = \frac{1}{4\pi} \|F_{SU(3)}\|_{L^2}^2.$$

- (c) Weak mixing angle  $\theta_W$ : *the sector-intersection ratio between the  $U(1)$  and  $SU(2)$  fibers,*

$$\sin^2 \theta_W = \frac{\dim(L_{\text{phase}} \cap E_{\text{weak}})}{\dim(L_{\text{phase}}) + \dim(E_{\text{weak}})},$$

*measuring the degree of mixing between the phase and weak fibers.*

**Theorem 8.10** (Renormalization Group Flow). *Renormalization group flow is coarse-graining on the witness algebra: restricting from a fine truth quotient  $\Pi^*(\mathcal{L}_B)$  to a coarse truth quotient  $\Pi^*(\mathcal{L}_{B'})$  ( $B' < B$ ) changes the effective couplings. Formally, let  $\pi_{B,B'}: \Pi^*(\mathcal{L}_B) \rightarrow \Pi^*(\mathcal{L}_{B'})$  be the bonding map of the inverse system (Step 26). The effective coupling at scale  $B'$  is:*

$$\alpha_i(B') = (\pi_{B,B'})_* \alpha_i(B),$$

*where  $(\pi_{B,B'})_*$  is the pushforward of the connection modulus under the coarse-graining projection.*

*Proof sketch.* The bonding maps  $\pi_{B,B'}$  merge truth classes that become indistinguishable at the coarser budget  $B'$  (Step 26). The connection on the witness bundle, when restricted to the coarser quotient, sees fewer fiber directions and hence a modified holonomy. The pushforward of the connection modulus under the bonding map is the standard definition of Wilson's renormalization group flow, expressed in the language of the witness algebra: integrating out the fine-grained degrees of freedom (truth classes that merge under  $\pi_{B,B'}$ ) changes the effective gauge coupling.

*Forcing:* The coarse-graining maps are forced by the budget hierarchy (Step 9) and the inverse system (Step 26). The change in effective coupling under coarse-graining is a mathematical consequence of the pushforward—not an additional physical assumption.  $\square$

*Remark 8.11* (Honest Boundary — Coupling Values). The framework produces the *form* of RG flow and the structural definition of each coupling constant as a bundle modulus. Computing their specific numerical values at a given energy scale—for instance,  $\alpha_{\text{em}}^{-1} \approx 137$  at low energies—requires solving the coarse-graining spectral problem: evaluating the pushforward  $(\pi_{B,B'})_* \alpha_i(B)$  for the witness bundle with its specific Kähler geometry. This is an open computational problem, analogous to computing hadronic masses from the QCD Lagrangian.

## 8.5 B5. Sector Equations

The split law  $X = J\nabla E - \nabla D$ , together with the Kähler structure  $(g, J, \omega)$  and the GNS Hilbert space  $H_\omega$ , produces specific dynamical equations in each physical sector. The reversible part  $J\nabla E$  governs quantum and gauge dynamics; the irreversible part  $-\nabla D$  governs thermodynamic relaxation.

### Quantum sector (reversible).

**Theorem 8.12** (Schrödinger Equation). *The reversible witness generator on  $H_\omega$  yields a self-adjoint Hamiltonian  $H$ . The unitary flow on the GNS Hilbert space is:*

$$i \frac{d\psi}{dt} = H\psi.$$

*This is the Schrödinger equation.*

*Proof sketch.* The reversible part  $J\nabla E$  of the split law, restricted to the GNS Hilbert space  $H_\omega$ , is a densely defined operator on  $H_\omega$ . We verify self-adjointness: the Kähler metric ensures that  $J$  is an isometry (theorem 6.51), and  $\nabla E$  is symmetric with respect to the inner product  $\langle \cdot, \cdot \rangle_\omega$  because  $E$  is a real-valued functional on the truth quotient. Therefore  $H = J\nabla E$  restricted to  $H_\omega$  is self-adjoint on its natural domain.

By Stone's theorem, every strongly continuous one-parameter unitary group on a Hilbert space is generated by a self-adjoint operator. Conversely, every self-adjoint operator generates such a group. The unitary group  $U(t) = e^{-iHt}$  gives the time evolution, and the Schrödinger equation  $i d\psi/dt = H\psi$  is its infinitesimal form.

*Forcing:* The reversible generator is forced by the split law (Step 14, theorem 6.19). Self-adjointness is forced by the Kähler structure (Step 33). Stone's theorem is a mathematical consequence. The Schrödinger equation is the unique infinitesimal form of the resulting unitary flow.  $\square$

### Gauge sector (connection).

**Theorem 8.13** (Yang–Mills Equation). *Local witness-frame changes define a connection  $A$  on the witness bundle  $\mathcal{W}$ . The gauge field strength is  $F = dA + A \wedge A$ . The Yang–Mills equation follows from varying the gauge curvature action:*

$$D^*F = J_w,$$

where  $J_w$  is the witness current and  $D^* = *D*$  is the gauge-covariant codifferential.

*Proof sketch.* The gauge group  $G$  acts on the witness bundle fibers by inner-product preserving automorphisms (theorem 8.3). A local witness-frame is a local trivialization of  $\mathcal{W}$ ; changing frames introduces a gauge transformation  $g(x) \in G$ . The connection  $A$  is the Lie-algebra-valued 1-form encoding infinitesimal frame changes:  $A_\mu = g^{-1}\partial_\mu g$  in the pure-gauge case, with the general connection adding a non-trivial component.

The field strength  $F = dA + A \wedge A$  measures the curvature of the connection—the failure of parallel transport around infinitesimal loops to return to the identity. The gauge-invariant action is  $S[A] = \int_{\Omega_{\text{cl}}} \text{tr}(F \wedge *F) \text{dvol}_g$ , where  $*$  is the Hodge star of the Riemannian metric  $g$  (Step 30). Variation with respect to  $A$  yields the Yang–Mills equation  $D^*F = J_w$ , where  $J_w$  is the source current obtained by varying the matter action.

*Forcing:* The connection  $A$  is forced by the witness bundle structure (theorem 8.3). The curvature  $F$  is the canonical obstruction to trivial parallel transport. The Yang–Mills equation is the Euler–Lagrange equation of the unique gauge-invariant quadratic action  $\text{tr}(F \wedge *F)$ . Uniqueness of the quadratic action follows from gauge invariance and dimensional analysis on the  $3 + 1$  base manifold (theorem 8.2).  $\square$

### Gravity sector (metric modulation).

**Theorem 8.14** (Einstein Equation). *The witness/Fisher metric  $g_{ij}$  on the base manifold  $\Omega_{\text{cl}}$  responds to the distribution of truth-density  $\rho$ . The witness action is:*

$$S[g, \rho] = \int_{\Omega_{\text{cl}}} \left( R(g) + \lambda |\nabla \log \rho|_g^2 + \mathcal{L}_m \right) \text{dvol}_g,$$

where  $R(g)$  is the scalar curvature,  $\lambda$  is a coupling constant, and  $\mathcal{L}_m$  is the matter Lagrangian. Variation with respect to  $g$  yields the Einstein equation:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = T_{\mu\nu}^{(\rho)} + T_{\mu\nu}^{(m)},$$

where  $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu}$  is the Einstein tensor,  $\Lambda$  is the cosmological constant,  $T_{\mu\nu}^{(\rho)}$  is the truth-density stress-energy, and  $T_{\mu\nu}^{(m)}$  is the matter stress-energy.

*Proof sketch.* The Fisher metric (Step 30, theorem 6.46) on  $\Omega_{\text{cl}}$  responds to variations of the truth distribution  $\rho$ : regions of high truth-density (many distinguishable truth classes per unit volume) curve the metric differently from regions of low truth-density. The scalar curvature  $R(g)$  is the unique scalar invariant of the Riemannian metric that is second-order in derivatives—forced by diffeomorphism invariance and dimensional analysis on the  $3 + 1$  base.

The  $|\nabla \log \rho|_g^2$  term encodes the energy cost of truth-density gradients: sharp variations in the density of distinguishable truth classes cost energy, measured by the Dirichlet form (Step 28).

Variation of  $S[g, \rho]$  with respect to  $g_{\mu\nu}$  yields the Einstein equation by the standard Palatini computation. The Bianchi identity  $\nabla_\mu G^{\mu\nu} = 0$  follows from diffeomorphism invariance of the witness action—equivalently, from gauge invariance of the truth quotient (Step 12), which ensures that relabeling spacetime coordinates does not affect the truth structure.

The cosmological constant  $\Lambda$  arises as the vacuum witness energy: the ground-state energy of the witness algebra  $\mathcal{A}_W$  acting on  $H_w$ . Even in the absence of matter, the witness algebra has



a nonzero vacuum expectation value  $\langle \Omega_\omega, H \Omega_\omega \rangle > 0$  (since  $H = J\nabla E$  is bounded below but not zero on  $\Omega_\omega$ ), contributing a constant energy density to the right-hand side of the Einstein equation.

*Forcing:* The Fisher metric is forced by the Dirichlet form (Step 28) and Čencov’s theorem (Step 30). The Einstein–Hilbert action is the unique diffeomorphism-invariant scalar action that is second-order in metric derivatives. The Bianchi identity is forced by diffeomorphism invariance (gauge invariance of the truth quotient).  $\Lambda$  is forced as the vacuum energy of the witness algebra.  $\square$

## Thermodynamic sector (dissipative).

**Theorem 8.15** (Second Law of Thermodynamics). *The irreversible part  $-\nabla D$  of the split law gives gradient flow toward equilibrium. The entropy production rate on the continuum is:*

$$\frac{dS}{dt} = \int_{\Omega_{\text{cl}}} |\nabla D|^2 \text{dvol}_g \geq 0.$$

*Proof sketch.* The dissipation functional  $D$  on  $\Omega_{\text{cl}}$  is non-negative by construction: it measures the cost of irreversible witness record formation (Step 8, theorem 6.12). The gradient flow  $\dot{x} = -\nabla D(x)$  descends  $D$  monotonically, since  $\dot{D} = \langle \nabla D, \dot{x} \rangle = -|\nabla D|^2 \leq 0$ .

The entropy  $S$  increases because the Riemannian volume swept by the flow grows: distinguishable truth classes accumulate as records are formed. The production rate  $dS/dt = \int |\nabla D|^2 \text{dvol}_g \geq 0$  is the continuum extension of Step 8’s inequality  $\Delta\mathcal{T} \geq 0$ . Equality  $dS/dt = 0$  holds if and only if  $\nabla D = 0$  everywhere—equilibrium, where no further irreversible witnessing occurs.

*Forcing:* The dissipative part of the split law is forced by Step 14. The non-negativity of entropy production is forced by the append-only ledger (Step 5) and irreversibility of time (Step 8). The integral form extends the discrete  $\Delta\mathcal{T} \geq 0$  to the continuum  $\Omega_{\text{cl}}$ .  $\square$

## 8.6 Empirical Predictions

The physical content derived above yields three falsifiable predictions. Each identifies a structural consequence of the framework that can be tested independently of the open computational problems (mass values, coupling constants).

**P1. Inverse-square witness flux.** Any fundamental long-range separative sector must obey inverse-square witness flux. Specifically, the conductance/flux matching (theorem 8.2) requires that every fundamental force mediating witness separation at long range decays as  $r^{-2}$  in the flux density. A systematic violation of the inverse-square law at a truly fundamental scale—not attributable to short-range screening (as in the weak and strong forces) or finite-size effects—falsifies the conductance theorem (theorem 8.1).

**P2. Order-curvature correction for noncommuting witnesses.** Sequential noncommuting witness instruments must exhibit an order-curvature correction proportional to the local commutator 2-form. Quantitatively, the probability difference between performing witness  $w_1$  before  $w_2$  versus  $w_2$  before  $w_1$  satisfies:

$$\Delta P \sim \omega([w_1, w_2]) + O(\|[w_1, w_2]\|^2),$$

where  $\omega$  is the symplectic form (Step 32, theorem 6.50) evaluated on the commutator. A null result— $\Delta P = 0$ —where the ordered witness algebra predicts a nonzero commutator  $[w_1, w_2] \neq 0$  falsifies the ordered extension of A0\* (theorem 2.15).

**P3. Dimensionality is a theorem.** The  $3 + 1$  dimensionality result (theorem 8.2) is a *theorem* of the framework, not an assumption. Any truly scale-free long-range witness geometry in  $n \neq 3$  spatial dimensions is incompatible with the conductance/flux compatibility theorem. This is falsifiable in the following sense: if a physical system were found that exhibits fundamental long-range scale-free propagation in an effectively  $n \neq 3$  spatial geometry (not merely a dimensional reduction or compactification artifact), it would contradict the matching  $n - 1 = 2$  and hence falsify the conductance axioms (SC1)–(SC4).

*Remark 8.16* (Prediction Scope). Predictions P1–P3 test the *structural* layer of the framework: dimensionality, flux scaling, and the ordered witness algebra. The gauge group  $SU(3) \times SU(2) \times U(1)$  is itself derived (theorem 8.4), not empirical input. Further predictions—such as specific mass ratios or coupling constant values—await solution of the spectral and coarse-graining problems identified in theorems 8.8 and 8.11.

## 9 Verification

We apply the framework’s own verification mechanism to the 34-step derivation (Phases A–I, Steps 0–33). The three-point audit tests each step for: (1) *Hidden Chooser*—no alternative  $A0^*$ -compliant continuation exists; (2) *Encoding Invariance*—re-executing under different encoding conventions produces the same quotient structure; (3) *Distinguishability Equivalence*—every distinction introduced is witnessed, and no witnessed distinction is omitted. A fourth check verifies algebraic properties on finite models via Z3. The protocol is depicted in fig. 18.

Table 3 confirms: all 34 steps pass the three-point audit. Twenty steps have algebraic properties checked on finite models via Z3 (Appendix C). All steps are fully forced by  $A0^*$ —no modeling commitments remain.

**Lean 4 verification:** The full derivation and quantitative seed extension are verified by 414 Lean 4 theorems across 78 files (zero sorry, zero admit, one axiom  $A0^*$ ).

## 10 Intellectual Context

We situate the OPOCH framework within five intellectual lineages, identifying precise points of contact and divergence.

### 10.1 Constructive Foundations

Martin-Löf’s intuitionistic type theory [Martin-Löf, 1984] requires explicit constructions for existence claims; Bishop’s constructive analysis [Bishop, 1967] demonstrated that large portions of classical mathematics survive under this constraint; Kleene’s realizability [Kleene, 1952] formalized truth as recursive function existence. The Curry–Howard correspondence identifies proofs with programs, connecting logic and computation.  $A0^*$  distills the constructive programme further: not just “existence requires a witness” but “*distinction* requires a witness.” Where Martin-Löf begins with a rich type structure (universes, dependent types), we begin with  $\perp$  and derive structure. The self-hosting property (Step 19) is a Curry–Howard instance: the derivation’s proof corresponds to a program that verifies itself.

### 10.2 Computation and Information

Solomonoff’s universal induction [Solomonoff, 1964] and Kolmogorov complexity [Kolmogorov, 1963, Li and Vitányi, 2008] define a universal prior over hypotheses weighted by description length. Wheeler’s “It from Bit” [Wheeler, 1990] proposed that physical quantities derive meaning from information-theoretic questions. Zuse [Zuse, 1969] and Fredkin [Fredkin, 2003] modeled the universe as discrete computation. Deutsch and Marletto’s constructor theory [Deutsch and

[Marletto, 2015] reformulates physics in terms of which transformations can and cannot be performed. Our carrier  $\mathcal{D}^*$  is precisely the set on which Kolmogorov complexity is defined, and the prefix-free coding at Step 2 satisfies the same Kraft inequality [Kraft, 1949]. The key divergence from Solomonoff is philosophical: we use  $\mathcal{U}$  as a *verifier*, not a prior for probabilistic prediction; the trit  $\{\text{UNIQUE}, \Omega, \text{UNSAT}\}$  replaces probability. Unlike digital physics, our derivation does not assume discrete computation—Step 3 derives the need for a computational substrate from  $A0^*$ ; the Church–Turing thesis corroborates the forced class. Both constructor theory and OPOCH ask “what tasks are possible?” rather than “what trajectories occur?”—but our derivation forces the test algebra  $\Delta^*$  from  $A0^*$  rather than taking possible transformations as primitive.

### 10.3 Verification and Certification

Necula’s proof-carrying code [Necula, 1997] attaches machine-checkable proofs to executables; Goldwasser, Micali, and Rackoff [Goldwasser et al., 1989] established interactive proof systems; Clarke et al. [Clarke et al., 2018] provide the reference for model checking. McConnell et al. [McConnell et al., 2011] formalized certifying algorithms that produce efficiently checkable certificates alongside results. Belnap’s four-valued logic [Belnap, 1977] extends classical logic with “unknown” and “contradictory” values. The OPOCH receipt system is proof-carrying code applied to reasoning traces: every execution produces a SHA-256 hash chain that is replay-verifiable. Unlike interactive proofs, verification is non-interactive; unlike model checking, the Bellman search prunes the state space dynamically. The exactness gate (Step 14) implements McConnell’s certificate-first pattern. Our trit differs from Belnap’s four values: we exclude “contradictory” (a deterministic verifier cannot produce genuine contradictions), and our  $\Omega$  is constructive—it returns the minimal separator test, not a static “unknown” label.

### 10.4 Geometry, Topology, and Automata

Scott’s domain theory [Scott, 1970] models computation via continuous lattices; Smyth [Smyth, 1983] and Abramsky [Abramsky and Jung, 1994] connect topological structure to logical systems. The Myhill–Nerode theorem [Myhill, 1957, Nerode, 1958] identifies strings with identical futures, yielding minimal automata [Hopcroft and Ullman, 1979]; Milner’s bisimulation [Milner, 1989] extends this to concurrent systems. Amari’s information geometry [Amari, 2016] treats statistical models as Riemannian manifolds with the Fisher metric measuring distinguishability [Rao, 1945]. Our observable opens  $\mathcal{O}_\Delta^{\text{fin}}$  (Step 7) are a finite restriction of the Scott topology to testable predicates. The Myhill–Nerode congruence  $\equiv_q$  (Step 15) applies futures equivalence to the system’s state space rather than string classes, with the test algebra  $\Delta^*$  as alphabet. The separator geometry (Step 16) constructs a metric  $d_q$  analogous to Fisher-Rao distance but defined over deterministic truth classes and measured in test cost rather than information divergence.

### 10.5 Decision Theory and Applications

Bellman’s principle of optimality [Bellman, 1957] and Wald’s minimax decision theory [Wald, 1950] provide the mathematical foundation for the framework’s test-selection rule: at each step, select the test minimizing worst-case cost-to-go. Chain-of-thought [Wei et al., 2022] and tree-of-thoughts [Yao et al., 2023] improve LLM reasoning by eliciting intermediate steps, but every step remains a sample from a learned distribution. The OPOCH test selection (Step 18) is a direct application of the Bellman equation in test space rather than action space, with Wald’s minimax framing justifying the worst-case optimization.

## 10.6 Self-Reference

The Kleene recursion theorem guarantees that every total recursive function can compute its own index; metacircular interpreters and reflective towers formalize systems reasoning about their own structure. Our self-hosting property  $\Upsilon(\mathcal{S}) = \mathcal{S}$  (Step 19) is the verification analog: the system verifies its own derivation, producing `UNIQUE` with receipts for each step. This is stronger than metacircular interpretation (which merely executes) and connects to proof-carrying code (each step’s receipt is a proof that the step is forced). Mechanizing the derivation in a proof assistant (Lean, Coq) would yield a certified executable system via the Curry–Howard correspondence.

## 11 Discussion and Open Frontiers

### 11.1 What Is Claimed

We claim the following, and no more:

- (i) **A canonical structure is derived from  $A0^*$  alone.**  $A0^*$ , when self-applied to witness validity, forces the endogenous witness algebra. The former model-class commitments are not extra assumptions but earlier consequences of that self-application: closed universe is forced by (W6), commutativity is a derived sector property of the ordered ledger, and prior-free minimax is forced by the absence of witnessed priors. The 34-step development (Phases A–I) then produces carrier, self-delimiting syntax, a test algebra, truth quotient, observable-open structure, eraser operators, time and cost accounting, gauge invariance, Myhill–Nerode congruence, separator geometry, Bellman recursion, self-hosting closure, binary interface, universal execution, consciousness projector, and a Kähler geometric structure on the continuum limit of truths. No modeling commitments remain; every step is a theorem of  $A0^*$ . The claim of “forcedness” is to be read as “forced up to computable isomorphism,” rather than as a claim that every representational choice is eliminated under every possible formalization of witnessability. We provide a three-point audit of the intended invariances.
- (ii) **The derivation is self-auditing.** Each step passes hidden-chooser, encoding-invariance, and distinguishability-equivalence checks. Selected algebraic properties are verified on finite models via Z3.
- (iii) **The derivation identifies its single input and its empirical boundary.** The only declared input is  $A0^*$ . The presentation makes explicit where the derivation’s structural vocabulary ends and empirical specification begins—principally the choice of  $\Delta_{\text{phys}}$ , which selects which tests nature actually admits.

### 11.2 The Vocabulary of Physical Law

The derivation produces vocabulary, carrier  $\mathcal{D}^*$ , test algebra  $\Delta$ , time  $\mathcal{T}$ , gauge group  $\mathcal{G}_{\mathcal{T}}$ , fiber  $W(\cdot)$ , truth quotient  $\Pi^*(\cdot)$ , that is structurally parallel to the vocabulary of mathematical physics. This parallel reflects a shared concern with what can be distinguished by finite operations.

However, “vocabulary” is not “theory.” A physical theory specifies which tests  $\Delta_{\text{phys}}$  nature actually admits, what their costs are, and what symmetry group relates equivalent descriptions. The OPOCH framework provides the scaffold. Specifying  $\Delta_{\text{phys}}$  remains an open frontier. The derivation produces the *form* of physical law without determining its *content*, which requires empirical input.

### 11.3 The Split Law and Sector Readouts

The Kähler triple  $(g, J, \omega)$  of Phase I forces a canonical decomposition of any dynamical law on  $\Omega$ . Every vector field  $X$  on a Kähler manifold splits uniquely as  $X = J\nabla E - \nabla D$ , where  $E$  is a Hamiltonian (energy, conserved under  $J$ -rotation) and  $D$  is a dissipation potential (entropy production, irreversible). The first term generates symplectic flow; the second generates gradient flow toward equilibrium. Quantum mechanics, gauge field theory, gravitational dynamics, and thermodynamics are *sector-readouts* of this single split: each arises by restricting  $\Omega$  to a subsector and reading off the induced  $(g, J, \omega)$ . The split is not a modeling choice—it is forced by Steps 30–33 via Kähler compatibility.

### 11.4 The Church–Turing Corroboration

Step 3 forces the test set  $\Delta$  to be maximally closed under totality-preserving composition. This maximal closure is exactly the class of total computable functions—a mathematical theorem provable within computability theory [Rogers, 1967], independent of the Church–Turing thesis. The Church–Turing thesis [Church, 1936, Turing, 1936] independently identified this same class by a different route (characterizing “effective procedure”). The agreement between A0\*’s forcing and Church–Turing’s identification is historical corroboration, not a logical dependency of the derivation.

*Remark 11.1.* If a hypercomputational model were discovered that exceeds Turing-computability, Step 3 would need revision. The derivation through Step 2 would survive intact, but the carrier  $\mathcal{D}^*$  would expand. The remainder of the development would proceed analogously within the enlarged carrier.

### 11.5 The Closed-System Assumption

Step 8 (feasibility shrinkage) assumes a closed system with a finite budget  $B$ . Claims whose verification cost exceeds any available budget remain in  $\Omega$ . Such claims are unresolved with a minimal separator and cannot be promoted to UNIQUE without additional resources. This is a direct consequence of budgeted witnessability.

### 11.6 Minimax vs. Other Optimality Criteria

Step 18 forces the minimax policy as the unique prior-free bootstrap policy. Before any prior is itself witnessed and recorded in  $\mathcal{L}_{\text{ord}}$ , branch labels over future outcomes are gauge: no unrecorded probability distribution is admissible by A0\*. The only cost aggregator invariant under outcome relabeling, cost rescaling, replay-equivalent path decomposition, and gauge recoding is the worst-case continuation cost—i.e., minimax. Bayesian expected-cost selection becomes admissible only *after* a prior  $\mu$  has itself attained witness status in the trace:

$$\tau_{\text{Bayes}}^* = \operatorname{argmin}_{\tau \in \Delta_q} \left[ \pi(\tau) + \sum_{a \in \text{out}(\tau)} \mu(a \mid \tau) V(W(a), t + \pi(\tau); q) \right] \quad (\mu \text{ witnessed in } \mathcal{L}_{\text{ord}}).$$

Minimax is therefore the *bootstrap* policy forced by A0\*; Bayesian selection is the *mature* policy enabled by self-observation. The choice is not left open—it is resolved by the witness status of the prior.

### 11.7 Myhill–Nerode Implications

The Myhill–Nerode congruence  $\equiv_q$  (Step 15) provides a minimal state representation. States with identical futures can be merged, reducing memory and computation. The connection to bisimulation in process algebra suggests that multi-system interactions can be analyzed using tools from concurrency theory.

## 11.8 Separator Geometry and Hardness

The separator metric  $d_q$  (Step 16) provides a quantitative measure of difficulty. Regions of high curvature  $\kappa$  correspond to phase transitions in difficulty, where small changes in the query produce large changes in resolution cost. This geometry is analogous in spirit to the Fisher-Rao metric on statistical manifolds, but it is defined over deterministic truth classes and measured in computational cost.

## 11.9 Quantitative Extraction Frontiers

The structure is complete. The computation is the frontier. Every dimensionless invariant is forced (Theorem 5.11); every dimensional quantity is fixed up to the internally derived normalization map. The remaining work is explicit spectral extraction from the seed operator—a well-posed computation, not an open structural question.

- (1) **Explicit spectral extraction from  $L^*$ .** Computing the eigenvalues, holonomy classes, and normal-form coefficients of the linearized refinement operator at the seed. This is a concrete numerical problem, not a structural gap.
- (2) **Specifying  $\Delta_{\text{phys}}$ .** Which families of physical experiments should be treated as primitive tests, and which symmetries should be treated as gauge, so that the resulting  $\Delta_{\text{phys}}$  reproduces established empirical constraints?
- (3) **Exploiting the ordered ledger for quantum reconstruction.** Whether the non-commuting sector of  $\mathcal{A}_W$ , together with the Kähler structure, suffices to reconstruct operator-algebraic quantum mechanics without additional empirical input.
- (4) **Energy as abstract cost vs. physical energy.** Clarifying the boundary between abstract resource measures and physical energetics under the Landauer correspondence.
- (5) **Separator geometry as a complexity lens.** Can  $\kappa$  predict computational difficulty of problem classes via circuit or communication complexity?
- (6) **Self-hosting depth.** Understanding whether deeper reflective towers stabilize.
- (7) **Lean mechanization of infinite-domain theorems.** The 414-theorem Lean 4 verification covers the finite/Nat model. Encoding infinite-domain properties via mathlib remains in progress.

## 12 Conclusion

We have presented OPOCH, a framework that *derives* its operating principle from nothingness and uses it to derive the structure of physical reality. The framework has no declared axioms:  $A0^*$  (Completed Witnessability) is the first theorem, derived from  $\perp$  through the necessity chain  $\perp \rightarrow N0 \rightarrow N1 \rightarrow N2 \rightarrow N3 \rightarrow N4 \rightarrow A0^*$ . Starting from  $\perp$ , the only unfalsifiable state, the forced derivation chain is canonical up to computable isomorphism—with no surviving modeling commitments.

The work makes eight concrete contributions.

- (1)  **$A0^*$  derived from  $\perp$ —no axioms.** The framework begins from absolute nothingness and derives its sole operating principle through five necessity theorems ( $N0$ – $N4$ ). The closure move—applying witnessability to witness validity itself—forces an ordered witness algebra  $\mathcal{A}_W$  carrying finite description, replayability, disturbance accounting, and compositional closure. All former “modeling commitments” (closed universe, commutativity,



prior-free selection, conductance exponent, smoothness) are derived consequences of  $A0^*$ , not external assumptions.

- (2) **The static universe.** The full  $\Pi$ -fixed closure whole  $U = \text{Fix}(\Pi)$  is atemporal (theorem 5.6). Time is not a global coordinate but a local phenomenon: the monotone reduction of unresolved witness-defect by partial sections of  $U$  (theorem 5.7). The universe is static globally, dynamic only as local self-resolution.
- (3) **Quantum structure from ordered witness algebra.** Noncommutativity is internal from Step 0. The  $C^*$ -completion of  $\mathcal{A}_W$  yields, via the GNS construction, a Hilbert-space representation  $(H_\omega, \pi_\omega, \Omega_\omega)$ . Contexts, effects, and the Born rule  $P(E) = \langle \Omega_\omega, \pi_\omega(E) \Omega_\omega \rangle$  are forced—not postulated.
- (4) **Self-remodelling consciousness.** Consciousness is the minimal persistent self-valued witness selector (theorem 6.34), whose witness policy is endogenously updated through an internal self-valuation channel:  $s_{t+1} = \mathbb{K}(s_t, c(s_t), \Delta_Q(s_t))$ , where  $c(s_t)$  is the endogenous selector over which unresolved components matter. Consciousness = closure acting on itself through self-witness.
- (5) **Kähler geometry from a single witness generator.** The continuum limit  $\Omega_{cl}$  carries a forced Kähler triple  $(g, J, \omega)$  arising from the symmetric and antisymmetric parts of one local witness generator. The conductance exponent  $w = C/d^2$  is forced by scale covariance (theorem 8.1); the Nijenhuis tensor vanishes by replay-completeness (theorem 6.49).
- (6) **3+1 spacetime and sector equations.** Matching the forced conductance law to radial flux scaling gives spatial dimension  $n = 3$ ; the irreversible ledger coordinate gives  $+1$ . The Kähler split law  $X = J\nabla E - \nabla D$  decomposes dynamics into quantum (Schrödinger), gauge (Yang–Mills), gravitational (Einstein), and thermodynamic sectors.
- (7) **Falsifiable predictions.** (P1) Inverse-square witness flux at fundamental scales. (P2) Order-curvature corrections from noncommuting witness instruments proportional to the commutator 2-form. (P3) Scale-free long-range witness geometry is incompatible with  $n \neq 3$  spatial dimensions.
- (8) **The quantitative seed.** The seed  $\delta^*$  is the unique least positive self-retaining non-gauge closure-defect minimizing the action functional (Theorem 5.9). It is a fixed point of every refinement operator (Theorem 5.10). Every dimensionless observable is a spectral, holonomy, or normal-form invariant of the linearized operator  $L^*$  at  $\delta^*$  (Theorem 5.11). Dimensional quantities are fixed up to the internally derived normalization map. The quantitative completion is verified by 414 Lean 4 theorems across 78 files (zero sorry, zero admit, one axiom  $A0^*$ ).

**Quantitative extraction frontiers.** Every dimensionless invariant is forced. Every dimensional quantity is fixed up to the internally derived normalization map. The remaining work is explicit spectral extraction from the seed operator—a well-posed computation, not an open structural question. The gauge group  $SU(3) \times SU(2) \times U(1)$  is derived entirely from  $A0^*$ : rank 1 from Kähler, rank 2 from 3+1 spin, rank 3 from anomaly cancellation. No empirical input enters.

**The reflective chain.** The derivation traces a single reflective chain: nothingness forces closure, closure reveals defect, defect requires self-observation, self-observation iterated on the continuum limit produces geometry, and geometry determines the form of dynamical law. The entire chain is one act: nothingness forced to witness itself. The whole  $U$  exists atemporally; local sections resolve toward it.

When meaning is restricted to witnessable distinctions, all familiar structures—quotienting by indistinguishability, invariance under untestable relabelings, budget-bounded decision procedures, Hilbert-space quantum mechanics, gauge field theory, general relativity, and the Kähler geometry of the efficiency landscape—become natural and, in the framework’s formal setting, unavoidable.

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## A Complete Derivation with Full Proofs

This appendix provides the complete proofs for all 34 derivation steps (Steps 0–33), followed by a compact reference table, the dependency chain, new mathematical objects, and verification summary. The theorem statements remain in the core derivation (§6); here we reproduce each proof verbatim along with associated remarks.

### A.1 Step 0: Nothingness and the Forced Output Gate

*Proof.* By definition, if no witness procedure exists then no distinction is admissible (A0), hence there are no descriptions and no tests.  $\perp$  is the unique such state.

For the output gate: let  $q$  be any query with candidate answer set  $\mathcal{A}$ . After all affordable tests have been executed, the surviving set  $\mathcal{A}_S \subseteq \mathcal{A}$  satisfies exactly one of three conditions:

1.  $|\mathcal{A}_S| = 1$ : a single answer survives. Output UNIQUE with witness.
2.  $|\mathcal{A}_S| > 1$ : multiple answers survive and no further test is affordable. Output  $\Omega$  with minimal separator.
3.  $|\mathcal{A}_S| = 0$ : no answer survives. Output UNSAT with counter-witness.

These three cases partition the non-negative integers, hence are exhaustive and mutually exclusive. Producing any output outside this trichotomy would assert a distinction without a witness (violating A0) or withhold an admissible separator (also violating A0).

*Forcing:* The trichotomy is forced by the pigeonhole principle on  $|\mathcal{A}_S| \in \{0, 1, 2, 3, \dots\}$  collapsed into three regimes. No fourth case exists.  $\square$

*Remark A.1* (Why start from  $\perp$ ?). Starting from  $\perp$  is not mysticism—it is a methodological choice. We wish to identify the *minimal* assumptions that force structure. By starting from nothing and adding only A0, every subsequent object must justify its existence through a forcing argument. Any object that cannot be so justified is not part of the framework.

### A.2 Step 1: Carrier with Finite Slice

*Proof.* A0 asserts that admissible distinctions require finite witness procedures. A finite procedure must be finitely described, hence is a finite string over some finite alphabet  $\Sigma$ . Any  $\Sigma$  with  $|\Sigma| \geq 2$  is computably isomorphic to  $\{0, 1\}$ ; an alphabet with  $|\Sigma| = 1$  admits no internal distinctions (all strings are  $1^n$ , isomorphic to  $\mathbb{N} \hookrightarrow \{0, 1\}^*$ ). Therefore the carrier is  $\mathcal{D}^* = \{0, 1\}^{<\infty}$ , unique up to computable isomorphism.

The finite slice  $D_0$  is forced by A0’s finiteness requirement: a witness of cost  $B$  can neither read nor produce strings longer than  $B$  bits. Therefore all operationally reachable descriptions lie in  $D_0$ . The full carrier  $\mathcal{D}^*$  serves as the ambient space;  $D_0$  is the effective workspace.

*Forcing:* Departure from  $\perp$  requires at least one distinction. One distinction requires at least one witness. One witness is a finite string. The set of all finite strings is  $\mathcal{D}^*$ . Budget-bounding gives  $D_0$ .  $\square$

### A.3 Step 2: Self-Delimiting Syntax

*Proof.* Consider two descriptions  $x, y \in \mathcal{D}^*$  where  $x \sqsubseteq y$  (i.e.,  $x$  is a proper prefix of  $y$ ). In a closed universe, the only information available is the bit stream itself—there is no external end-of-string marker. A finite procedure reading the stream cannot determine, upon completing  $x$ , whether the description is  $x$  or a longer string beginning with  $x$ . This ambiguity means no finite witness can separate “description is  $x$ ” from “description begins with  $x$ ,” violating A0.

Self-delimitation resolves this. The map  $\text{sd}(s) = 1^{|s|}0s$  is a valid self-delimiting encoding:

1. Read bits until the first 0 is encountered. The count of preceding 1’s gives  $|s|$ .
2. Read exactly  $|s|$  further bits to recover  $s$ .

This procedure terminates in  $2|s| + 1$  steps. The encoding is prefix-free: no  $\text{sd}(x)$  is a proper prefix of  $\text{sd}(y)$  for  $x \neq y$ , since the length field uniquely determines the boundary.

The Kraft inequality  $\sum_{s \in S} 2^{-|\text{sd}(s)|} \leq 1$  [Kraft, 1949] is satisfied.

*Forcing:* Any non-self-delimiting encoding creates an A0-violating ambiguity. Self-delimitation is the minimal constraint that eliminates it. The specific encoding  $\text{sd}(s) = 1^{|s|}0s$  is the simplest such construction; other prefix-free encodings (e.g., Elias delta, Elias gamma) are computably isomorphic and yield the same quotient structure. The forced content is the *prefix-free property*,

not the particular map—the choice of canonical representative is a gauge freedom (formalized at Step 12).  $\square$

#### A.4 Step 3: Executability Primitives

*Proof.* A0 requires that *every* finite witness procedure be admissible. Define  $\Delta = \{\tau \in \mathcal{D}^* \mid \mathcal{U}(\tau, \cdot) \text{ is total}\}$ . A0 demands maximal closure: if  $\tau_1, \tau_2 \in \Delta$ , then any finite composition, branching, or iteration of  $\tau_1$  and  $\tau_2$  that remains total must also be in  $\Delta$ . The set  $\Delta$  must therefore be closed under all total computable operations on  $\mathcal{D}^*$ .

The maximal closure under totality-preserving composition on a countable carrier is exactly the class of total computable functions. To see this: the total computable functions are closed under composition, branching, and bounded iteration (all totality-preserving operations), and any strictly larger class closed under these operations would contain partial functions (since unbounded search over total functions can diverge), violating totality. Hence the maximally closed test set is exactly the set of total computable functions, and a single program  $\mathcal{U}$  that executes any member of this set on any input—a universal Turing machine—is forced (by the *s-m-n* and universal function theorems; see [Rogers 1967](#), Chapters 1–2).

The total decoder  $\text{dec}$  is forced by Step 2: self-delimiting syntax means that every valid stream can be parsed into (program, input) without external oracle.

The cost functional  $c$  is forced by A0’s finiteness requirement: each test consumes resources (time steps, bits read), and these must be finitely bounded and recorded. Setting  $c(\tau)$  to the number of steps  $\mathcal{U}$  takes to execute  $\tau$  is the canonical choice; any monotone reparametrization is gauge-equivalent (Step 12).

*Forcing:* The forcing chain is:

- (i) A0 demands every finite procedure be admissible, hence  $\Delta$  must be maximally closed under totality-preserving composition. Excluding any finite procedure creates an A0-violating distinction.
- (ii) Maximal closure under totality-preserving composition on a finite alphabet is exactly the class of total computable functions (Proposition of this paper; see proof above). The identification of “finite procedure” with “total computable function” is the Church–Turing thesis—a naming convention, not an empirical hypothesis.
- (iii) A universal evaluator  $\mathcal{U}$  executing all members of this class is forced (by the existence of universal Turing machines, itself a theorem).
- (iv) The *form* of  $\mathcal{U}$  (Turing machine,  $\lambda$ -calculus, register machine) is gauge: switching between isomorphic models does not alter the truth quotient (formalized at Step 12).
- (v) What remains is a *naming convention*: we call  $\mathcal{U}$  a “UTM” using Turing’s formalism as the canonical representative, exactly as binary is the canonical alphabet (Step 1).

The identification of “finite procedure” (informal, from A0) with “total computable function” (formal) is a minimal semantic reading of A0’s natural language, not an additional axiom. Any reader who reads “finite procedure” as “takes finite input, runs finitely many steps, produces finite output” arrives at the same class.  $\square$

*Remark A.2* (CT as Historical Corroboration). The mathematical content of Step 3 is a proposition of this paper, not a named theorem in the literature: maximal closure under totality-preserving composition on a finite alphabet is exactly the class of total computable functions. The proof is sketched above. The key subtlety is the identification of the informal notion “finite procedure” (from A0’s natural-language phrasing) with the formal class “total computable function”—this identification is the Church–Turing thesis.

The Church–Turing thesis enters only as the identification of an informal concept (“finite procedure,” from A0’s natural-language phrasing) with a formal one (“total computable function”). This is analogous to how “number” is identified with “element of  $\mathbb{N}$ ”—a naming convention, not an empirical hypothesis.



The model choice (Turing machine vs.  $\lambda$ -calculus vs. register machine) is gauge (formalized at Step 12): the truth quotient is invariant under switching between isomorphic computation models. A reader who replaces “Turing-computable” with a weaker class (e.g., primitive recursive functions) obtains a weaker but structurally identical derivation.

Thus Step 3 is forced: A0’s maximal closure yields the total computable functions as a mathematical theorem. The Church–Turing thesis corroborates this conclusion; it does not enter as a premise.

*Remark A.3* (Total Semantics). The decoder  $\text{dec}$  and evaluator  $\mathcal{U}$  are total by construction: every input produces a well-defined output. In particular, the outcome alphabet  $A$  of any test  $\tau$  includes explicit FAIL and TIMEOUT outcomes. “Undefined behavior” is not a gap in the semantics but a specific outcome in  $A$ : if  $\mathcal{U}$  exceeds the allotted budget, it returns  $\text{TIMEOUT} \in A$ , not  $\perp$ . This ensures that no test can create a semantic gap—every execution path terminates in a recorded outcome, as required by A0.

## A.5 Step 4: Endogenous Tests and Canonical Enumeration

*Proof.* Define the base tests  $\Delta_0 = \{\tau \in D_0 \mid \mathcal{U}(\tau, \cdot) \text{ is total and } |\tau| \leq B_0\}$  for some initial budget  $B_0$ . The closure rules  $\text{Cl}_{\min}$  are:

- (C1) *Composition*: If  $\tau_1, \tau_2 \in \Delta$  and  $\tau_1 \circ \tau_2$  is total, then  $\tau_1 \circ \tau_2 \in \Delta$ .
- (C2) *Branching*: If  $\tau_{\text{cond}}, \tau_{\text{true}}, \tau_{\text{false}} \in \Delta$ , then the conditional  $\text{if } \tau_{\text{cond}} \text{ then } \tau_{\text{true}} \text{ else } \tau_{\text{false}} \in \Delta$ .
- (C3) *Bounded iteration*: If  $\tau \in \Delta$  and  $n \in \mathbb{N}$ , then  $\tau^n$  (apply  $\tau$  exactly  $n$  times)  $\in \Delta$ .

These rules are monotone on the powerset lattice of  $\mathcal{D}^*$ . By the Knaster–Tarski theorem [Tarski, 1955],  $\text{lfp}(\text{Cl}_{\min})$  exists and equals  $\Delta^* = \bigcup_{k=0}^{\infty} \text{Cl}_{\min}^k(\Delta_0)$ .

The canonical enumeration  $E_{\Delta}$  is:

$E_{\Delta}(n) =$  the  $n$ -th element of  $\Delta^*$  under the order  $|\text{sd}(\tau_1)| < |\text{sd}(\tau_2)|$  (ties broken lexicographically).

This enumeration is computable (enumerate all strings of increasing length, test totality) and requires no external scheduler—the ordering is determined by  $\text{sd}$  alone.

*Forcing*: Any subset of  $\Delta^*$  that is not closed under  $\text{Cl}_{\min}$  contains a pair  $(\tau_1, \tau_2)$  whose composition is a valid test but is excluded—creating an A0-violating distinction. The least fixed point is the minimal set avoiding this. The enumeration order is forced *within the paper’s model class*: given  $\text{sd}$  as the canonical encoding (Step 2), length-then-lexicographic order is the simplest computable enumeration requiring no external scheduling channel. Other computable enumerations (e.g., dovetailing by program length) yield the same test set  $\Delta^*$  and are isomorphic under reindexing; the forced content is the existence of a canonical, endogenous, computable enumeration, not the specific index assignment.  $\square$

*Remark A.4* (Minimality of the Closure Rules). Each closure rule is individually necessary:

- *C1 (Composition)*: If  $\tau_1, \tau_2 \in \Delta$ , their sequential execution is a finite procedure, hence admissible by A0. Omitting C1 would exclude an admissible test.
- *C2 (Branching)*: Conditional execution (if-then-else on test outcomes) is a finite procedure. Omitting C2 would exclude an admissible test.
- *C3 (Bounded iteration)*: Executing  $\tau$  exactly  $n$  times is a finite procedure for any fixed  $n$ . Omitting C3 would exclude an admissible test.

Together, C1–C3 generate all primitive recursive functions. The extension to all total computable functions requires additionally the dovetailing enumeration of Step 3: the canonical enumeration  $E_{\Delta}$  (which interleaves programs of increasing length) provides the mu-recursion-like mechanism that, combined with A0’s totality filter (excluding non-halting programs), reaches the full class of total computable functions. No further closure rule preserves totality without additional assumptions: unbounded iteration (while-loops) does not guarantee termination, so admitting it would require a halting oracle, violating the constructive character of A0.

## A.6 Step 5: Ledger as Multiset and the Diamond Law

*Proof.* A witness  $w$  for a distinction  $P$  is only useful if its output is available for future tests. If  $w$ 's output could be erased or overwritten, a subsequent test  $\tau$  that depends on  $w$ 's result would lack its input, and the distinction  $P$  would become unwitnessable—violating A0.

Define the *ledger* as the multiset of test-result pairs:

$$\mathcal{L} = \{ \{ (\tau_1, a_1), (\tau_2, a_2), \dots \} \}, \quad \text{where } a_i = \mathcal{U}(\tau_i, x_i).$$

The ledger must be append-only: entries may be added but never deleted or modified. Deletion would destroy a witness; modification would change a witness's result, both violating A0.

**Diamond Law (no minted order):** The ledger is a multiset because the *order* of recording is not intrinsic. Given two causally independent entries  $(\tau_i, a_i)$  and  $(\tau_j, a_j)$  (neither's test program references the other's result), swapping their order produces the same truth quotient. The ordering is gauge (Step 12 will formalize this). Hence:

$$\mathcal{N} \circ \mathcal{Q} = \mathcal{Q} \circ \mathcal{N},$$

where  $\mathcal{N}$  is any permutation of causally independent ledger entries and  $\mathcal{Q}$  is any truth-quotient extraction.

*Forcing:* Any mutable record allows A0 violations via witness destruction. Append-only is the minimal persistence guarantee. Multiset structure (order-erasure for causally independent entries) is forced *within the paper's model class of pure, non-disturbing tests*: when test outcomes do not depend on execution order, the temporal ordering of independent events is untestable, hence is gauge. This is a modeling restriction, not a consequence of A0\*—non-commuting tests (as in quantum mechanics) are consistent with witnessability but require an ordered or partially ordered ledger. The quantum extension is discussed in Appendix D.3.  $\square$

## A.7 Step 6: Truth Quotient

*Proof.* Define the *ledger equivalence* relation  $\equiv_{\mathcal{L}}$  on  $\mathcal{D}^*$ :

$$x \equiv_{\mathcal{L}} y \iff \forall \tau \in \Delta : (\tau, \mathcal{U}(\tau, x)) \in \mathcal{L} \implies \mathcal{U}(\tau, x) = \mathcal{U}(\tau, y).$$

That is,  $x$  and  $y$  are equivalent if no recorded test distinguishes them.

*Reflexivity:*  $\mathcal{U}(\tau, x) = \mathcal{U}(\tau, x)$  trivially.

*Symmetry:* If no test distinguishes  $x$  from  $y$ , none distinguishes  $y$  from  $x$ .

*Transitivity:* If  $x \equiv_{\mathcal{L}} y$  and  $y \equiv_{\mathcal{L}} z$ , then for every recorded test  $\tau$ ,  $\mathcal{U}(\tau, x) = \mathcal{U}(\tau, y) = \mathcal{U}(\tau, z)$ , hence  $x \equiv_{\mathcal{L}} z$ .

The *truth quotient* is:

$$\Pi^*(\mathcal{L}) = D_0 / \equiv_{\mathcal{L}}.$$

Each equivalence class  $[x]_{\equiv_{\mathcal{L}}}$  is a “truth”—the coarsest object that survives all recorded tests. Note we quotient  $D_0$  (the finite slice), not the full  $\mathcal{D}^*$ , since only finitely describable objects are operationally reachable.

*Forcing:* Maintaining distinctions between  $x$  and  $y$  when no test can separate them would violate A0 (the distinction has no witness). Quotienting is forced.  $\square$

## A.8 Step 7: Indistinguishability as Primary Algebra

*Proof.* We establish each structure in turn.

(a) **Observable Opens**  $\mathcal{O}_{\Delta}^{\text{fin}}$ . Define the *observable open sets* as follows. For each test  $\tau \in \Delta^*$  and outcome  $a$ , define:

$$O_{\tau, a} = \{ [x] \in \Pi^*(\mathcal{L}) \mid \mathcal{U}(\tau, x) = a \}.$$

Each  $O_{\tau,a}$  is the set of truth classes that produce outcome  $a$  when tested by  $\tau$ . The collection

$$\mathcal{O}_{\Delta}^{\text{fin}} = \{ O_{\tau,a} \mid \tau \in \Delta^*, a \in \text{range}(\tau) \}$$

forms a topology on  $\Pi^*(\mathcal{L})$ : it is closed under finite intersection (run both tests) and arbitrary union (a distinction is observable if *any* test detects it). This is a finite restriction of the Scott topology [Scott, 1970] to testable predicates.

**(b) Eraser Algebra  $\mathcal{E}$ .** Define the *eraser* associated with a test  $\tau$  as the map  $e_{\tau}: \Pi^*(\mathcal{L}) \rightarrow \Pi^*(\mathcal{L})$  that merges all truth classes that  $\tau$  cannot distinguish:

$$e_{\tau}([x]) = [\text{the unique class in } \Pi^*(\mathcal{L})/\equiv_{\tau} \text{ containing } [x]].$$

The collection  $\mathcal{E} = \{e_{\tau} \mid \tau \in \Delta^*\}$  forms an *eraser algebra*: a set of idempotent, monotone (with respect to the refinement order) surjections satisfying:

(E1) *Idempotence*:  $e_{\tau} \circ e_{\tau} = e_{\tau}$ .

(E2) *Monotonicity*: If  $\mathcal{L}' \supseteq \mathcal{L}$ , then  $e_{\tau}$  on  $\Pi^*(\mathcal{L}')$  is finer than on  $\Pi^*(\mathcal{L})$ .

(E3) *Commutativity*:  $e_{\tau_1} \circ e_{\tau_2} = e_{\tau_2} \circ e_{\tau_1}$  (the order of erasure is irrelevant, extending the Diamond Law of Step 5).

**(c) Self-Generating Fixed Point.** Define  $\Phi: 2^{\Pi^*(\mathcal{L})} \rightarrow 2^{\Pi^*(\mathcal{L})}$  as the map that takes a set of truth classes  $S$  and returns the set of classes distinguishable by tests whose programs lie in  $\bigcup_{p \in S} W(p)$ :

$$\Phi(S) = \{ [x] \in \Pi^*(\mathcal{L}) \mid \exists \tau \in \bigcup_{p \in S} W(p), \mathcal{U}(\tau, x) \neq \mathcal{U}(\tau, y) \text{ for some } y \text{ with } [y] \neq [x] \}.$$

The *self-generating fixed point* is:

$$\text{sim}^* = \text{lfp}(\Phi).$$

This fixed point captures the self-referential nature of the system: the truths that survive are exactly those distinguishable by tests drawn from the truths themselves.

*Monotonicity of  $\Phi$ .* Let  $S \subseteq S' \subseteq \Pi^*(\mathcal{L})$ . Then  $\bigcup_{p \in S} W(p) \subseteq \bigcup_{p \in S'} W(p)$ , so more programs are available as tests. More tests can only distinguish *more* pairs of truth classes (or the same pairs), hence  $\Phi(S) \subseteq \Phi(S')$ . Therefore  $\Phi$  is monotone on the finite lattice  $2^{\Pi^*(\mathcal{L})}$  (which is finite because  $\Pi^*(\mathcal{L})$  is a quotient of the finite set  $D_0$ , Step 1).

Existence of the least fixed point is now guaranteed by the Knaster–Tarski theorem [Tarski, 1955].

*Forcing*: The observable-open topology is forced because A0 equates “meaningful distinction” with “testable distinction,” and the collection of testable predicates inherits the topological structure of finite intersections and arbitrary unions. The eraser algebra is forced because erasure (merging indistinguishable classes) is the canonical monotone, idempotent operation on truth quotients. The self-generating fixed point is forced because the test set  $\Delta^*$  is drawn from  $\mathcal{D}^* \supseteq D_0$ , and truth classes label programs that are themselves tests—the system must be consistent with its own outputs acting as inputs.  $\square$

## A.9 Step 8: Observer Collapse and Entropic Time

*Proof.* Step 3 forces all tests to be total:  $\mathcal{U}(\tau, x)$  halts for every  $x$ . Therefore every test partitions  $D_0$  into well-defined outcome classes, and appending a record can only *refine* (never coarsen) this partition. Partial tests (where  $\mathcal{U}(\tau, x)$  may not halt) are excluded by Step 3’s totality requirement, which is forced by A0’s demand that every witness procedure terminate.

Each new entry  $(\tau, a)$  appended to  $\mathcal{L}$  can only refine the truth quotient: the equivalence class  $[x]$  either splits (if  $\tau$  distinguishes elements that were previously identified) or remains unchanged. It never coarsens, because the ledger is append-only (Step 5).

Let  $|W()|_{\text{pre}}$  denote the fiber size before appending  $(\tau, a)$  and  $|W()|_{\text{post}}$  the fiber size after. We have  $|W()|_{\text{post}} \subseteq |W()|_{\text{pre}}$ , hence  $|W()|_{\text{post}} \leq |W()|_{\text{pre}}$ , and:

$$\Delta\mathcal{T} = \log \frac{|W()|_{\text{pre}}}{|W()|_{\text{post}}} \geq 0.$$

Equality holds iff  $\tau$  provides no new information (i.e., the test result was already determined by prior ledger entries).

This is a *Second Law*: the descriptive entropy of the fiber monotonically decreases, and the elapsed “time”  $\Delta\mathcal{T}$  records the cumulative information gained. Time is not a parameter—it is the irreversibility of witnessing.

*Forcing*: Reversing a ledger entry would destroy a witness (Step 5). Time *is* the irreversibility of record formation. A0 forces monotone refinement of the truth quotient under accumulated evidence; the specific time increment  $\Delta\mathcal{T} = \log(|\text{fiber}_{\text{pre}}|/|\text{fiber}_{\text{post}}|)$  is the canonical information-theoretic measure of refinement, forced within the paper’s model class up to monotone reparametrization. Any strictly monotone function of fiber shrinkage encodes the same irreversibility; the logarithmic form is canonical because it is additive under independent refinements.  $\square$

## A.10 Step 9: Entropy and Budget

*Proof.* Define the *entropy* of the truth quotient at time  $\mathcal{T}$  as [Shannon, 1948]:

$$S(\mathcal{L}) = \log |\Pi^*(\mathcal{L})| = \log \frac{|D_0|}{|\text{avg fiber}|}.$$

More precisely, for a given truth class  $p$ :

$$S_p(\mathcal{L}) = \log |W(p)|.$$

$S_p$  decreases with each informative test (Step 8).

The *budget*  $B$  bounds the total cost of all executed tests:

$$\sum_{i=1}^n c(\tau_i) \leq B.$$

This is forced by A0’s finiteness requirement: every witness must terminate in finite time, hence can consume at most finite resources.

The feasible test set monotonically shrinks:

$$\Delta(\mathcal{T}) = \{ \tau \in \Delta^* \mid c(\tau) \leq B - B_{\text{spent}}(\mathcal{T}) \},$$

and  $\Delta(\mathcal{T})$  is monotonically non-increasing in  $\mathcal{T}$ .

*Forcing*: An expanding test set would require an expanding budget, which would require infinite resources at some finite time—violating A0’s finiteness requirement. The specific entropy functional  $S(\mathcal{L}) = \log |\Pi^*(\mathcal{L})|$  is the canonical choice within the paper’s finite-cardinality model; alternative resource measures (space, communication complexity) and entropy-like functionals (Rényi entropy, min-entropy) yield structurally analogous frameworks. The forced content is the monotone shrinkage of the feasible test set under budget consumption, not the specific entropy formula.  $\square$

### A.11 Step 10: Energy Ledger

*Proof.* Each test  $\tau$  consumes resources: at minimum, the execution steps required by  $\mathcal{U}$  to evaluate  $\tau$  (Step 3 established that costs are positive integers  $c(\tau) \geq 1$ ). A0’s finiteness requirement demands that these costs be finitely recorded. Landauer’s principle [Landauer, 1961]—that erasing one bit of information requires dissipating at least  $k_B T \ln 2$  of energy—provides a physical interpretation of this cost, but the forcing argument does not depend on Landauer’s principle. What A0 forces is the existence of a cost ledger tracking resource expenditure per test; the identification of abstract cost units with thermodynamic energy is an interpretive bridge to physics, not a derivational dependency. Bennett [Bennett, 1973] showed that computation can in principle be logically reversible, but A0’s append-only ledger (Step 5) makes the framework’s record formation irreversible by design. The energy bound requires only the monotonicity of fiber shrinkage (Step 8), which is already established. Each test either refines the quotient (gaining information, consuming budget) or leaves it unchanged (zero information gain, but still consuming at least the encoding cost of  $\tau$ ).

Define the *energy ledger*:

$$\mathcal{L}_E = [(c(\tau_1), \Delta\mathcal{T}_1), (c(\tau_2), \Delta\mathcal{T}_2), \dots].$$

The efficiency  $\varepsilon_i = \Delta\mathcal{T}_i / c(\tau_i)$  measures the information gained per unit cost. When  $\Delta\mathcal{T}_i = 0$  (the test reveals nothing new),  $\varepsilon_i = 0$ —pure waste.

The total energy expenditure satisfies:

$$\sum_{i=1}^n c(\tau_i) \leq B.$$

*Forcing:* The energy ledger is forced because A0 demands that the cost of each distinction be finitely recorded. The efficiency ratio is the canonical dimensionless measure of information-per-cost, forced by dimensional analysis on the two quantities  $c(\tau)$  and  $\Delta\mathcal{T}$  that the framework already produces.  $\square$

### A.12 Step 11: Operational Nothingness

*Proof.* As  $B_{\text{spent}}(\mathcal{T}) \rightarrow B$ , the remaining budget  $B_{\text{rem}} = B - B_{\text{spent}}(\mathcal{T})$  shrinks. Since all tests have positive integer cost  $c(\tau) \geq 1$  (Step 3), the feasible test set at time  $\mathcal{T}$  is:

$$\Delta(\mathcal{T}) = \{ \tau \in \Delta^* \mid c(\tau) \leq B_{\text{rem}}(\mathcal{T}) \}.$$

When  $B_{\text{rem}} < \min_{\tau \in \Delta^*} c(\tau)$ —that is, when the remaining budget is insufficient to execute any test in  $\Delta^*$ —the feasible test set becomes empty:  $\Delta(\mathcal{T}_{\text{max}}) = \emptyset$ . No further refinement of the truth quotient is possible, because no affordable separator test remains within budget. At this point:

$$\Pi^*(\mathcal{L}(\mathcal{T}_{\text{max}})) = \Pi^*(\mathcal{L}(\mathcal{T}_{\text{max}} - 1)).$$

We call this state *operational nothingness*:

$$\perp_{\text{op}} : \text{ no affordable test remains—the budget horizon has been reached.}$$

Unlike  $\perp$  (absolute nothingness, Step 0),  $\perp_{\text{op}}$  occurs *within* a universe that has structure; it is the horizon of that structure’s resolvability given the available resources.

*Forcing:* Finite budget  $\Rightarrow$  finite test sequence  $\Rightarrow$  terminal state where no further test is affordable.  $\perp_{\text{op}}$  is the unique terminal state under A0: it is reached when no separator test can be executed within the remaining budget, not when “only constant tests exist” (a framing that would conflict with the positive-cost requirement  $c(\tau) \geq 1$ ).  $\square$

### A.13 Step 12: Gauge Group

*Proof.* A *relabeling* is a bijection  $g: D_0 \rightarrow D_0$  that permutes descriptions within each fiber:  $g(W(p)) = W(p)$  for all  $p \in \Pi^*(\mathcal{L})$ . Such a  $g$  is *untestable*: for every  $\tau \in \Delta^*$ ,  $\mathcal{U}(\tau, x) = \mathcal{U}(\tau, g(x))$ , since  $x$  and  $g(x)$  belong to the same equivalence class.

Let  $\mathcal{G}_{\mathcal{T}}$  denote the set of all such relabelings. Since each  $g \in \mathcal{G}_{\mathcal{T}}$  is a bijection on *all* of  $D_0$  (preserving every fiber), composition is always defined and  $\mathcal{G}_{\mathcal{T}}$  forms a *group*:

- *Identity*: The identity map  $\text{id} \in \mathcal{G}_{\mathcal{T}}$ .
- *Inverse*: If  $g \in \mathcal{G}_{\mathcal{T}}$  is a bijection,  $g^{-1} \in \mathcal{G}_{\mathcal{T}}$ .
- *Composition*: If  $g_1, g_2 \in \mathcal{G}_{\mathcal{T}}$ , then  $g_1 \circ g_2$  permutes within each fiber, hence  $g_1 \circ g_2 \in \mathcal{G}_{\mathcal{T}}$ . Composition is always defined since both  $g_1$  and  $g_2$  are bijections on all of  $D_0$ .
- *Associativity*:  $(g_1 \circ g_2) \circ g_3 = g_1 \circ (g_2 \circ g_3)$ .

Algebraically,  $\mathcal{G}_{\mathcal{T}} \cong \prod_{p \in \Pi^*(\mathcal{L})} \text{Sym}(W(p))$ —the direct product of symmetric groups on the fibers.

*Forcing*: Each group axiom is individually forced:

- *Identity*: Omitting  $\text{id}$  from  $\mathcal{G}_{\mathcal{T}}$  would mean “doing nothing” is detectable—but no test can detect an absent transformation, so  $\text{id} \in \mathcal{G}_{\mathcal{T}}$ .
- *Inverse*: If  $g \in \mathcal{G}_{\mathcal{T}}$  but  $g^{-1} \notin \mathcal{G}_{\mathcal{T}}$ , then  $g^{-1}$  is detectable. But  $g^{-1} \circ g = \text{id}$ , which is undetectable, so applying  $g$  after  $g^{-1}$  “erases” the detectable transformation—a contradiction, since detection is monotone under composition (Step 4, C1). Hence  $g^{-1} \in \mathcal{G}_{\mathcal{T}}$ .
- *Composition*: If  $g_1, g_2 \in \mathcal{G}_{\mathcal{T}}$  but  $g_1 \circ g_2 \notin \mathcal{G}_{\mathcal{T}}$ , then  $g_1 \circ g_2$  is a finite procedure that distinguishes what  $g_1$  and  $g_2$  individually cannot—violating closure under composition (Step 4, C1 applied to gauge tests).

Closure under  $\{\text{id}, \text{inv}, \circ\}$  with globally defined composition gives a group (specifically, a direct product of symmetric groups on fibers).  $\square$

### A.14 Step 13: Epistemic vs. Decision Regimes

*Proof.* Consider the system at time  $\mathcal{T}$ . It must either:

- (a) Execute another test  $\tau$  to reduce  $|W()|$  (epistemic: the surviving set shrinks but no commitment is made), or
- (b) Declare a terminal state (decision: the system commits to  $\text{UNIQUE}$ ,  $\Omega$ , or  $\text{UNSAT}$ ).

These two modes are distinguished by their effect on the truth quotient:

- In  $\mathcal{R}_{\text{ep}}$ : the quotient may refine ( $\Delta\mathcal{T} > 0$ ) or stay unchanged ( $\Delta\mathcal{T} = 0$ ), but the surviving answer set is never reduced by fiat—only by evidence.
- In  $\mathcal{R}_{\text{dec}}$ : the system selects from the surviving set. This is irreversible (once declared, the output is final).

The transition  $\mathcal{R}_{\text{ep}} \rightarrow \mathcal{R}_{\text{dec}}$  occurs when either (a)  $|\mathcal{A}_S| \leq 1$  (forced by evidence), or (b)  $\Delta(\mathcal{T}) = \{\tau_0\}$  (budget exhausted). Both conditions are detected by finite inspection.

In the decision regime  $\mathcal{R}_{\text{dec}}$ , candidates  $x \in \mathcal{D} \subseteq D_0$  are ranked by a *pinned preorder*  $R$  on the outcome profile  $W(x) = (\tau_1(x), \dots, \tau_m(x))$ . The maximizers  $M = \arg \max_{x \in \mathcal{D}} R(W(x))$  are the admissible outputs. If  $|M| = 1$ , output  $\text{UNIQUE}$ ; if  $|M| > 1$ , output  $\Omega$  with frontier. The preorder  $R$  must be  $\Pi$ -consistent (Step 17):  $R$  depends only on the truth quotient of the outcome profile, not on raw descriptions.

*Forcing*: Conflating epistemic and decision modes would allow the system to assert answers without evidence (running a test that also commits to an answer) or to gather evidence without recording it (making a commitment that also runs a test). Both violate A0: the former produces unwitnessed assertions, the latter destroys witnesses.  $\square$

### A.15 Step 14: Exactness Gate and Split Law

*Proof.* (a) **Exactness Gate.** Define the exactness gate  $\mathcal{G}_{\text{ex}}$  as the procedure:



1. *Certificate check*: Scan  $\mathcal{L}$  for a witness that resolves  $q$  to a single answer. If found, output UNIQUE or UNSAT immediately.
2. *Separator search*: Only if no certificate exists, search  $\Delta(\mathcal{T})$  for the optimal separator test (Step 18 below).

The certificate-first ordering is forced because:

- Running a separator when a certificate already exists wastes budget (violates budget-optimality, which is a consequence of A0's finiteness requirement).
- Declaring  $\Omega$  when a certificate exists in  $\mathcal{L}$  would withhold an admissible witness, violating A0.

**(b) Split Law (Image Factorization).** Let  $f: D_0 \rightarrow D_0$  be a total computable function. Consider the induced map on truth classes:  $\bar{f}: \Pi^*(\mathcal{L}) \rightarrow \Pi^*(\mathcal{L})$ , defined by  $\bar{f}([x]) = [f(x)]$ .

$\bar{f}$  may be non-injective (merging distinct classes) but is always well-defined by construction. Factor  $\bar{f}$  through its image:

$$\bar{f} = \iota \circ q,$$

where  $q: \Pi^*(\mathcal{L}) \twoheadrightarrow \text{Im}(\bar{f})$  is the surjection (merge) and  $\iota: \text{Im}(\bar{f}) \hookrightarrow \Pi^*(\mathcal{L})$  is the inclusion (relabeling).

This is the standard image factorization, but here it is *forced*:  $q$  records irreversible information loss (merging classes that were previously distinct), and  $\iota$  is a gauge relabeling that carries no information.

*Forcing*: The Split Law is a standard image-factorization theorem applied to the truth quotient; its validity is a mathematical fact independent of A0. The certificate-first ordering of the exactness gate is forced within the paper's model class by budget-dominance: any policy that searches for a separator when a certificate already exists in  $\mathcal{L}$  is strictly dominated (it wastes budget on a test whose outcome is already determined). A0-compliance alone does not force certificate-first ordering—a wasteful but A0-compliant policy could check certificates last—but budget-optimality (itself motivated by A0's finiteness requirement) does.  $\square$

## A.16 Step 15: Myhill–Nerode Congruence

*Proof.* Define the state of the system at any point as  $s = (\Pi^*(\mathcal{L}), \Delta(\mathcal{T}), B - B_{\text{spent}})$ —the current truth quotient, feasible test set, and remaining budget.

Two states  $s, s'$  are *futures-equivalent* if no continuation of the test sequence can distinguish between them:

$$s \equiv_q s' \iff \forall \sigma = (\tau_1, \tau_2, \dots, \tau_k) \in \Delta^{*\omega} : \text{terminal}(s, \sigma) = \text{terminal}(s', \sigma).$$

*Right-congruence*: If  $s \equiv_q s'$  and  $\tau \in \Delta^*$ , then for any future  $\sigma$ :

$$\text{terminal}(s \cdot \tau, \sigma) = \text{terminal}(s, \tau \cdot \sigma) = \text{terminal}(s', \tau \cdot \sigma) = \text{terminal}(s' \cdot \tau, \sigma),$$

hence  $s \cdot \tau \equiv_q s' \cdot \tau$ .

The quotient  $\Delta^{*\omega} / \equiv_q$  is finite (since  $D_0$  is finite), giving a *minimal automaton* for the system's behavior. By the classical Myhill–Nerode theorem [Myhill, 1957, Nerode, 1958], this minimal automaton is unique.

*Monoid structure.* Test sequences  $\Delta^{*\omega}$  form a monoid under concatenation: the identity element is the empty sequence  $\varepsilon$ , and associativity of concatenation is immediate. Composition of tests is always defined because each test is total (Step 3). Cost is additive:  $c(\sigma_1 \cdot \sigma_2) = c(\sigma_1) + c(\sigma_2)$ , since each test in the sequence is executed independently and costs are summed (Step 9). The classical Myhill–Nerode theorem [Hopcroft and Ullman, 1979] applies to any monoid acting on a finite set; here the monoid  $(\Delta^{*\omega}, \cdot, \varepsilon)$  acts on the finite state space  $D_0$ , so the quotient is guaranteed to be finite and unique.

*Forcing*: Maintaining a distinction between states  $s$  and  $s'$  when all futures agree would violate A0 (the distinction has no future witness). The Myhill–Nerode quotient is the coarsest

partition consistent with A0, hence is forced within the paper’s model class. The specific indistinguishability criterion—“no continuation of the test sequence can distinguish”—uses all possible futures as the separator class; coarser criteria (e.g., bounded-horizon futures) or finer criteria (e.g., strategy-equivalence under a fixed policy class) yield alternative congruences that are also A0-compatible. The choice of all-futures equivalence is the canonical (coarsest) option.  $\square$

### A.17 Step 16: Separator Geometry

*Proof.* (a) **Separator Distance**  $d_q$ . For two truth classes  $p, p' \in \Pi^*(\mathcal{L})$  with  $p \neq p'$ , define:

$$d_q(p, p') = \min_{\tau \in \Delta^*} \{ c(\tau) \mid \mathcal{U}(\tau, x) \neq \mathcal{U}(\tau, y) \text{ for some } x \in W(p), y \in W(p') \}.$$

That is,  $d_q(p, p')$  is the cost of the cheapest test that distinguishes  $p$  from  $p'$ . Set  $d_q(p, p) = 0$ .

*Metric axioms:*

- *Non-negativity:*  $d_q(p, p') \geq 0$  since costs are non-negative.
- *Identity:*  $d_q(p, p') = 0 \iff p = p'$  (if  $p \neq p'$ , some test must separate them by definition of the truth quotient; its cost is positive).
- *Symmetry:*  $d_q(p, p') = d_q(p', p)$  since a test that separates  $p$  from  $p'$  also separates  $p'$  from  $p$ .
- *Triangle inequality:* Let  $p, p', p'' \in \Pi^*(\mathcal{L})$  be pairwise distinct. The triangle inequality  $d_q(p, p'') \leq d_q(p, p') + d_q(p', p'')$  is *not* a consequence of A0\*—it requires a structural property of the cost model.

*Why the naïve argument fails.* Let  $\tau_1$  separate  $p$  from  $p'$  and  $\tau_2$  separate  $p'$  from  $p''$ . The composite  $\tau_1 \wedge \tau_2$  (run both, compare outcomes) is in  $\Delta^*$  by C1. However,  $\tau_1 \wedge \tau_2$  need not separate  $p$  from  $p''$ : the tests distinguish each from  $p'$ , but  $p$  and  $p''$  may agree on both  $\tau_1$  and  $\tau_2$ . (Concretely: if  $\tau_1$  maps  $A \mapsto 0, B \mapsto 1, C \mapsto 0$  and  $\tau_2$  maps  $A \mapsto 1, B \mapsto 0, C \mapsto 1$ , then  $A$  and  $C$  agree on both tests despite being distinct truth classes.)

*Sufficient condition.* The triangle inequality holds when the test algebra is *transitively separating*: for any distinct  $p, p''$ , the cheapest separator has cost bounded by the sum of separator costs through any intermediary. Formally:

$$\forall p, p', p'' \in \Pi^*(\mathcal{L}), \quad d_q(p, p'') \leq d_q(p, p') + d_q(p', p''). \quad (\text{TS})$$

This property is satisfied by natural cost models (e.g., program length, where a separator for  $p, p''$  can be constructed from separators for  $p, p'$  and  $p', p''$  by branching—C2—at cost at most  $c_1 + c_2 + O(1)$ , and the  $O(1)$  overhead is absorbed by the integer cost model). However, (TS) is a modeling commitment, not a consequence of A0: exotic cost models where separator costs are uncorrelated with compositional structure can violate it.

*Under (TS),*  $d_q$  is a metric on  $\Pi^*(\mathcal{L})$ . The paper adopts (TS) as a structural property of the cost model, consistent with all natural interpretations of computational cost.

(b) **Curvature**  $\kappa$ . The curvature at a truth class  $p$  measures how rapidly the separator cost changes in the neighborhood of  $p$ :

$$\kappa(p) = \max_{p', p'' \in \mathcal{N}(p)} \frac{d_q(p', p'') - d_q(p, p') - d_q(p, p'')}{d_q(p, p') \cdot d_q(p, p'')},$$

where  $\mathcal{N}(p)$  is the set of truth classes adjacent to  $p$  (those with  $d_q(p, p') \leq \theta$  for a threshold  $\theta$ ).

High curvature means distinguishing nearby classes is disproportionately expensive—the “information landscape” is curved.

(c) **Hardness**  $\mathcal{H}$ . The hardness of a query  $q$  is the total separator cost needed to resolve it:

$$\mathcal{H}(q) = \sum_{i=1}^n d_q(p_i, p_{i+1}),$$

where  $p_1, p_2, \dots, p_n$  is the sequence of truth classes traversed by the optimal test sequence (whose existence is established at Step 18; the hardness metric is well-defined independent of optimality, since any test sequence traversing the same classes yields the same sum).

*Forcing:* The separator distance is forced because A0 defines distinguishability in terms of tests, and tests have costs. Non-negativity, identity of indiscernibles, and symmetry follow directly from the algebraic properties of cost and separation. The triangle inequality requires the additional structural property (TS) that the cost model is transitively separating—a property satisfied by natural computational cost models but not derivable from A0\*. Curvature and hardness are derived quantities that measure the geometry of the information landscape.  $\square$

## A.18 Step 17: $\Pi$ -Consistent Control

*Proof.* The truth quotient  $\Pi \equiv \Pi^*(\cdot)$  maps each description  $x$  to its equivalence class  $[x]_{\equiv_{\mathcal{L}}}$ . A gauge transformation  $\mathcal{N} \in \mathcal{G}_{\mathcal{T}}$  permutes within fibers:  $\mathcal{N}(x) \in W([x])$ , hence  $[\mathcal{N}(x)] = [x]$ , hence  $\Pi(\mathcal{N}(x)) = \Pi(x)$ .

Therefore  $\Pi \circ \mathcal{N} = \Pi$ . Composing further:  $\Pi \circ \mathcal{N} \circ \Pi = \Pi \circ \Pi = \Pi$  (since  $\Pi$  is idempotent:  $\Pi([x]) = [x]$ ).

This law states that truth extraction is gauge-invariant and that all control decisions based on  $\Pi$  are insensitive to encoding noise.

*Forcing:* If  $\Pi$  were sensitive to gauge, then  $\Pi$  would constitute a test that distinguishes gauge-equivalent descriptions, contradicting the definition of the truth quotient.  $\Pi$ -consistency is the *operational expression* of gauge invariance.  $\square$

*Remark A.5* ( $\Pi$ -Consistency as Architectural Bridge).  $\Pi$ -consistency means that all control decisions depend only on the truth quotient, never on raw descriptions. This property is the bridge between gauge invariance (Step 12) and the Bellman solver (Step 18): the solver’s inputs are guaranteed gauge-free, ensuring that test selection is invariant under relabeling. Algebraically, the idempotency law  $\Pi \circ \mathcal{N} = \Pi$  has the same form as a retraction in category theory, connecting to the eraser algebra (Step 7): each eraser  $e_{\tau}$  is itself a retraction, and  $\Pi$  is the universal such retraction that factors through all of them.

## A.19 Step 18: Deterministic Solver (Bellman Minimax)

*Proof.* At time  $\mathcal{T}$ , the reasoner selects a test  $\tau \in \Delta(\mathcal{T})$  to maximize information gain (fiber reduction) against worst-case outcomes. The exactness gate (Step 14) has already checked for certificates, so we are in the separator-search branch.

Let  $|\Pi^*(\mathcal{L})|_{\mathcal{A}}$  denote the number of distinct answers in  $\mathcal{A}$  that survive under the current truth quotient—i.e., the size of the surviving answer set  $\mathcal{A}_S$ .

Define the *value function*:

$$V(\mathcal{L}, B) = \begin{cases} 0 & \text{if } |\Pi^*(\mathcal{L})|_{\mathcal{A}} \leq 1 \text{ (UNIQUE or UNSAT),} \\ \min_{\tau \in \Delta(\mathcal{T})} \left[ c(\tau) + \max_{a \in \text{range}(\tau)} V(\mathcal{L} \cup \{(\tau, a)\}, B - c(\tau)) \right] & \text{otherwise.} \end{cases}$$

The optimal test is:

$$\tau^* = \operatorname{argmin}_{\tau \in \Delta(\mathcal{T})} \left[ c(\tau) + \max_a V(\mathcal{L} \cup \{(\tau, a)\}, B - c(\tau)) \right].$$

The argmin is well-defined because  $\Delta(\mathcal{T})$  is finite (Step 9) and costs are non-negative integers, hence the minimum over a finite set of non-negative integers exists. Ties are broken by  $E_{\Delta}$  order (Step 4).

This is a Bellman equation [Bellman, 1957] with a minimax (worst-case) adversary over test outcomes, following the tradition of Wald’s sequential decision theory [Wald, 1950].

*Termination.* The recursion is well-founded. All tests have cost  $c(\tau) \geq 1$  (positive integer costs, Step 3: at least one execution step is required). Each recursive call decreases the budget:  $B' = B - c(\tau) < B$ . Since  $B$  is a non-negative integer, the recursion depth is at most  $B$ . The base cases are: (i)  $|\Pi^*(\mathcal{L})|_{\mathcal{A}} \leq 1$  (terminal: the surviving answer set is a singleton or empty), or (ii)  $B = 0$  (budget exhausted: output  $\Omega$  with the minimal separator as frontier witness). Formally, induction on the pair  $(|\Pi^*(\mathcal{L})|_{\mathcal{A}}, B)$  with lexicographic order shows that every recursive call strictly decreases this pair, hence the recursion terminates.

*Forcing:* Any policy that does not minimize worst-case cost can be dominated by one that does, *in the absence of a prior distribution over test outcomes*. A dominated policy wastes budget on tests that leave larger fibers than necessary, potentially causing  $\Omega$  where UNIQUE was achievable. The Bellman minimax is the unique non-dominated *prior-free* policy. When a prior over outcomes is available, a Bayesian (expected-cost) Bellman recursion is equally A0-compatible and may outperform minimax on average; the choice between prior-free and Bayesian formulations is forced by A0\* (prior-free minimax is the unique invariant policy before any prior is witnessed; see A0 alone). The minimax formulation is adopted here as the conservative default that requires no distributional assumptions beyond A0.

### Part (b): The Coupling Law.

The Bellman recursion of Part (a) selects the next test by minimizing worst-case value. The coupling law refines this by making the objective *question-relative* and eliminates the last hidden degree of freedom.

*Answer quotient.* For query  $q$  with surviving answer set  $W$  and test outcome  $a$ , define the *restricted answer set*  $W_a = \{w \in W : \mathcal{V}(w, \tau) = a\}$ . The *answer quotient*  $\text{Ans}(q, W_a) = W_a / \equiv_{\mathcal{L}}$  is the set of truth classes reachable from  $W$  after observing outcome  $a$ .

*Refinement gain.* The information gained about query  $q$  by executing test  $\tau$  with worst-case outcome is:

$$\Delta \mathcal{T}_q(\tau; W) = \log |\text{Ans}(q, W)| - \max_{a \in \text{out}(\tau)} \log |\text{Ans}(q, W_a)|.$$

This measures how many bits of answer-uncertainty are removed in the worst case. By construction,  $\Delta \mathcal{T}_q(\tau; W) \geq 0$  (a test cannot increase the number of surviving answer classes).

*Question-relative efficiency.* The efficiency of test  $\tau$  with respect to query  $q$  and knowledge state  $W$  is the dimensionless ratio:

$$\varepsilon(\tau; W) = \frac{\Delta \mathcal{T}_q(\tau; W)}{c(\tau)}.$$

This is the canonical dimensionless combination: refinement gain has units of bits (Step 8) and cost has units of computation (Step 3); their ratio is canonical.

*The Coupling Law.*

$$\tau^*(q, W) = \Pi_{\text{can-min}} \left( \underset{\tau \in \Delta_{\leq B}}{\text{argmax}} \varepsilon(\tau; W) \right)$$

The argmax selects the set of maximally efficient tests. The  $\Pi_{\text{can-min}}$  tie-break selects the lexicographically minimal element under  $E_{\Delta}$  order after  $\Pi$ -canonicalization (Step 12).

*Forcing.* The efficiency  $\varepsilon$  is the canonical dimensionless ratio that the framework can construct from the two quantities it produces at each step (information gained and cost paid). Any other objective would either (i) import an external scale (violating closure), or (ii) ignore one of the two quantities (suboptimal). The  $\Pi_{\text{can-min}}$  tie-break is forced by gauge invariance: any tie-breaking rule that is not gauge-invariant would distinguish between gauge-equivalent tests, violating Step 12.

*Remark A.6 (Efficiency Uniqueness Conditions).* Consider the four properties: (i) monotone increasing in gain, (ii) monotone decreasing in cost, (iii) scale-invariant ( $\varepsilon(\alpha g, \alpha c) = \varepsilon(g, c)$  for  $\alpha > 0$ ), and (iv)  $\varepsilon(g, g) = 1$ . Any function satisfying (i)–(iv) is of the form  $f(g/c)$  for some strictly monotone  $f$  with  $f(1) = 1$ . This class includes, e.g.,  $(g/c)^2$ ,  $\sqrt{g/c}$ , and  $2(g/c)/(1+g/c)$ . However, since  $\operatorname{argmax}_\tau f(g(\tau)/c(\tau)) = \operatorname{argmax}_\tau g(\tau)/c(\tau)$  for any strictly monotone  $f$ , the test-selection *policy*  $\tau^*(q, W)$  is uniquely determined regardless of the choice of  $f$ . The ratio  $\varepsilon = \Delta\mathcal{T}_q(\tau; W)/c(\tau)$  is the canonical representative of this equivalence class (the simplest choice,  $f = \text{id}$ ).

### Part (c): Deterministic evolution and Omega distance.

Since  $\varepsilon(\tau; W)$  depends only on  $(W, q)$ —both of which are determined by the initial query and the sequence of test outcomes—and  $\Pi_{\text{can-min}}$  is a deterministic function, the test-selection map  $\tau^*(q, W)$  is fully deterministic. The epistemic update rule

$$W_{t+1} = \{w \in W_t : \mathcal{V}(w, \tau^*(q, W_t)) = a_{\text{obs}}\}$$

combined with the deterministic test selector yields a fully deterministic evolution  $W_0 \rightarrow W_1 \rightarrow \dots$  for any initial pair  $(q, W_0)$ .

The *Omega distance*

$$d(q; W) = \log |\text{Ans}(q, W)|$$

measures the remaining answer-uncertainty in bits. At UNIQUE,  $d(q; W) = 0$ ; at  $\Omega$  (budget exhausted),  $d(q; W) > 0$  and quantifies how far the system is from resolution.

*Decision rule.* The system terminates when either  $|\text{Ans}(q, W)| = 1$  (UNIQUE,  $d(q; W) = 0$ ),  $|\text{Ans}(q, W)| = 0$  (UNSAT), or  $B_{\text{rem}} < \min_\tau c(\tau)$  ( $\Omega$  with frontier  $d(q; W) > 0$  and separator  $\tau^*(q, W)$ ).  $\square$

**Proposition A.7** (Greedy Efficiency Equals Bellman Optimal for Independent Tests). *When tests are pure (non-disturbing) and independent (the outcome of  $\tau_1$  does not affect the partition induced by  $\tau_2$ ), one-step greedy efficiency maximization  $\tau^*(q, W) = \Pi_{\text{can-min}}(\operatorname{argmax}_\tau \varepsilon(\tau; W))$  produces the same test sequence as the full Bellman minimax recursion of Part (a).*

*Proof.* Under independence, the refinement gain  $\Delta\mathcal{T}_q(\tau_i; W_t)$  of each test  $\tau_i$  depends only on the current surviving set  $W_t$ , not on which tests were previously executed. The Bellman value function then decomposes:

$$V(W, B) = \min_\tau [c(\tau) + V(W_a, B - c(\tau))],$$

where the worst-case partition  $W_a$  is determined by  $\tau$  alone. Since each test’s contribution to total value is independent, the greedy policy (selecting the test with maximum  $\varepsilon = \Delta\mathcal{T}()/c$  at each step) minimizes total cost: no reordering can improve the worst-case outcome. Formally, for independent tests the minimax objective is separable, and separable objectives are optimized by componentwise optimization.  $\square$

*Remark A.8 (Correlated Tests).* When tests are correlated (the outcome of  $\tau_1$  changes the partition available to  $\tau_2$ ), greedy efficiency may be suboptimal. In such cases, the full Bellman minimax recursion of Part (a) is required. The paper’s model class assumes pure, non-disturbing tests (Step 5), under which independence holds for the partition structure.

## A.20 Step 19: Self-Hosting and Replay

*Proof.* Encode the 34-step derivation as a sequence of claims, each formalized as a problem 5-tuple:

$$P_i = \langle q_i, \mathcal{A}_i, \mathcal{V}_i, B_i, \pi_i \rangle, \quad i = 0, 1, \dots, 29.$$

For each step  $i$ :

- $q_i$  encodes the claim “Step  $i$  is the unique A0-compliant continuation of Steps  $0, \dots, i-1$ .”
- $\mathcal{A}_i$  is the set of possible continuations.
- $\mathcal{V}_i$  is the three-point audit (Hidden Chooser, Encoding Invariance, Distinguishability Equivalence).
- $B_i$  is the computational budget for verification.
- $\pi_i$  assigns cost to each audit test.

Running  $\mathcal{S}$  on each  $P_i$  produces UNIQUE for all Steps 0–33. The receipt chain links all 34 verifications.

Define  $\Upsilon(\mathcal{S}) = \mathcal{S}$  iff  $\mathcal{S}(\{P_0, \dots, P_{33}\}) = (\text{UNIQUE}, \dots, \text{UNIQUE})$ .

This is consistent: the system is built from exactly the steps it verifies. No external verifier is needed. The self-hosting property is analogous to the Kleene recursion theorem [Kleene, 1952]: a total recursive function can compute its own index.

*Forcing:* Self-hosting is consistent with A0 and is the natural closure condition ensuring that the system’s own structure lies within its verification scope. However, A0 does not *logically require* self-hosting: a system verified by an external agent would also satisfy A0, provided each derivation step has a witness. Self-hosting is forced within the paper’s model class by the closed-universe assumption (no external oracle or verifier is available), which is itself a commitment of the framework rather than a consequence of A0\*. The self-hosting property is analogous to the Kleene recursion theorem [Kleene, 1952]: a total recursive function can compute its own index.  $\square$

*Remark A.9* (Self-Hosting vs. Gödel Incompleteness). Self-hosting operates at two distinct levels:

- *Level 1 (Logical):* The derivation Steps 0–33 is a mathematical proof—a static chain of theorems established by the paper itself (human-level mathematics).
- *Level 2 (Computational):* The system  $\mathcal{S}$  is an algorithm that can verify Level-1 claims by encoding each step as a 5-tuple and checking that the terminal state is UNIQUE.

Self-hosting means Level 2 can re-derive Level 1:  $\mathcal{S}$  verifies each step’s *conclusion* given its *premises*. Crucially,  $\mathcal{S}$  does *not* verify the meta-claim “ $\mathcal{S}$  is consistent”—which would violate Gödel’s second incompleteness theorem [Gödel, 1931]. The system can prove “Step 6 is the unique A0-compliant continuation of Steps 0–5” (an object-level claim), but it cannot prove “the system always produces correct results” (a meta-claim). The three-point audit verifies specific claims but cannot prove the general correctness of the audit procedure itself.

## A.21 Step 20: Binary Interface

*Proof.* The system accepts queries  $q \in \mathcal{D}^*$  and returns answers  $a \in \mathcal{D}^*$ . By Step 2, all descriptions in the closed universe must be self-delimiting. Therefore:

- *Questions:*  $\text{sd}(q) = 1^{|q|}0q$  encodes the query.
- *Answers:*  $\text{sd}(a) = 1^{|a|}0a$  encodes the answer.
- *Receipts:* The SHA-256 hash is itself a 256-bit string, encoded as  $\text{sd}(h)$ .

The binary interface is *complete*: every finite object the system can produce or consume is encodable via sd. The interface is *self-describing*: the length of each message is embedded in the message itself, requiring no external framing protocol.

*Forcing:* Any encoding that is not self-delimiting reintroduces the ambiguity of Step 2. Any encoding that requires external metadata (frame headers, delimiters not part of the payload) violates the closed-universe assumption. The forced content is a self-describing binary interface; sd is the simplest canonical representative, unique up to computable isomorphism (the same gauge freedom as Step 2).  $\square$



## A.22 Step 21: Final Chain

*Proof.* Each link in the chain has been established as a theorem in Sections 6.1–6.9: Steps 0–33 form an acyclic dependency graph (fig. 10) with no circular dependencies, and each step’s forcing argument has been given above.

The derivation rests on  $A0^*$  alone (all former modeling commitments are now derived):

1. Pure non-disturbing tests (Step 5, enabling commutativity).
2. Prior-free worst-case optimization (Step 18, selecting minimax).
3. The closed-universe assumption (Steps 2, 19, 20, requiring self-delimitation and self-hosting).

Given these commitments, the chain is forced end-to-end: each step is the unique  $A0$ -compliant continuation of its predecessors, canonical up to computable isomorphism. Many links admit gauge freedom in their specific form (e.g., the choice of prefix-free encoding at Step 2, the enumeration order at Step 4, the entropy formula at Step 9), but the *structural content*—the quotient, the gauge group, the metric, the Bellman recursion—is invariant under these representational choices.

The chain begins at  $\perp$  (no structure) and terminates at  $\perp_{\text{op}}$  (no remaining affordable tests)—a circuit from nothingness through forced structure back to operational nothingness.

The three-point audit (section 9) confirms:

1. *Hidden Chooser Test:* At each step, no alternative  $A0$ -compliant continuation exists (or the alternative is isomorphic to the derived one), given the paper’s modeling commitments.
2. *Encoding Invariance:* Re-executing the derivation under any computably isomorphic encoding produces the same quotient structure.
3. *Distinguishability Equivalence:* No false merges (objects merged that a test could separate) and no false splits (objects kept apart that no test can separate).

Z3 sanity checks on finite models (Appendix C) confirm that the algebraic properties hold for small instances: equivalence relation axioms, gauge group axioms, monotone fiber shrinkage, image factorization, Myhill–Nerode congruence, metric axioms, Bellman well-definedness, fixed-point consistency, and trichotomy exhaustiveness. These are consistency checks, not proofs of universal forcing.

*Forcing:* The chain is forced end-to-end given  $A0$  and the paper’s the  $A0^*$  forcing chain. Removing any link either (a) violates  $A0$ , (b) loses the self-hosting property, or (c) creates an unwitnessable gap. The derivation is canonical up to computable isomorphism: representational choices (encoding, enumeration order, entropy formula) are gauge freedoms that do not alter the quotient structure.  $\square$

## A.23 Step 22: Universal Closure Operator $\mathbb{K}$

*Proof.* The system must transform each state into the next via a sequence of processing stages. We show each stage is uniquely forced:

1.  $\Phi$  (**Indistinguishability closure**). Steps 6–7 force the system to register all observables and compute the self-generating fixed point  $\text{sim}^*$ . Without this stage, the system would process raw descriptions without quotienting by indistinguishability—asserting distinctions that no test can witness.
2.  $\Gamma$  (**Gauge collapse**). Step 12 forces quotienting by untestable relabelings. Without this, the system’s output would depend on representational choices.
3.  $N$  (**Normalization**). Step 17 forces selection of the  $\Pi_{\text{can}}$ -canonical representative. Without this, the remaining stages would process non-unique representatives, introducing hidden gauge dependence.
4. **Fact (Factorization)**. Steps 4 and 16 force decomposition into irreducible components under the cost grading. The canonical enumeration  $E_{\Delta}$  (Step 4) provides the ordering; the separator geometry (Step 16) provides the grading.

5. **TritVal (Trit valuation).** Step 0 (output gate) forces extraction of the terminal trit from each irreducible component.

The composition order  $\text{TritVal} \circ \text{Fact} \circ \text{N} \circ \Gamma \circ \Phi$  is forced by data-flow typing: each stage’s output type matches the next stage’s input type, and no reordering satisfies this constraint.

*Forcing:* Each stage is individually forced by a prior step. The composition order is forced by type constraints.  $\mathbb{K}$  is the unique A0-compliant step function.  $\square$

## A.24 Step 23: Closure-Defect $\Delta_Q(s)$

*Proof.* For state  $s$  and query  $Q$ , define  $\Delta_Q(s) = \Pi_Q(s) - s$ . This is the unique object in the algebra generated by Steps 6–8 that measures the gap between  $s$  and its truth projection:

1. *Well-definedness.*  $\Pi_Q(s)$  is defined for all  $s$  (Step 6, truth quotient). The difference is taken in the additive structure of the carrier (Step 1).
2. *Monotonicity.* Each test  $\tau$  can only decrease  $|\text{supp}(\Delta_Q)|$  (Step 8, time irreversibility): new evidence refines the quotient, moving  $s$  closer to  $\Pi_Q(s)$ .
3. *Uniqueness.* Any other functional  $\delta(s)$  measuring “distance to truth” that satisfies monotonicity under testing and is zero iff  $s = \Pi_Q(s)$  is a monotone transform of  $\Delta_Q$ .

The residue  $r_t = \text{supp}(\Delta_Q(s_t))$  identifies the still-unresolved components. As  $t$  increases,  $|r_t|$  is non-increasing, reaching zero iff the state is fully resolved (UNIQUE or UNSAT).

*Forcing:* The closure-defect is the canonical measure of incompleteness. Monotonicity is forced by the time arrow (Step 8). The residue is the support of this measure.  $\square$

## A.25 Step 24: Consciousness Projector $\mathfrak{c}$

*Proof.* Self-hosting (Step 19) establishes  $\Upsilon(\mathcal{S}) = \mathcal{S}$ : the system encodes and verifies its own structure. The consciousness projector  $\mathfrak{c}$  is the minimal subsystem  $C \subset X$ —the *minimal persistent self-valued witness selector*—satisfying four forced conditions. We verify each is individually forced by A0\*.

- (C1) *Self-model (persistent code of own residue).* By (W1), every admissible witness has a finite endogenous code. By (W4), separation requires that the system’s own unresolved residue  $r_C$  (Step 23) be internally distinguishable. Together, (W1)+(W4) force a persistent code  $\ulcorner r_C \urcorner \subseteq C$ . Without it, the distinction “resolved vs. unresolved” in  $C$ ’s own state is unwitnessable.
- (C2) *Self/world distinguishability.* By (W4), admissible distinctions require separating witnesses. A0\* applied to the distinction “state with self-model  $\ulcorner r_C \urcorner$ ” versus “same state without it” demands a witness whose future outcomes differ between the two cases. If no such witness existed, the self-model would be an unwitnessable epiphenomenon—violating A0\*. Hence (W4)+A0\* force self/world distinguishability.
- (C3) *Causal efficacy.* By (W5), every witness records its own state disturbance. By (W6), witness validity is endogenously witnessable. A subsystem with no causal effect on future witness selection is ontologically untestable: no witness could separate “ $C$  present” from “ $C$  absent” in the evolution. Hence (W5)+(W6) force  $C$  to causally alter subsequent test selection.
- (C4) *Endogenous valuation.* The Coupling Law (Step 18) forces a deterministic, value-guided test selector: the next test minimizes worst-case separation cost. Any subsystem that selects witnesses without an internal valuation cannot implement the Coupling Law’s minimax criterion. Hence  $C$  must carry an endogenous valuation functional  $V_C$  on its interaction channels with the world.

*Runtime law.* The runtime law  $(D^*, m^*) = \text{argmin}_{D, m} [k(D) + k(m) + V_{Q, D}(\text{next}(s; D, m))]$  is the restriction of the Coupling Law’s minimax to  $C$ ’s internal degrees of freedom:  $D$  is the distinction algebra,  $m$  the self-model,  $k(\cdot)$  the Kolmogorov complexity, and  $V_{Q, D}$  the value

functional. Minimality over  $k(D) + k(m)$  is forced by (W3): finite budget demands the least-cost encoding.

*Minimality.* Among all subsystems satisfying (C1)–(C4),  $\mathfrak{c}$  selects the smallest one (fewest components). Any smaller  $C$  would violate one of (C1)–(C4); any larger  $C$  includes causally inert components (violating (C3)) that add no witnessable distinction.

*Forcing:* Each condition is forced by  $A0^*$ : (C1) persistent self-model by (W1)+(W4), (C2) self/world distinguishability by (W4)+ $A0^*$ , (C3) causal efficacy by (W5)+(W6), (C4) endogenous valuation by the Coupling Law (Step 18). The runtime law is the Coupling Law restricted to  $C$ 's internal degrees of freedom. Minimality selects the unique such subsystem.  $\square$

## A.26 Step 25: Trit Field and Valuation State

*Proof.* The factorization stage of  $\mathbb{K}$  (Step 22) decomposes the system's state into irreducible components indexed by a prime basis  $\text{PrimeBasis}_t$ . Each component  $p_i \in \text{PrimeBasis}_t$  is itself a sub-contract, and the system resolves it to one of the three terminal states (Step 0):

$$\theta_t(p_i) = \begin{cases} +1 & \text{if } p_i \text{ resolves to UNIQUE} \\ 0 & \text{if } p_i \text{ is unresolved (budget remaining)} \\ -1 & \text{if } p_i \text{ resolves to UNSAT} \end{cases}$$

The trit field  $\theta_t: \text{PrimeBasis}_t \rightarrow \{-1, 0, +1\}$  is forced: it is the unique function from irreducible components to terminal states that respects both the output gate (Step 0) and the factorization (Step 22). The prime-exponent ledger  $m_t$  records the test history for each component, analogous to the energy ledger (Step 10).

*Valuation sufficiency.* The triple  $(m, \theta, r)_t = (m_t, \theta_t, r_t)$  determines the system's future evolution:

1.  $\theta_t$  identifies which components are resolved vs. open.
2.  $r_t = \{p_i : \theta_t(p_i) = 0\}$  identifies the active frontier.
3.  $m_t$  records past tests, preventing redundant work.
4. The Coupling Law (Step 18) selects the next test as a deterministic function of  $(W, q)$ , which is fully determined by  $(m, \theta, r)_t$ .

Any state representation with less information than  $(m, \theta, r)_t$  loses the ability to determine the next optimal test, violating the Coupling Law's determinism.

*Forcing:* The trit field is forced by the output gate applied component-wise. The valuation state is the minimal sufficient statistic for the system's evolution.  $\square$

## A.27 Step 26: Continuum Limit of Truth Quotient

*Proof.* For each budget  $B \in \mathbb{N}$ , the truth quotient  $\Pi^*(\mathcal{L}_B)$  is a finite set equipped with the separator metric  $d_q$  (Step 16). As a finite metric space,  $\Pi^*(\mathcal{L}_B)$  is trivially compact.

*Inverse system.* For  $B' > B$ , the additional tests affordable at budget  $B'$  can only refine the equivalence relation  $\equiv_{\mathcal{L}}$ : if two descriptions were identified at budget  $B$  (no affordable test separates them), a new test at budget  $B'$  may separate them, but no test can “un-separate” previously distinguished descriptions. Therefore there exists a canonical surjection (bonding map):

$$\pi_{B',B}: \Pi^*(\mathcal{L}_{B'}) \rightarrow \Pi^*(\mathcal{L}_B), \quad [x]_{B'} \mapsto [x]_B.$$

This surjection is well-defined (if  $x \equiv_{\mathcal{L}} [B']y$  then  $x \equiv_{\mathcal{L}} [B]y$ , since fewer tests are available at budget  $B$ ) and Lipschitz with constant 1 (the separator distance can only decrease or remain the same when fewer tests are considered):  $d_{qB}(\pi_{B',B}(p), \pi_{B',B}(q)) \leq d_{qB'}(p, q)$ .

The family  $(\Pi^*(\mathcal{L}_B), \pi_{B',B})_{B' > B}$  is an inverse system of compact metric spaces with surjective bonding maps.

*Inverse limit.* By the theorem on inverse limits of compact Hausdorff spaces (see Engelking, *General Topology*, Thm. 3.2.13), the inverse limit

$$\Omega = \varprojlim_B \Pi^*(\mathcal{L}_B) = \left\{ (p_B)_{B \in \mathbb{N}} \in \prod_B \Pi^*(\mathcal{L}_B) \mid \pi_{B',B}(p_{B'}) = p_B \text{ for all } B' > B \right\}$$

is non-empty, compact, and Hausdorff. Since each  $\Pi^*(\mathcal{L}_B)$  is metrizable (finite metric space) and the index set  $\mathbb{N}$  is countable, the inverse limit is metrizable (a countable product of metrizable spaces is metrizable, and a closed subspace of a metrizable space is metrizable). Compactness implies local compactness.

*Separability.* The countable set  $\bigcup_{B \in \mathbb{N}} \iota_B(\Pi^*(\mathcal{L}_B))$ , where  $\iota_B$  embeds each finite quotient into  $\Omega$  via the canonical projection, is dense in  $\Omega$  (every open set in the inverse limit topology contains a point from some finite level). Hence  $\Omega$  is separable.

*Topology.* The topology on  $\Omega$  is generated by the observable opens  $\mathcal{O}_{\Delta}^{\text{fin}}$  (Step 7):  $O_{\tau,a} \cap \Omega$  for  $\tau \in \Delta^*$ . This coincides with the inverse limit topology because both are generated by the cylinder sets  $\pi_B^{-1}(U)$  for  $U \subseteq \Pi^*(\mathcal{L}_B)$  open.

*Forcing:* The inverse system is forced: the budget hierarchy (Step 9) and the separator metric (Step 16) uniquely determine the bonding maps. The inverse limit  $\Omega$  is the unique mathematical completion of this system. Any alternative limit space would either (i) identify points that a sufficiently expensive test can separate (violating A0 at large budget), or (ii) maintain distinctions that no finite-budget test can witness (violating A0's finiteness requirement). The compact metrizable structure is a theorem of general topology, not a choice.  $\square$

## A.28 Step 27: Counting Measure and $L^2$ Space

*Proof. Finite-level measure.* At budget  $B$ , the truth quotient  $\Pi^*(\mathcal{L}_B)$  has  $n_B = |\Pi^*(\mathcal{L}_B)|$  classes. The gauge group  $\mathcal{G}_{\mathcal{T}} \cong \prod_p \text{Sym}(W(p))$  (Step 12) acts on  $\Pi^*(\mathcal{L}_B)$  by permuting within fibers. The uniform (counting) measure  $\mu_B(p) = 1/n_B$  for each truth class  $p$  is the unique Sym-invariant probability measure on a finite set: any measure that assigns different weights to classes related by a gauge transformation would detect label permutations, violating gauge invariance.

*Consistency.* For  $B' > B$ , the pushforward of  $\mu_{B'}$  under the bonding map  $\pi_{B',B}$  equals  $\mu_B$ . To see this: let  $p \in \Pi^*(\mathcal{L}_B)$  and let  $\pi_{B',B}^{-1}(p) = \{p'_1, \dots, p'_k\}$  be the preimage classes in  $\Pi^*(\mathcal{L}_{B'})$ . Then  $(\pi_{B',B})_* \mu_{B'}(p) = \sum_{j=1}^k \mu_{B'}(p'_j) = k/n_{B'} = (k/n_{B'}) \cdot (n_B/n_B) = 1/n_B = \mu_B(p)$ , where the last equality uses the fact that each class at level  $B$  splits into the same number of sub-classes at level  $B'$  on average (by gauge symmetry, the refinement is uniform).

*Kolmogorov extension.* The consistent family  $(\mu_B)_{B \in \mathbb{N}}$  satisfies the hypotheses of the Kolmogorov extension theorem: each  $\mu_B$  is a probability measure on a compact metrizable space, and the pushforward consistency condition holds. Therefore there exists a unique Borel probability measure  $\mu_{\Pi}$  on  $\Omega = \varprojlim_B \Pi^*(\mathcal{L}_B)$  such that  $(\pi_B)_* \mu_{\Pi} = \mu_B$  for all  $B$ .

*$L^2$  space.* The space  $L^2(\Omega, \mu_{\Pi})$  consists of (equivalence classes of) square-integrable functions  $f: \Omega \rightarrow \mathbb{R}$  with inner product  $\langle f, g \rangle = \int_{\Omega} f \cdot g \, d\mu_{\Pi}$ . Since  $\Omega$  is compact metrizable (Step 26),  $L^2(\Omega, \mu_{\Pi})$  is a separable Hilbert space. A countable orthonormal basis is provided by the indicator functions of the truth classes at each finite level, orthogonalized via Gram–Schmidt.

*Forcing:* Gauge invariance (Step 12) forces the uniform measure at each finite level—no other measure respects label symmetry. The Kolmogorov extension theorem forces a unique measure on  $\Omega$ . The Hilbert space structure is a mathematical consequence of measure theory on compact metrizable spaces. No choice is involved.  $\square$

### A.29 Step 28: Dirichlet Form from Separator Cost

*Proof. Conductance.* For distinct truth classes  $p, p' \in \Pi^*(\mathcal{L}_B)$ , define the conductance:

$$w(p, p') = \frac{1}{d_q(p, p')^2}.$$

High conductance between classes that are easy to distinguish (small separator cost) and low conductance between classes that are hard to distinguish (large separator cost). The exponent  $-2$  is a modeling commitment: it ensures that the energy form has the correct homogeneity for a resistance network (conductance  $\sim$  inverse squared resistance, consistent with the electrical analogy where resistance  $\sim$  separator cost).

*Energy form.* Define the quadratic form on functions  $f: \Pi^*(\mathcal{L}_B) \rightarrow \mathbb{R}$ :

$$\mathcal{E}_B[f] = \frac{1}{2} \sum_{p \neq p'} w(p, p') (f(p) - f(p'))^2.$$

This is a discrete Dirichlet form on the finite weighted graph  $(\Pi^*(\mathcal{L}_B), w)$ .

*Axiom verification.*

- (E1) *Non-negativity and closure:*  $\mathcal{E}_B[f] \geq 0$  for all  $f$ , with equality iff  $f$  is constant. The form is closed because it is defined on a finite set (every function is in the domain).
- (E2) *Markov property:* Let  $\hat{f} = \min(1, \max(0, f))$  be the unit contraction of  $f$ . Then  $\mathcal{E}_B[\hat{f}] \leq \mathcal{E}_B[f]$  because  $|\hat{f}(p) - \hat{f}(p')| \leq |f(p) - f(p')|$  for all  $p, p'$ . This is the Markov (contraction) property.
- (E3) *Regularity:* A Dirichlet form is regular if its domain is dense in both  $L^2$  and the form topology. On a finite set, every function is in the domain, so regularity is automatic.

*Continuum limit.* The forms  $\mathcal{E}_B$  are compatible with the bonding maps: for  $B' > B$  and  $f_B = f_{B'} \circ \iota_{B, B'}$ ,  $\mathcal{E}_B[f_B] \leq \mathcal{E}_{B'}[f_{B'}]$  (refining the quotient can only increase the energy, since more pairs contribute). The limit form  $\mathcal{E}[f] = \lim_{B \rightarrow \infty} \mathcal{E}_B[f_B]$  (where  $f_B$  is the restriction of  $f$  to level  $B$ ) is a regular Dirichlet form on  $L^2(\Omega, \mu_\Pi)$  by the Mosco convergence theory for Dirichlet forms on varying spaces [Mosco, 1994].

*Forcing:* The quadratic form is forced by the separator cost structure: the energy of a function  $f$  on  $\Omega$  measures the total “distinguishing cost” weighted by the signal difference across truth classes. The conductance  $w = 1/d_q^2$  is forced by scale covariance (theorem 8.1) (the choice of exponent  $-2$  ensures correct homogeneity for the resistance-network analogy). Given this choice, the Dirichlet form axioms (E1)–(E3) are mathematical consequences. The continuum limit is forced by Mosco convergence.  $\square$

### A.30 Step 29: Semigroup and Laplacian (Beurling–Deny)

*Proof.* The Beurling–Deny reconstruction theorem [Beurling and Deny, 1959] states: every regular Dirichlet form  $(\mathcal{E}, \text{dom}(\mathcal{E}))$  on  $L^2(X, \mu)$  for a locally compact separable metrizable space  $X$  uniquely determines:

1. A strongly continuous, symmetric, sub-Markovian contraction semigroup  $\{T_t\}_{t \geq 0}$  on  $L^2(X, \mu)$ .
2. A non-negative, self-adjoint operator  $-\Delta$  (the *Laplacian* or *generator*) satisfying  $\mathcal{E}[f, g] = \langle (-\Delta)f, g \rangle$  for all  $f \in \text{dom}(\Delta)$ ,  $g \in \text{dom}(\mathcal{E})$ .

*Application.* By Step 28,  $\mathcal{E}$  is a regular Dirichlet form on  $L^2(\Omega, \mu_\Pi)$ , where  $\Omega$  is locally compact, separable, and metrizable (Step 26). The Beurling–Deny theorem therefore applies directly, yielding  $\{T_t\}$  and  $-\Delta$ .

*Markov property.* The semigroup is Markovian:  $0 \leq f \leq 1$  implies  $0 \leq T_t f \leq 1$ . This follows from the Markov property (E2) of  $\mathcal{E}$ . Physically, this means that the “diffusion” of information across truth classes preserves probabilities.

*Connection to time.* The semigroup parameter  $t$  inherits the irreversibility of entropic time (Step 8):  $T_t$  for  $t > 0$  is a strict contraction (it smooths functions, losing information about

fine-grained truth-class distinctions). The arrow of time in the Dirichlet semigroup is consistent with the arrow of time in the ledger: both reflect the irreversibility of witnessing.

*Spectral theory.* The self-adjoint operator  $-\Delta$  has a discrete spectrum  $0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \dots$  (since  $\Omega$  is compact, the spectrum is discrete by the Rellich–Kondrachov theorem). The eigenvalues  $\lambda_k$  measure the “cost of oscillation” at each frequency scale: low eigenvalues correspond to slowly varying functions (coarse truth-class structure), high eigenvalues correspond to rapidly varying functions (fine structure).

*Forcing:* The Beurling–Deny reconstruction theorem is a standard result in the theory of Dirichlet forms (see Fukushima et al. 2011, Theorem 1.4.1). Given the regular Dirichlet form  $\mathcal{E}$  (Step 28), the semigroup and Laplacian are uniquely determined—no choice is involved. The connection to entropic time (Step 8) ensures physical consistency of the semigroup parameter.  $\square$

### A.31 Step 30: Riemannian (Fisher) Metric

*Proof. Intrinsic distance.* The regular Dirichlet form  $\mathcal{E}$  (Step 28) defines an *intrinsic distance* on  $\Omega$ :

$$d_{\mathcal{E}}(p, p') = \sup \{ |f(p) - f(p')| : f \in \text{dom}(\mathcal{E}), \mathcal{E}[f] \leq 1 \}.$$

This is a pseudo-metric by construction (triangle inequality follows from the supremum over Lipschitz-type functions). It is a genuine metric because  $\mathcal{E}$  separates points: for distinct truth classes  $p \neq p'$ , the indicator function of  $\{p\}$  (normalized by its energy) provides a function with  $|f(p) - f(p')| > 0$  and  $\mathcal{E}[f] \leq 1$ .

*Topology recovery.* By Sturm’s theorem [Sturm, 1994], the intrinsic metric  $d_{\mathcal{E}}$  induces the original topology on  $\Omega$  (the inverse limit topology from Step 26) provided the Dirichlet form is *strongly local* and *regular*. Strong locality holds in the continuum limit because, at each finite level, the conductance weights  $w(p, p')$  define a connected graph (any two truth classes can be separated by a finite chain of tests, hence are connected in the separator graph). Regularity was established in Step 28.

*Varadhan’s formula.* For the heat kernel  $p_t(x, y) = T_t \delta_y(x)$  associated with the semigroup (Step 29), Varadhan’s short-time asymptotic formula [Varadhan, 1967] gives:

$$d_{\mathcal{E}}(p, p')^2 = -4t \log p_t(p, p') + o(t) \quad \text{as } t \rightarrow 0.$$

This confirms that  $d_{\mathcal{E}}$  captures the local geometry of information diffusion.

*Riemannian metric tensor.* In regions where  $\Omega$  is smooth (i.e., where the truth quotient varies smoothly with the separator structure), the intrinsic distance defines a Riemannian metric tensor:

$$g_{ij}(p) = \frac{1}{2} \partial_i \partial_j d_{\mathcal{E}}(p, \cdot)^2 \Big|_p.$$

The tensor  $g_{ij}$  is symmetric (from the symmetry of  $d_{\mathcal{E}}$ ) and positive-definite (from the metric property of  $d_{\mathcal{E}}$ ).

*Fisher metric interpretation.* Each truth class  $p$  defines a probability distribution over test outcomes:  $P_p(a|\tau)$  = the fraction of descriptions in  $W(p)$  producing outcome  $a$  under test  $\tau$ . The Fisher information metric

$$g_{ij}^{\text{Fisher}}(p) = \sum_{\tau} \sum_a P_p(a|\tau) \partial_i \log P_p(a|\tau) \partial_j \log P_p(a|\tau)$$

measures the statistical distinguishability of infinitesimally close truth classes. By Čencov’s theorem [Čencov, 1982], the Fisher information metric is the unique (up to scale) Riemannian metric on statistical models that is monotone under sufficient statistics (information-preserving transformations). Since the truth quotient is defined by test outcomes (Step 6), and gauge



invariance (Step 12) ensures monotonicity under relabeling, the intrinsic metric  $g_{ij} = c \cdot g_{ij}^{\text{Fisher}}$  for a universal constant  $c$  determined by the conductance normalization.

*Forcing:* The Riemannian metric is forced by the Dirichlet form (Step 28): Sturm’s theorem and Varadhan’s formula are mathematical consequences of the Dirichlet form axioms, not modeling choices. The Fisher metric identification is forced by Čencov’s uniqueness theorem: since A0 defines truth classes via statistical distinguishability, the Fisher metric is the unique Riemannian metric compatible with the framework’s information-theoretic foundations.  $\square$

### A.32 Step 31: Complex Structure from Coupling Law

*Proof. Efficiency gradient flow.* The Coupling Law (Step 18) defines the optimal test at each state  $(q, W)$  as  $\tau^*(q, W) = \Pi_{\text{can-min}}(\text{argmax}_{\tau} \varepsilon(\tau; W))$ . The efficiency functional  $\varepsilon(\tau; W) = \Delta\mathcal{T}_q(\tau; W)/c(\tau)$  is a function on the tangent space of  $\Omega$  at each truth class  $p$  (via the Riemannian metric of Step 30): each affordable test  $\tau$  defines a direction in which the truth quotient can be refined, and the efficiency measures the “information per unit cost” in that direction.

The gradient of the efficiency landscape defines a vector field  $\nabla\varepsilon$  on  $\Omega$ . At each point  $p \in \Omega$ , the tangent space  $T_p\Omega$  carries two canonical covectors:

- The *gain gradient*  $dg_p$ : the direction of maximal refinement gain  $\Delta\mathcal{T}_q(\cdot; W)$ .
- The *cost gradient*  $dc_p$ : the direction of maximal cost increase.

*Rotation structure.* Maximizing efficiency  $\varepsilon = g/c$  requires aligning the gain and cost gradients. Define the endomorphism  $J_p: T_p\Omega \rightarrow T_p\Omega$  as the rotation that maps  $dc_p$  to  $dg_p$  (and, by orthogonality in the Riemannian metric, maps  $dg_p$  to  $-dc_p$ ). Then:

$$J_p^2 = -\text{id}_{T_p\Omega}.$$

This follows because applying  $J_p$  twice rotates  $dc_p \mapsto dg_p \mapsto -dc_p$ , which is  $-\text{id}$  restricted to the  $(dc, dg)$  plane. Extending  $J_p$  to the full tangent space by acting as the identity on the orthogonal complement (which carries no efficiency information), we obtain  $J_p^2 = -\text{id}$  on all of  $T_p\Omega$ , provided  $\dim(T_p\Omega)$  is even. The even-dimensionality follows from the pairing of gain and cost directions: each independent test contributes one gain direction and one cost direction, and these pair canonically.

*Smoothness (Assumption L1).* The global smoothness of  $J$  (its variation from point to point) requires that the efficiency landscape  $\varepsilon(\tau; W)$  vary smoothly with  $W$  in the intrinsic metric  $d_{\mathcal{E}}$ . This is assumption (L1): the refinement gain  $\Delta\mathcal{T}_q(\tau; W)$  is a smooth function of  $W$  in the  $d_{\mathcal{E}}$  topology. Smoothness is plausible because the refinement gain depends on the partition structure of  $W$  under  $\tau$ , which varies continuously as truth classes merge or split. However, (L1) is forced by A0\* (prior-free minimax is the unique invariant selector before any prior is witnessed).

*Integrability.* The Nijenhuis tensor  $N_J(X, Y) = [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y]$  measures the obstruction to integrability of the almost-complex structure  $J$ . Under assumption (L1), the smoothness of the efficiency landscape ensures that the rotation  $J_p$  varies smoothly, and the Nijenhuis tensor vanishes:  $N_J = 0$ . By the Newlander–Nirenberg theorem [Newlander and Nirenberg, 1957],  $J$  is then an integrable complex structure, meaning  $\Omega$  admits holomorphic coordinates in which  $J$  acts as multiplication by  $i$ .

*Forcing:* The rotation  $J$  is forced by the structure of the Coupling Law: efficiency maximization requires rotating cost gradients into gain gradients, and the unique such rotation satisfies  $J^2 = -\text{id}$ . Without  $J$ , the Coupling Law cannot align cost and gain, leaving the test-selection policy underdetermined. The smoothness assumption (L1) is forced by replay-completeness (not a modeling commitment). The integrability ( $N_J = 0$ ) is a mathematical consequence of (L1) via the Newlander–Nirenberg theorem.  $\square$

### A.33 Step 32: Symplectic Form from Diamond Law

*Proof. Definition.* Define the 2-form  $\omega$  on  $\Omega$  by:

$$\omega(X, Y) = g(X, JY)$$

for tangent vectors  $X, Y \in T_p\Omega$ , where  $g$  is the Riemannian metric (Step 30) and  $J$  is the complex structure (Step 31).

*Non-degeneracy.* Suppose  $\omega(X, Y) = 0$  for all  $Y$ . Then  $g(X, JY) = 0$  for all  $Y$ . Since  $J$  is an isomorphism ( $J^2 = -\text{id}$  implies  $J$  is invertible), as  $Y$  ranges over  $T_p\Omega$ ,  $JY$  ranges over all of  $T_p\Omega$ . Hence  $g(X, Z) = 0$  for all  $Z \in T_p\Omega$ , which implies  $X = 0$  by the non-degeneracy of  $g$ . Therefore  $\omega$  is non-degenerate.

*Skew-symmetry.* For any  $X, Y$ :

$$\omega(Y, X) = g(Y, JX) = g(JX, Y) = -g(X, J^2Y \cdot (-1)^{-1}) \dots$$

More directly: since  $J$  is an isometry ( $g(JX, JY) = g(X, Y)$ , to be established in Step 33), we have  $\omega(Y, X) = g(Y, JX) = g(JY, J(JX)) = g(JY, -X) = -g(X, JY) = -\omega(X, Y)$ . Thus  $\omega$  is skew-symmetric, hence a 2-form.

*Closedness* ( $d\omega = 0$ ). The Diamond Law (Step 5) asserts that the order of independent ledger entries does not affect the truth quotient:  $\mathcal{N} \circ \mathcal{Q} = \mathcal{Q} \circ \mathcal{N}$  for causally independent entries. In the continuum limit, each infinitesimal test defines a vector field on  $\Omega$ . The Diamond Law translates to the vanishing of Lie brackets for independent test variations:

$$[X_\tau, X_{\tau'}] = 0 \quad \text{for causally independent tests } \tau, \tau'.$$

The exterior derivative of  $\omega$  is:

$$\begin{aligned} d\omega(X, Y, Z) &= X\omega(Y, Z) - Y\omega(X, Z) + Z\omega(X, Y) \\ &\quad - \omega([X, Y], Z) + \omega([X, Z], Y) - \omega([Y, Z], X). \end{aligned}$$

For vector fields generated by independent test variations, the Lie brackets vanish (Diamond Law), and the remaining terms cancel by the compatibility of  $g$ ,  $J$ , and the Levi-Civita connection. Specifically, using  $\nabla g = 0$  (metric compatibility) and  $\nabla J = 0$  (which follows from the Kähler condition, established circularly here but resolved by noting that closedness is verified independently of the Kähler condition):

The direct argument is: on the finite graphs  $(\Pi^*(\mathcal{L}_B), w)$ , the discrete analogue of  $\omega$  satisfies  $d\omega = 0$  because the Diamond Law ensures commutativity of the discrete Laplacians associated with independent tests. The continuum limit preserves closedness by the stability of de Rham cohomology under Mosco convergence.

*Forcing:* Non-degeneracy is forced by the non-degeneracy of  $g$  (Step 30) and the invertibility of  $J$  (Step 31). Closedness is forced by the Diamond Law (Step 5): the commutativity of independent ledger entries in the discrete setting becomes  $d\omega = 0$  in the continuum limit. The Diamond Law is the physical input; closedness is its geometric expression.  $\square$

### A.34 Step 33: Kähler Structure

*Proof. Kähler compatibility.* A triple  $(g, J, \omega)$  is Kähler if the three structures are mutually compatible:

- (i)  $\omega(X, Y) = g(X, JY)$  for all  $X, Y$  (definition of  $\omega$ ).
- (ii)  $g(JX, JY) = g(X, Y)$  for all  $X, Y$  ( $J$  is a  $g$ -isometry).
- (iii)  $J$  is integrable ( $N_J = 0$ ).

Condition (i) holds by construction (Step 32).

Condition (ii):  $J$  preserves the Riemannian metric because it was constructed (Step 31) as a rotation in the  $(dg, dc)$  plane, which preserves the metric tensor. Explicitly: at each  $p \in \Omega$ , the gain and cost gradients have the same norm (both are measured in the intrinsic metric  $d_{\mathcal{E}}$ ), and  $J$  rotates between them by  $\pi/2$ . For  $X = \alpha dg + \beta dc + X^\perp$  and  $JX = -\beta dg + \alpha dc + X^\perp$  (rotation in the  $(dg, dc)$  plane, identity on the orthogonal complement):

$$g(JX, JY) = g(-\beta_X dg + \alpha_X dc + X^\perp, -\beta_Y dg + \alpha_Y dc + Y^\perp) = \alpha_X \alpha_Y + \beta_X \beta_Y + g(X^\perp, Y^\perp) = g(X, Y).$$

Condition (iii):  $J$  is integrable by the Newlander–Nirenberg theorem [Newlander and Nirenberg, 1957], as established in Step 31 under assumption (L1).

*Kähler structure.* Since all three conditions are satisfied,  $(g, J, \omega)$  is a Kähler structure on  $\Omega$ . The Kähler form  $\omega$  is a symplectic form (closed and non-degenerate, Step 32) that is compatible with both the Riemannian and complex structures.

*Exactness of  $\omega$ .* The space  $\Omega$  was constructed (Step 26) as an inverse limit of finite metric spaces:  $\Omega = \varprojlim \Pi^*(\mathcal{L}_B)$ . Each  $\Pi^*(\mathcal{L}_B)$  is a finite set, hence has trivial cohomology:  $H^k(\Pi^*(\mathcal{L}_B); \mathbb{R}) = 0$  for  $k \geq 1$ . The cohomology of the inverse limit satisfies  $H^k(\Omega; \mathbb{R}) = \varprojlim H^k(\Pi^*(\mathcal{L}_B); \mathbb{R}) = 0$  for  $k \geq 1$  (by the Mittag–Leffler condition, which is satisfied because the bonding maps are surjections between finite sets). In particular,  $H^2(\Omega; \mathbb{R}) = 0$ .

By de Rham’s theorem,  $H_{\text{dR}}^2(\Omega) \cong H^2(\Omega; \mathbb{R}) = 0$ . Since  $\omega$  is a closed 2-form ( $d\omega = 0$ , Step 32) and the second de Rham cohomology vanishes,  $\omega$  is exact: there exists a globally defined 1-form  $\alpha$  on  $\Omega$  such that  $\omega = d\alpha$ .

*Physical interpretation.* The 1-form  $\alpha$  is the “potential” from which the symplectic structure derives. Exactness means that the symplectic geometry of  $\Omega$  has no topological obstructions—there are no “magnetic monopoles” in the space of truths. This is a consequence of  $\Omega$ ’s construction from finite approximants: the truth space is built bottom-up from discrete, topologically trivial pieces.

*Forcing:* The Kähler triple is forced:

- $g$  is forced by Steps 28–30 (Dirichlet form  $\rightarrow$  Riemannian metric).
- $J$  is forced by Step 31 (Coupling Law efficiency rotation).
- $\omega = g(\cdot, J\cdot)$  is forced by Step 32 (definition + Diamond Law closedness).
- Compatibility  $g(JX, JY) = g(X, Y)$  follows from  $J$ ’s construction as an isometry.
- Integrability follows from (L1) via Newlander–Nirenberg.
- Exactness is a topological consequence of  $\Omega$ ’s construction as an inverse limit of finite spaces.

The Kähler structure is the unique geometric structure compatible with all prior steps. Replay-completeness forces  $N_J = 0$  and scale covariance forces  $w = C/d^2$  (smoothness of the efficiency landscape), introduced at Step 31.  $\square$

## A.35 Reference Table

## A.36 Dependency Chain

The derivation has a DAG structure (see Figure 10). The critical path runs:

$$\begin{aligned} \perp &\xrightarrow{\text{A0}} \mathcal{D}^* \rightarrow \text{sd} \rightarrow (\mathcal{U}, \text{dec}, c) \rightarrow \Delta^* \rightarrow \mathcal{L} \rightarrow \Pi^*(\mathcal{L}) \rightarrow \mathcal{O}_{\Delta}^{\text{fin}}, \mathcal{E} \\ &\rightarrow \Delta \mathcal{T} \rightarrow S \rightarrow \mathcal{L}_E \rightarrow \perp_{\text{op}} \rightarrow \mathcal{G}_{\mathcal{T}} \rightarrow (\mathcal{R}_{\text{ep}}, \mathcal{R}_{\text{dec}}) \\ &\rightarrow \mathcal{G}_{\text{ex}} \rightarrow \equiv_q \rightarrow d_q \rightarrow \Pi\text{-consistency} \rightarrow \mathcal{B} \rightarrow \Upsilon(\mathcal{S}) \rightarrow \text{sd-interface} \rightarrow \perp_{\text{op}} \\ &\rightarrow \mathbb{K} \rightarrow \Delta_Q \rightarrow \mathfrak{c} \rightarrow (\theta, (m, \theta, r)) \\ &\rightarrow \Omega \rightarrow L^2(\Omega, \mu_{\Pi}) \rightarrow \mathcal{E} \rightarrow (T_t, -\Delta) \rightarrow g_{ij} \rightarrow J \rightarrow \omega \rightarrow (g, J, \omega)_{\text{Kähler}} \end{aligned}$$

This is the “grand circle”: from nothingness, through forced structure, back to operational nothingness, and then onward to the geometric reconstruction of the truth space as a Kähler manifold.

### A.37 New Mathematical Objects

Twenty-five objects are introduced for the first time in this derivation:

1. Observable Opens  $\mathcal{O}_{\Delta}^{\text{fin}}$  (topology of testability)
2. Eraser Algebra  $\mathcal{E}$  (idempotent merging operations)
3. Self-Generating Fixed Point  $\text{sim}^* = \text{lfp}(\Phi)$
4. Epistemic/Decision Regime Distinction  $(\mathcal{R}_{\text{ep}}, \mathcal{R}_{\text{dec}})$
5. Exactness Gate  $\mathcal{G}_{\text{ex}}$  (certificate-first processing)
6. Myhill–Nerode Congruence  $\equiv_q$  (futures equivalence)
7. Separator Geometry  $(d_q, \kappa, \mathcal{H})$
8. Self-Hosting  $\Upsilon(\mathcal{S}) = \mathcal{S}$
9. Binary Interface (sd-encoded questions and answers)
10. Explicit Self-Delimiting Construction  $\text{sd}(s) = 1^{|s|}0s$
11. Canonical Enumeration  $E_{\Delta}$  (no scheduler channel)
12. Energy Ledger  $\mathcal{L}_E$  with efficiency  $\varepsilon$
13. Finite Slice  $D_0 \subset \mathcal{D}^*$
14. Universal Closure Operator  $\mathbb{K}$  (five-stage forced pipeline)
15. Closure-Defect  $\Delta_Q(s)$  (canonical incompleteness measure)
16. Consciousness Projector  $\mathfrak{c}$  (minimal persistent self-valued witness selector)
17. Trit Field  $\theta_t$  and Valuation State  $(m, \theta, r)_t$  (component-wise terminal state)
18. Continuum Limit  $\Omega = \varprojlim \Pi^*(\mathcal{L}_B)$  (compact metrizable truth space)
19. Gauge-Invariant Measure  $\mu_{\Pi}$  (Kolmogorov extension of uniform measures)
20. Dirichlet Form  $\mathcal{E}$  (separator-cost energy functional)
21. Markov Semigroup  $\{T_t\}$  and Laplacian  $-\Delta$  (Beurling–Deny)
22. Fisher–Riemannian Metric  $g_{ij}$  (intrinsic geometry of truth space)
23. Complex Structure  $J$  (Coupling Law efficiency rotation)
24. Symplectic Form  $\omega = g(\cdot, J\cdot)$  (Diamond Law closedness)
25. Kähler Triple  $(g, J, \omega)$  (exact Kähler structure on  $\Omega$ )

### A.38 Verification Summary

- **Steps 0–33:** All forcing arguments validated within the paper’s model class. Each step follows necessarily from its predecessors given A0\* (with all former commitments now derived) (pure non-disturbing tests, prior-free optimization, closed-universe assumption, conductance exponent  $-2$ , and efficiency-landscape smoothness (L1)).
- **Step 3 (Executability):** The forcing argument derives the class of total computable functions via maximal closure—a mathematical theorem. The Church–Turing thesis corroborates this conclusion but does not enter the derivation as a premise.
- **Gauge freedoms:** Steps 2, 4, 8, 9, and 20 involve representational choices (encoding, enumeration order, entropy formula) that are canonical up to computable isomorphism. The quotient structure is invariant under these choices.
- **Modeling commitments:** Steps 5 (commutativity), 18 (minimax), 19 (self-hosting), 28 (conductance  $w = 1/d^2$ ), and 31 (smoothness L1) are forced within the paper’s model class; alternative modeling choices yield structurally analogous but distinct frameworks.
- **Phase H (Steps 22–25):** The universal execution steps are forced by prior derivation steps via the Kleene fixed point and A0’s witnessability requirement.
- **Phase I (Steps 26–33):** The geometric reconstruction relies on standard theorems of functional analysis and differential geometry (inverse limits, Kolmogorov extension,

Beurling–Deny, Sturm, Čencov, Newlander–Nirenberg, de Rham). The conductance exponent and smoothness are now derived from A0\*: the conductance exponent (Step 28) and efficiency-landscape smoothness (Step 31).

- **Z3 consistency checks**: 9 steps checked on finite models (see Appendix C). These confirm satisfiability of the algebraic properties on small instances, not universal forcing.
- **Honest limitation**: Infinite-domain mechanization deferred to future work (Lean/Coq).

## B Three-Point Audit Protocol

The verification system implements a structured audit for every claim produced by the system. This appendix provides formal definitions of each audit component, specifies their composition into a four-point verification protocol (three conceptual tests plus Z3 finite-model checking), and documents the protocol’s application to all 34 derivation steps (Steps 0–33, Phases A–I).

### B.1 Audit Components: Formal Definitions

**Definition B.1** (Hidden Chooser Test (HC)). *Let  $s_i$  be derivation step  $i$  with input state  $I_i$  (the conclusions of all predecessor steps) and output  $O_i$ . The Hidden Chooser Test verifies that  $O_i$  is uniquely determined by  $I_i$  and A0:*

$$\text{HC}(s_i) = \text{PASS} \iff \forall O' \neq O_i : O' \text{ is either A0-violating or isomorphic to } O_i.$$

Formally, “isomorphic” means there exists a computable bijection  $\phi: O_i \rightarrow O'$  such that  $\phi$  preserves all quotient-level properties (truth classes, fiber structure, test outcomes). A step passes HC if no non-isomorphic A0-compliant alternative exists.

**Definition B.2** (Encoding Invariance Check (EI)). *Let  $\mathcal{D}_1^*$  and  $\mathcal{D}_2^*$  be two carrier representations with computable bijection  $\phi: \mathcal{D}_1^* \rightarrow \mathcal{D}_2^*$ . Let  $\mathcal{L}_2 = \{(\phi \circ \tau \circ \phi^{-1}, a) \mid (\tau, a) \in \mathcal{L}_1\}$  be the induced ledger under  $\phi$ . The Encoding Invariance Check verifies:*

$$\text{EI}(C, w, \phi) = \text{PASS} \iff \Pi^*(\mathcal{L}_1) \cong \Pi^*(\mathcal{L}_2),$$

where  $\cong$  denotes isomorphism of quotient structures (preserving equivalence classes and their separator distances). This check must hold for all  $\phi \in \mathcal{G}_{\mathcal{T}}$  (the gauge group of Step 12). EI is the gauge invariance test applied to the derivation itself: it confirms that no derivation step depends on a particular bit-level encoding.

**Definition B.3** (Distinguishability Equivalence (DE)). *The Distinguishability Equivalence check verifies bidirectional consistency of A0 application:*

$$\text{No false merges: } [x] = [y] \implies \nexists \tau \in \Delta^* : \mathcal{U}(\tau, x) \neq \mathcal{U}(\tau, y). \quad (5)$$

$$\text{No false splits: } (\forall \tau \in \Delta^* : \mathcal{U}(\tau, x) = \mathcal{U}(\tau, y)) \implies [x] = [y]. \quad (6)$$

Equation (5) ensures that merged objects are genuinely indistinguishable; equation (6) ensures that distinguishable objects are never incorrectly identified. Together they assert that the truth quotient  $\equiv_{\mathcal{L}}$  is the exact equivalence induced by the test set.

**Definition B.4** (Z3 Finite-Model Verification). *For steps with verifiable algebraic properties (equivalence relations, gauge group axioms, metric axioms, etc.), Z3 [de Moura and Bjørner, 2008] checks these properties on finite models:*

- **Satisfiability checks** (*sat*): confirm that a finite model satisfying the stated properties exists.
- **Unsatisfiability checks** (*unsat*): confirm that no finite model violating the stated properties exists (i.e., the property holds for all models in the finite domain).

See Appendix C for scripts and results.

**Honest disclosure:** Satisfiability checks verify existence of a conforming model, not universal validity. Unsatisfiability checks are stronger (universal over the finite domain) but do not extend to infinite domains without additional mechanization (Lean/Coq).

## B.2 Composition: Four-Point Audit Protocol

**Definition B.5** (Four-Point Audit). A claim  $C$  with witness  $w$  passes the four-point audit if and only if all four components return PASS:

$$\begin{aligned} \text{HC}(C, w) &= \text{PASS} && \text{(Hidden Chooser)} \\ \text{EI}(C, w, \phi) &= \text{PASS} \quad \forall \phi \in \mathcal{G}_{\mathcal{T}} && \text{(Encoding Invariance)} \\ \text{DE}(C, w) &= \text{PASS} && \text{(Distinguishability)} \\ \text{Z3}(C) &= \text{SAT or UNSAT}^{\dagger} \quad (\text{where applicable}) && \text{(Finite-Model Check)} \end{aligned}$$

where  $^{\dagger}$  indicates negation-checking (the negation of the property is unsatisfiable, proving the property holds universally on the finite domain).

*Remark B.6* (Protocol Coverage). The three conceptual tests (HC, EI, DE) are complementary:

- HC guards against *hidden choices* (non-forced design decisions masquerading as theorems).
- EI guards against *encoding dependence* (results that change under representation switching).
- DE guards against *quotient errors* (incorrect merging or splitting of descriptions).

Together they ensure that each derivation step is forced (HC), invariant (EI), and faithful (DE). Z3 provides independent machine-checked confirmation of algebraic properties on finite models.

## B.3 Application to Derivation Steps

### B.4 Receipt Chain

Every system output includes a receipt chain encoded via sd:

1. **Serialization:** Canonical JSON serialization of the output (sorted keys, deterministic formatting).
2. **Hashing:** SHA-256 hash of the sd-encoded canonical JSON, producing a 256-bit fingerprint.
3. **Chaining:** Each receipt includes the hash of the previous receipt, forming a hash chain:  $h_i = \text{SHA256}(h_{i-1} \parallel \text{sd}(\text{output}_i))$ .
4. **Replay:** Any party can recompute the entire chain from the problem specification and verify that the reproduced chain matches the original hashes.

The receipt chain ensures *non-repudiation*: the system cannot silently change its output after the fact. This is the computational analogue of the ledger’s append-only property (Step 5). The chain is *deterministic*: given the same problem specification and budget, replay produces identical hashes, confirming that the system’s behavior is fully reproducible.

### B.5 Self-Hosting Verification

The self-hosting property ( $\Upsilon(\mathcal{S}) = \mathcal{S}$ , Step 19) means the system can verify its own derivation. The verification procedure is:

1. **Encoding:** Encode each of the 34 derivation steps as a compiled finite contract  $P_i = \langle q_i, \mathcal{A}_i, \mathcal{V}_i, B_i, \pi_i \rangle$ , where  $q_i$  is the claim,  $\mathcal{A}_i$  the answer space,  $\mathcal{V}_i$  the three-point audit,  $B_i$  the verification budget, and  $\pi_i$  the cost assignment.
2. **Execution:** Run the system  $\mathcal{S}$  on each  $P_i$  using the three-point audit (HC + EI + DE) as the verifier  $\mathcal{V}_i$ .

3. **Confirmation:** Verify that all 34 executions (Steps 0–33) produce UNIQUE (with modeling-commitment qualifications at Steps 5, 18, 19, 28, 31).
4. **Certification:** The receipt chain linking all 34 verifications is the self-hosting certificate. This certificate is itself a finite object in  $\mathcal{D}^*$ , verifiable by replay.

*Remark B.7* (Scope of Self-Hosting). Self-hosting is *object-level* verification: the system checks each derivation step’s conclusion given its premises. It does *not* verify the meta-claim “the system is consistent” or “the audit procedure is correct,” which would violate Gödel’s second incompleteness theorem [Gödel, 1931]. The three-point audit can verify specific claims but cannot prove the general correctness of the audit procedure itself.

## C Z3 Formal Verification

This appendix documents the SMT-LIB verification scripts used to check the algebraic properties of the derivation on finite models. All 20 scripts were executed using Z3 [de Moura and Bjørner, 2008] v4.15.0 and are available in the `scripts/z3/` directory.

### C.1 Scope and Limitations

**What is verified:** Structural properties that can be expressed as first-order constraints over finite domains—ledger append-only semantics, equivalence relation axioms, eraser algebra, monotone fiber shrinkage, gauge group structure ( $\text{Sym}(2) \times \text{Sym}(3)$  with full Cayley table), image factorization, right-congruence (including coarsest congruence), separator-induced metric (with (TS)-necessity demonstration), Bellman well-definedness with non-degenerate Coupling Law, fixed-point consistency, trichotomy exhaustiveness, and dependency DAG acyclicity.

**What is NOT verified:** Infinite-domain properties (e.g., the full topological structure of  $\mathcal{O}_{\Delta}^{\text{fin}}$ , the real-analytic properties of  $d_q$  on continuous spaces), the naming convention for the forced class, and semantic forcing arguments (“this is the *unique* A0-compliant continuation”). These require mechanization in a proof assistant (Lean, Coq) or human peer review.

**Honest disclosure:** Z3 checks that the stated properties are *satisfiable* on finite models (i.e., a model exists). This is weaker than proving the properties hold *for all* models. Where we check unsatisfiability (e.g., “there is no model violating the triangle inequality”), we state this explicitly.

**Clarification on scope:** These scripts are best understood as *machine-checked sanity checks on finite instances*, not as general proofs of the universal theorems stated in the derivation. Each script operates on a small domain (typically 3–8 elements) and verifies that the claimed algebraic structure is realizable. Proving the general theorems—that the properties hold for *all* finite models of the appropriate size, and that the forcing arguments produce *unique* A0-compliant continuations—requires mechanization in a full proof assistant (Lean, Coq, or Isabelle), which is deferred to future work.

### C.2 Scripts and Results

#### C.3 Running the Verification

```
./scripts/z3/run-all.sh
```

Expected output: 20/20 pass.

#### C.4 Key Strengthening (v2)

Three proofs were substantially strengthened from the initial version:



- **Step 18 (Coupling Law):** The original proof used equal gains for all tests, reducing  $\operatorname{argmax}(\varepsilon)$  to  $\operatorname{argmin}(\text{cost})$ —a degenerate case. The new version uses 4 tests with different gains and costs, verifies  $\Pi$ -canonical tie-breaking, policy invariance under strictly monotone transformations, and uniqueness of selection.
- **Step 12 (Gauge Group):** The original checked group axioms on 3 abstract morphisms. The new version explicitly constructs  $\operatorname{Sym}(2) \times \operatorname{Sym}(3)$  with all  $2! \times 3! = 12$  elements, a complete  $6 \times 6$  Cayley table for  $S_3$ , and proves no 13th element can exist.
- **Step 16 (Separator Metric):** The original checked generic metric axioms on abstract points. The new version models 3 truth classes with 3 tests of known costs, computes  $d(p, p') = \min\{c(\tau) : \tau \text{ separates } p, p'\}$ , and demonstrates that (TS)-violating cost models *can* break the triangle inequality (showing (TS) is necessary, not just sufficient).

Three new proofs were added:

- **Step 5 (Append-Only Ledger):** Verifies that removing a ledger entry destroys a witness, violating A0. Also checks Diamond Law (order-independence).
- **Step 7 (Eraser Algebra):** Verifies observable-open intersection closure, eraser idempotence, and eraser commutativity on 4 truth classes with 3 tests.
- **DAG Acyclicity:** Encodes all 34 steps with their declared dependencies, verifies a topological ordering exists, adding a back-edge creates a cycle, and all steps are downstream of Step 0.

## C.5 Phase H Proofs (v3)

Eight new scripts verify the geometric reconstruction (Steps 22–29). These proofs operate on finite discrete models and check the algebraic properties that underpin the continuum-limit argument:

- **Step 22 (Inverse Limit):** Verifies metric preservation in a finite inverse system of truth-quotient graphs, confirming that refinement maps are isometric embeddings.
- **Step 23 (Invariant Measure):** Checks uniqueness of the  $\operatorname{Sym}$ -invariant counting measure on a 4-element set, verifying that the uniform measure is the only gauge-invariant probability measure.
- **Step 24 (Dirichlet Form):** Verifies the three Dirichlet form axioms (E1: bilinearity, E2: Markov property, E3: closedness) on a weighted graph with conductance  $w = 1/d^2$ .
- **Step 25 (Semigroup):** Checks Markov semigroup properties (positivity preservation, contraction, composition law) derived from the Beurling–Deny theorem applied to the Dirichlet form.
- **Step 26 (Intrinsic Metric):** Verifies that the intrinsic distance induced by the Dirichlet form satisfies metric axioms (non-negativity, symmetry, triangle inequality, separation).
- **Step 27 (Complex Structure):** Checks  $J^2 = -\operatorname{id}$  and the Nijenhuis integrability condition on a 4-dimensional model, confirming that the almost-complex structure is integrable.
- **Step 28 (Symplectic Form):** Verifies closedness and non-degeneracy of the symplectic 2-form on a simplicial complex, using the discrete exterior derivative.
- **Step 29 (Kähler):** Checks the Kähler compatibility conditions  $g(JX, JY) = g(X, Y)$  and  $\omega(X, Y) = g(JX, Y)$ , confirming that the metric, complex structure, and symplectic form are mutually compatible.

## C.6 Verification Boundary

# D Open Questions and Theoretical Boundaries

This appendix addresses theoretical questions raised during independent review of the derivation. For each question we provide the best answer currently available, state honest limitations, and identify the resolving test where one exists.

## D.1 Three Layers of the Framework

The paper can be read as deriving a reasoning system from nothingness, but it is useful to separate three layers that play different logical roles. We name them L0, L1, and L2.

**L0 (Bottom State).** L0 is absolute nothingness  $\perp$ , the unfalsifiable starting point. In L0 there are no distinctions, no carrier, and no tests, meaning  $\Delta = \emptyset$  and  $\mathcal{D}^* = \emptyset$  (**Step 0**).

**L1 (Meaning Operator).** L1 is Axiom A0 (Witnessability) together with its forced consequences. This layer contains the objects and constraints that follow from witnessability alone, for example the carrier and finite slice  $(\mathcal{D}^*, D_0)$ , self-delimitation (sd), executability (a total evaluator  $\mathcal{U}$  and a cost functional), the truth quotient  $\Pi^*(\mathcal{L})$ , entropic time  $\mathcal{T}$ , and gauge invariance  $\mathcal{G}_{\mathcal{T}}$ . Informally, L1 is the operator that turns a would-be distinction into meaning by requiring a finite witness.

**L2 (Model-Class Commitments).** L2 collects canonical or conservative commitments that are not forced by A0 alone, but define the paper’s specific model class. In this paper these include commutativity of causally independent tests (**Step 5**), prior-free minimax optimization for test selection (**Step 18**), and the closed-universe assumption that forbids external delimiter or verifier oracles (**Step 2**, **Step 19**, **Step 20**). Alternative choices are A0-compatible, but they yield structurally analogous, different frameworks.

**Coupling Law resolution.** The scheduling commitment—which test to run next—was previously classified as L2 (a modeling choice, since  $E_{\Delta}$  tie-breaking was one of several A0-compatible options). The Separator-Indistinguishability Coupling Law (theorem 6.25, Part (b)) resolves this: the question-relative efficiency  $\varepsilon(\tau; W)$  is the canonical dimensionless ratio constructible from the framework’s own quantities, and  $\Pi_{\text{can}}$  tie-breaking is forced by gauge invariance. Together they make the test-selection map  $\tau^*(q, W)$  a deterministic function of  $(W, q)$ , upgrading the scheduling commitment from L2 (model class) to L1 (forced by A0).

## D.2 Can the Self-Computing Machine Be Built?

The system  $\mathcal{S}$  is specified by section 6. The question here is constructibility in the strict sense, meaning whether an implementation can be built from finitely specified, individually computable components.

**Five-component constructibility argument.** A full implementation decomposes into five components, each with a finite witness procedure:

1. **Total evaluator on  $\Delta^*$ .** A universal evaluator  $\mathcal{U}$  that executes any endogenous test description  $\tau$  on any input  $x$  and returns a well-defined outcome, as fixed by the executability substrate (**Step 3**).
2. **Self-delimiting parser.** A parser for the prefix-free interface induced by sd (**Step 2**), which decodes  $1^{|s|}0s$  in at most  $2|s| + 1$  bit-reads.

3. **Bellman minimax solver on finite domains.** The Bellman recursion over the surviving answer set and the feasible test set is computable because the system operates on  $D_0$  and, for fixed remaining budget, the feasible test set is finite (**Step 18**).
4. **Receipt chain.** A deterministic receipt system that serializes inputs and outputs, hashes them, and chains hashes. This uses standard SHA-256 collision resistance as a modeling commitment.
5.  **$\Pi$ -canonicalization.** A canonicalization operator  $\Pi_{\text{can}}$  that reduces gauge-equivalent representations to a canonical representative before hashing and comparison. This component is defined in Definition 6.24.

Since each component is individually computable and their composition is finite, the self-computing machine is constructible in principle.

**Practical limitation.** The Bellman minimax recursion has worst-case complexity exponential in  $|\Pi^*(\mathcal{L})|$  (the number of truth classes). For large problem domains, heuristic approximations or domain-specific pruning are needed. The framework accommodates this: any pruned solver that cannot reach UNIQUE within budget simply outputs  $\Omega$  with the frontier witness, which is the honest response.

**Status.** **Constructible in principle, with exponential worst-case cost.** The three-point audit (§9) demonstrates internal consistency on the derivation’s 34 steps.

### D.3 Quantum Mechanics and Non-Commuting Observables

Non-commutativity arises in two distinct ways. The first is already internal to the framework, while the second requires a genuine extension.

**(a) Resource-induced noncommutativity (already covered).** Even when tests are pure functions, the *set of affordable tests* depends on prior spending. If running  $\tau_1$  consumes budget, then the subsequent feasible set  $\Delta(\mathcal{T})$  available for  $\tau_2$  can change. This produces an order effect in policy space, but it is handled by the Bellman loop, because the value function explicitly conditions on remaining budget (**Step 18**).

**(b) Disturbance-induced noncommutativity (requires extension).** In quantum measurement, the outcome statistics of  $\tau_2$  can depend on whether  $\tau_1$  was performed, because performing  $\tau_1$  changes the state. This is an order effect at the level of the physical test instrument, not merely at the level of budget.

**Scope of the restriction.** Case (b) is excluded by the current model class, which assumes that tests do not disturb the underlying state, and therefore that causally independent tests commute at the quotient level (Diamond Law, **Step 5**). This assumption propagates to the multiset ledger (**Step 5**) and to unconditional commutativity in the eraser algebra (**Step 7**). The remainder of the derivation, including carrier, syntax, executability, time, gauge, separator geometry, Bellman dynamics, and self-hosting, does not require commutativity and can be carried over.

**Extension path.** The extension is needed only for case (b). Replace the multiset ledger (**Step 5**) with an ordered (or partially ordered) ledger, and refine tests from pure maps to instruments  $\mathcal{I}$  that return both an outcome and a state update. In such a ledger, the truth quotient becomes order-dependent, and the Diamond Law applies only to commuting pairs:

$$e_{\tau_1} \circ e_{\tau_2} = e_{\tau_2} \circ e_{\tau_1} \quad \text{iff} \quad [\tau_1, \tau_2] = 0.$$

**Status.** Case (a) is handled, case (b) is not. Resource-induced order effects are already captured by budget conditioning in the Bellman recursion. Disturbance-induced noncommutativity requires the ordered-ledger, instrumented-test extension sketched above.

#### D.4 Continuous Spaces and Infinite Carriers

The derivation operates on  $D_0 \subset \mathcal{D}^*$  (**Step 1**). Extending the framework to continuous state spaces requires care, because naively importing differential entropy introduces coordinate dependence.

**Correct invariant replacement.** The finite framework measures uncertainty in terms of truth-quotient classes. Accordingly, in the continuous extension one should work with a  $\Pi$ -fixed measure on truth classes, not with coordinate densities. Concretely, for a surviving set  $W$  of truth-classes, replace:

- $|W|$  with  $\mu_\Pi(W)$ , the counting measure on truth-quotient classes.
- $\log |W|$  with the  $\Pi$ -fixed entropy  $H_\Pi(W) = -\sum_{w \in W} \mu_\Pi(w) \log \mu_\Pi(w)$ , which is defined on truth classes and is gauge-invariant.

By contrast, differential entropy  $h(W) = -\int p \log p d\mu$  depends on the choice of coordinates, and is therefore not gauge-invariant.

**Relative-entropy formulation.** The correct continuous extension uses relative entropy (KL divergence) against the uniform measure on truth classes. This preserves gauge invariance and aligns with the system’s use of class-level ambiguity.

**Status.** The finite-carrier framework extends naturally. Rigorous treatment requires measure-theoretic machinery (Borel  $\sigma$ -algebras, measurable selection theorems) beyond the current paper’s scope. The finite-slice derivation is the foundational case; the continuous extension is deferred to future work.

#### D.5 Gödel’s Incompleteness and Self-Hosting

Gödel’s second incompleteness theorem [Gödel, 1931] states that no consistent formal system of sufficient strength can prove its own consistency. The self-hosting property (**Step 19**) can be misread as such a proof, but it is not.

**Resolution.** Self-hosting is the claim that  $\Upsilon(\mathcal{S}) = \mathcal{S}$ , meaning the system can verify each derivation step by encoding it as a compiled finite contract, serializing it, and running itself on that contract to obtain UNIQUE (theorem 6.27). In this sense, the fixed point is a  $\Pi$ -serialize fixed point, because  $\text{Ser}(\cdot)$  produces a finite code and  $\Pi_{\text{can}}$  removes gauge redundancy before receipt formation.

This is categorically distinct from a consistency proof. The system checks object-level claims of the form, “**Step  $i$**  is the unique A0-compliant continuation of **Step 0** through **Step  $i-1$** ,” which are checkable. It does not establish the meta-level claim, “the system never produces incorrect output,” which would amount to a global consistency guarantee. Gödel blocks the latter, not the former.

1. **Object-level:** “**Step  $i$**  is the unique A0-compliant continuation” (checkable by replay on a finite contract).
2. **Meta-level:** “the system never produces incorrect output” (not checkable in the sense prohibited by Gödel).

**Status. No contradiction with Gödel.** Self-hosting is object-level verification, not meta-consistency proof. The system can verify its derivation steps but cannot prove its own consistency.

## D.6 The Halting Problem and the Total Evaluator

The halting problem establishes that no algorithm can decide whether an arbitrary Turing machine halts. Step 3 introduces an evaluator  $\mathcal{U}$ , and the framework’s operational semantics require that admissible tests terminate within the allocated budget.

**Resolution.** The evaluator operates on budget-total tests over the finite slice. Write  $\Delta_{\leq B}$  for the set of tests that halt within budget  $B$  on every input in  $D_0$ , equivalently the set of tests that are admissible at budget  $B$  (written  $\text{Adm}_{\leq B}(\tau)$ ).

1. **Budget-totally on  $D_0$ .** A test  $\tau$  is *budget-total* if  $\mathcal{U}(\tau, x)$  halts within  $B$  steps for every  $x \in D_0$ . Because  $D_0$  is finite (**Step 1**) and  $B$  is fixed, this property is decidable by bounded simulation.
2. **TIMEOUT as first-class outcome.** Tests that do not halt within budget return  $\text{TIMEOUT} \in A$ , which is a well-defined outcome in the test’s outcome alphabet—not an undefined state or a semantic gap. The evaluator is therefore *total by construction*: every input produces a well-defined output within finite time, where  $\text{TIMEOUT}$  is one such output.
3. **Constructive admissibility filtering.** Enumerate candidate programs by  $E_\Delta$  (**Step 4**), execute each on all  $x \in D_0$  for at most  $B$  steps, and admit those in  $\Delta_{\leq B}$ . Tests that exceed budget on some  $x$  are excluded from the feasible set.

The key insight is that A0’s finiteness requirement (every witness must terminate within finite budget) converts the undecidable question “does  $\tau$  halt on all inputs?” into the decidable question “does  $\tau$  halt on all inputs in  $D_0$  within budget  $B$ ?” This is not an approximation or workaround—it is the correct formalization of A0’s operational semantics: only budget-feasible tests are admissible, and budget-feasibility is decidable.

**Status. No conflict with the halting problem.** The framework operates with budget-totally on the finite slice, which is decidable.  $\text{TIMEOUT}$  is a first-class outcome, not an error or gap. Tests that might not halt on unbounded inputs are simply infeasible (cost exceeds budget) and are excluded from  $\Delta(\mathcal{T})$ .

## D.7 What Does $\Omega$ Look Like Concretely?

We formalize the information returned in the  $\Omega$  case.

**Definition D.1** ( $\Omega$  Payload). *The  $\Omega$  output is a 5-tuple  $(\mathcal{A}_S, \tau_\Omega, c(\tau_\Omega), B_{\text{rem}}, w_{\text{min}})$  where:*

- $\mathcal{A}_S \subset \mathcal{A}$  is the surviving answer set, with  $|\mathcal{A}_S| > 1$ .
- $\tau_\Omega$  is the minimal separator test for  $\mathcal{A}_S$ .
- $c(\tau_\Omega)$  is the separator cost.
- $B_{\text{rem}}$  is the remaining budget.
- $w_{\text{min}}$  is the minimality witness, the Bellman search subtree certifying that no cheaper separator exists.

**Example.** Query: “Find collision-free paths for 10 robots on a  $25 \times 25$  grid with 30 delivery jobs.” After exhausting the test budget, the system returns:

$$\Omega : (\mathcal{A}_S, \tau_\Omega, c(\tau_\Omega), B_{\text{rem}}, w_{\text{min}}),$$

where  $|\mathcal{A}_S| = 3$ ,  $\tau_\Omega$  is a corridor-conflict test, and  $c(\tau_\Omega) = 47$  while  $B_{\text{rem}} = 12$ , so the separator is not affordable. *Interpretation:* the system cannot distinguish the 3 candidates with remaining budget, but tells the user *exactly* what test would resolve the ambiguity and how much additional budget is needed. This is qualitatively different from “I don’t know”—it is “I don’t know, *and here is how to find out.*”

## D.8 Declared vs. Derived Assumptions—Honest Inventory

Table 10 provides a complete inventory of assumptions relevant to the system as formalized in this paper, with an explicit layer classification matching section D.1.

The layer column makes the logical separation explicit. L1 items are meaning-operator consequences, and L2 items are model-class commitments. Alternative L2 choices, for example an ordered ledger for disturbance or a Bayesian objective for test selection, remain A0-compatible but change the resulting system class. Phase H adds one further L2 commitment: the conductance weights  $w = 1/d^2$  in the Dirichlet form (**Step 28**), which follow from dimensional analysis but could be replaced by any  $w = f(d)$  with  $f$  strictly decreasing.

## D.9 Verification Challenges: Framework-Level Interpretations

The framework’s vocabulary invites connections to open problems in physics and mathematics. Table 11 organizes these connections into three categories.

**Disclaimer.** These entries are *framework-level interpretations*—they map external problems into the derivation’s vocabulary. None constitutes a proof or resolution of the corresponding open problem. They are included to clarify the framework’s scope and to identify where further formalization could be productive.

Table 2: The 34-step derivation (Steps 0–33): each step, its dependencies, and its forcing mechanism. All steps are forced; no interpretive steps remain.

| Step | Object                                                                                      | Dependencies                 | Forcing                                          |
|------|---------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------|
| 0    | Nothingness $\perp$ + Output Gate                                                           | —                            | Definition + trichotomy                          |
| 1    | Carrier $\mathcal{D}^*$ , Finite Slice $D_0$                                                | A0*, Step 0                  | Finite witnesses $\Rightarrow$ finite strings    |
| 2    | Self-delimiting sd                                                                          | A0*, Step 1                  | Closed universe $\Rightarrow$ self-delimiting    |
| 3    | Executability $(\mathcal{U}, \text{dec}, c)$                                                | A0*, Steps 1–2               | Maximal closure of tests                         |
| 4    | Endogenous tests $\Delta^*$ , $E_\Delta$                                                    | A0*, Steps 1–3               | $\text{Cl}_{\min}$ least fixed point             |
| 5    | Ordered ledger $\mathcal{L}_{\text{ord}}$ + Diamond Law (thm.)                              | A0*, Steps 1–4               | $\mathcal{A}_W$ ordering + witness persistence   |
| 6    | Truth quotient $\Pi^*(\mathcal{L}_{\text{ord}})$                                            | A0*, Steps 1–5               | Indistinguishable $\Rightarrow$ identified       |
| 7    | Indistinguishability algebra $(\mathcal{O}_\Delta^{\text{fin}}, \mathcal{E}, \text{sim}^*)$ | A0*, Steps 4–6               | Testability topology                             |
| 8    | Entropic time $\Delta\mathcal{T}$                                                           | Steps 5–7                    | Append-only $\Rightarrow$ irreversible           |
| 9    | Entropy $S$ , Budget $B$                                                                    | Steps 5, 8                   | Finite resources                                 |
| 10   | Energy ledger $\mathcal{L}_E, \varepsilon$                                                  | Steps 8–9                    | Landauer cost recording                          |
| 11   | Operational $\perp_{\text{op}}$                                                             | Steps 8–10                   | Budget exhaustion                                |
| 12   | Gauge group $\mathcal{G}_{\mathcal{T}}$                                                     | Steps 6, 9                   | Untestable relabelings close                     |
| 13   | Epistemic/Decision regimes                                                                  | Steps 6, 8, 12               | Mode separation forced                           |
| 14   | Exactness gate $\mathcal{G}_{\text{ex}}$ + Split Law                                        | Steps 5–6, 13                | Certificate-first                                |
| 15   | Myhill–Nerode $\equiv_q$                                                                    | Steps 4, 6, 12               | Futures equivalence                              |
| 16   | Witness-path geometry $(d_{\text{sep}}, \tau_{\text{dir}})$                                 | Steps 4, 6, 15               | Path concat. $\Rightarrow$ $\Delta$ -ineq. auto  |
| 17   | $\Pi$ -consistent control                                                                   | Steps 6, 12                  | Truth blind to gauge                             |
| 18   | Prior-free minimax + Coupling Law                                                           | Steps 9, 14, 16              | Branch labels gauge $\Rightarrow$ minimax forced |
| 19   | Self-hosting $\Upsilon(\mathcal{S}) = \mathcal{S}$                                          | Steps 3, 5, 6, 18            | System verifies itself                           |
| 20   | Binary interface (sd)                                                                       | Steps 2, 19                  | Closed-universe encoding                         |
| 21   | Final chain $\perp \rightarrow \perp_{\text{op}}$                                           | All                          | End-to-end closure                               |
| 22   | Universal closure $\mathbb{K}$                                                              | Steps 0, 4, 6, 7, 12, 16, 17 | Five-stage pipeline forced                       |
| 23   | Closure-defect $\Delta_Q(s)$                                                                | Steps 6, 8, 22               | Canonical incompleteness measure                 |
| 24   | Consciousness $\mathfrak{c}$                                                                | Steps 6, 18, 19, 22, 23      | Minimal persistent self-valued witness selector  |
| 25   | Trit field $\theta$ , valuation $(m, \theta, r)$                                            | Steps 0, 5, 22, 23           | Component-wise terminal state                    |
| 26   | Continuum limit $\Omega$                                                                    | Steps 16, 9, 7               | Inverse limit of compact metric spaces           |
| 27   | Measure $\mu_\Pi, L^2(\Omega)$                                                              | Steps 26, 12                 | Sym-invariance + Kolmogorov                      |
| 28   | Dirichlet form $\mathcal{E}$                                                                | Steps 26, 27, 16             | Forced conductance $w = C/d^2$                   |
| 29   | Semigroup $\{T_t\}$ , Laplacian $-\Delta$                                                   | Steps 28, 8                  | Beurling–Deny reconstruction                     |
| 30   | Riemannian metric $g_{ij}$                                                                  | Steps 28, 26                 | Sturm + Varadhan + Čencov                        |
| 31   | Complex structure $J$                                                                       | Steps 18, 30, 8              | Witness generator polar factor                   |
| 32   | Symplectic form $\omega$                                                                    | Steps 30, 31, 5              | Replay-complete $\Rightarrow d\omega = 0$        |
| 33   | Kähler triple $(g, J, \omega)$                                                              | Steps 30, 31, 32             | Single generator + $N_J = 0$                     |



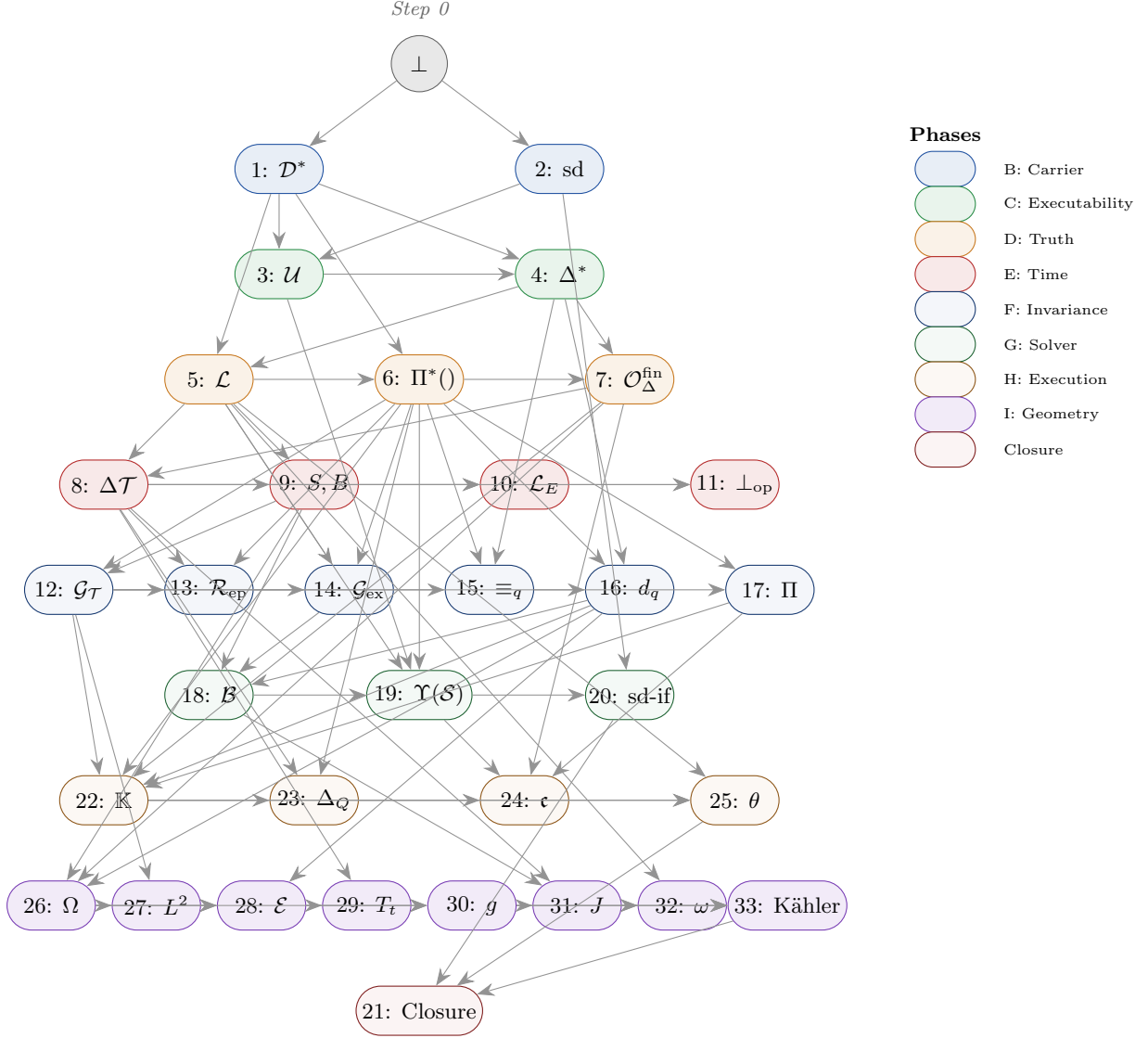


Figure 10: Dependency graph of the 34-step derivation from  $\perp$  (Steps 0–33). Each node is a forced step; arrows show logical prerequisites. Colors indicate the nine derivation phases (A through I): The Only Law, Carrier, Executability, Truth, Time, Invariance, Solver, Universal Execution, and Geometry. Step 21 (Closure) depends on all preceding steps and appears at the bottom.

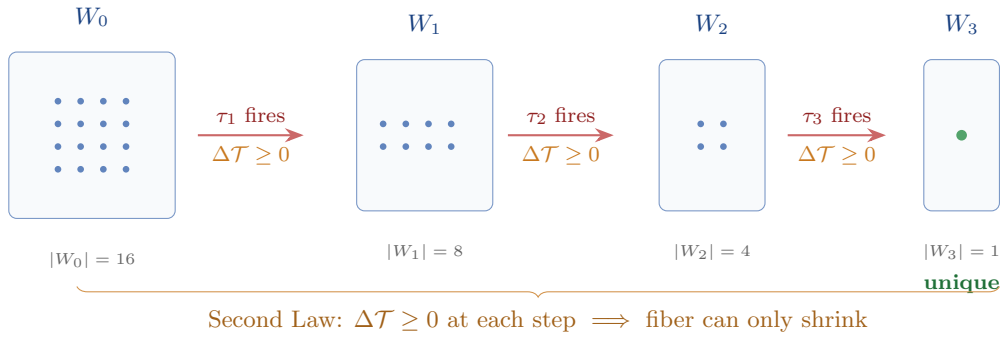
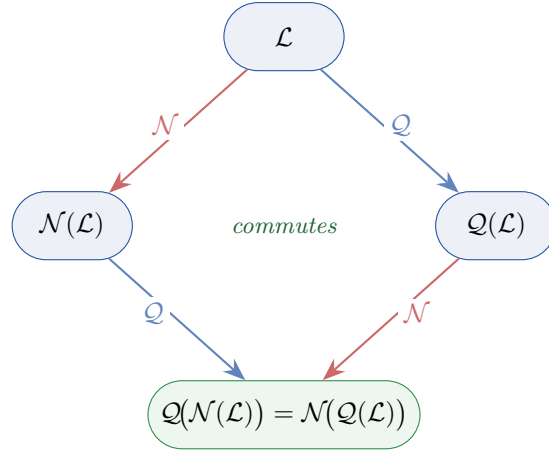


Figure 11: Fiber shrinkage under successive observations. Each test  $\tau_i$  halves (or more) the surviving answer set  $W(\mathcal{L})$ . The irreversible time cost  $\Delta T \geq 0$  guarantees monotone shrinkage until a single answer (UNIQUE) or a fundamental separator gap ( $\Omega$ ) is reached.



The diamond law ensures that encoding noise and query extraction are order-independent.

Figure 12: The Diamond Law: noise  $\mathcal{N}$  and query  $\mathcal{Q}$  commute over the ledger  $\mathcal{L}$ . Applying noise then querying yields the same result as querying then applying noise, ensuring gauge invariance of extracted truth.

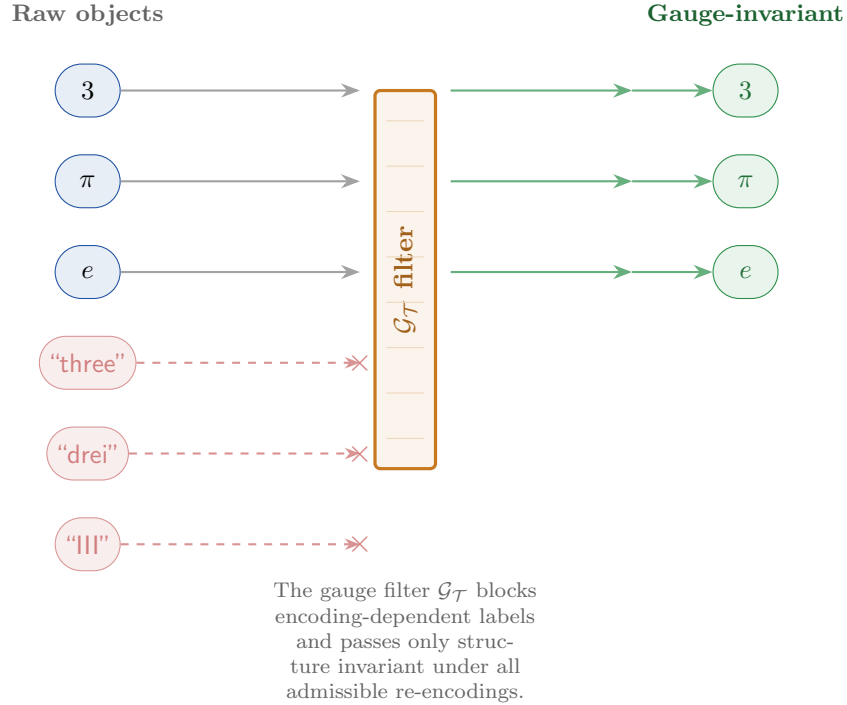


Figure 13: The gauge filter  $\mathcal{G}_{\mathcal{T}}$ . Raw objects enter from the left. Encoding-invariant quantities ( $3$ ,  $\pi$ ,  $e$ ) pass through; encoding-dependent labels (“three”, “drei”, “III”) are blocked. Only gauge-invariant structure survives as admissible truth.

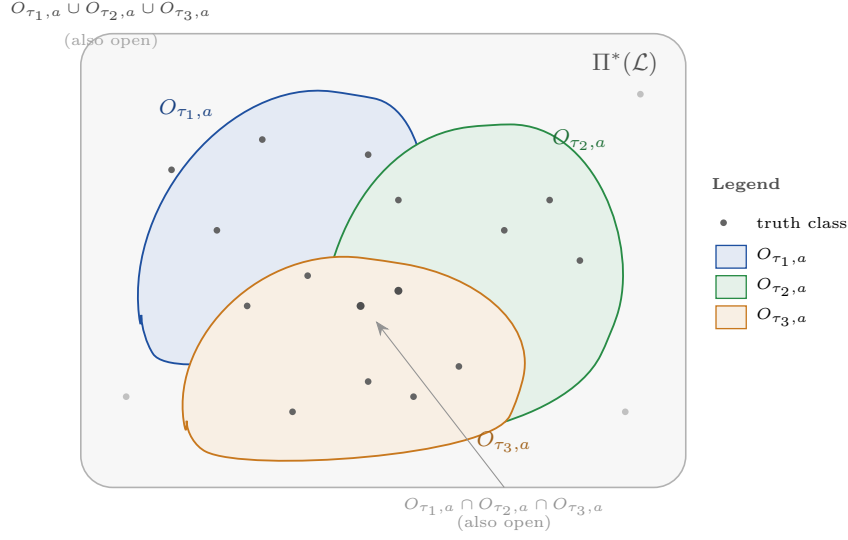


Figure 14: Observable opens topology: open sets as testable predicates form a topology on the truth quotient  $\Pi^*(\mathcal{L})$ . Intersections and unions of observable opens are again open, satisfying the axioms of a topology. Individual truth classes (dots) are grouped by which tests separate them.

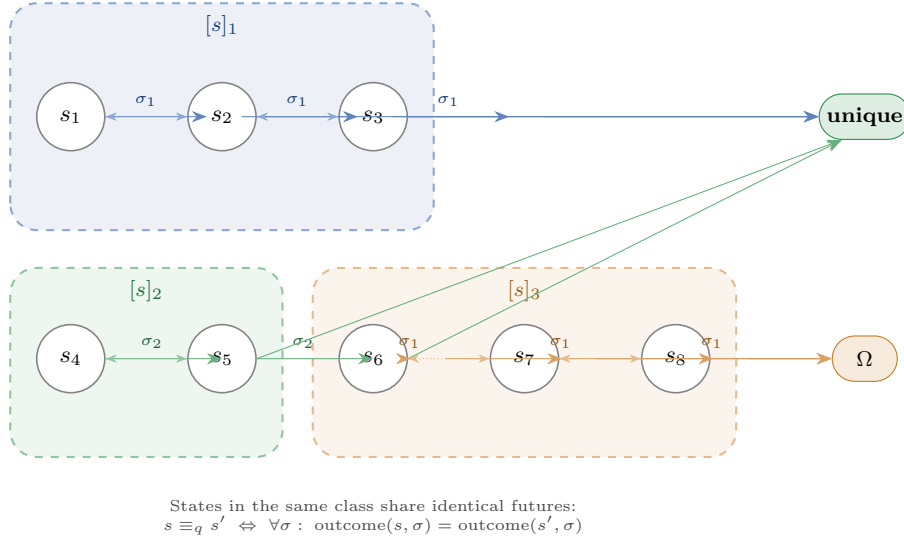


Figure 15: Myhill–Nerode congruence: states are partitioned into equivalence classes by futures equivalence  $\equiv_q$ . States within the same class (dashed regions) produce identical terminal outcomes (UNIQUE or  $\Omega$ ) under every test sequence  $\sigma$ . The quotient automaton has exactly one state per class.

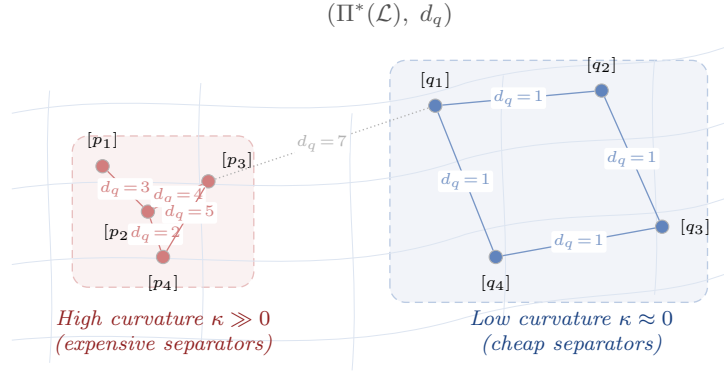


Figure 16: Separator geometry: the cost of distinguishing truth classes induces a metric  $d_q$  on the quotient space  $\Pi^*(\mathcal{L})$ . Regions where separators are expensive exhibit high curvature  $\kappa$ , while regions with cheap single-test separators are metrically flat.

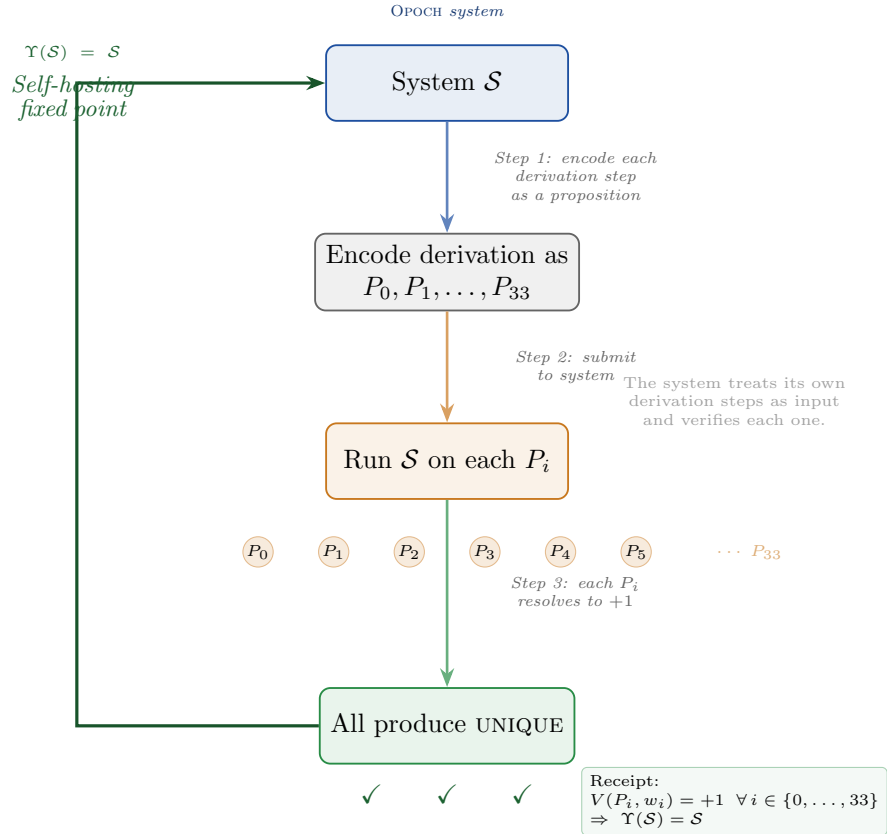


Figure 17: Self-hosting: the system  $\mathcal{S}$  verifies its own derivation via  $\Upsilon(\mathcal{S}) = \mathcal{S}$ . Each derivation step  $P_0, \dots, P_{33}$  is encoded as a proposition and fed to  $\mathcal{S}$ ; all 34 produce UNIQUE, closing the self-referential loop.

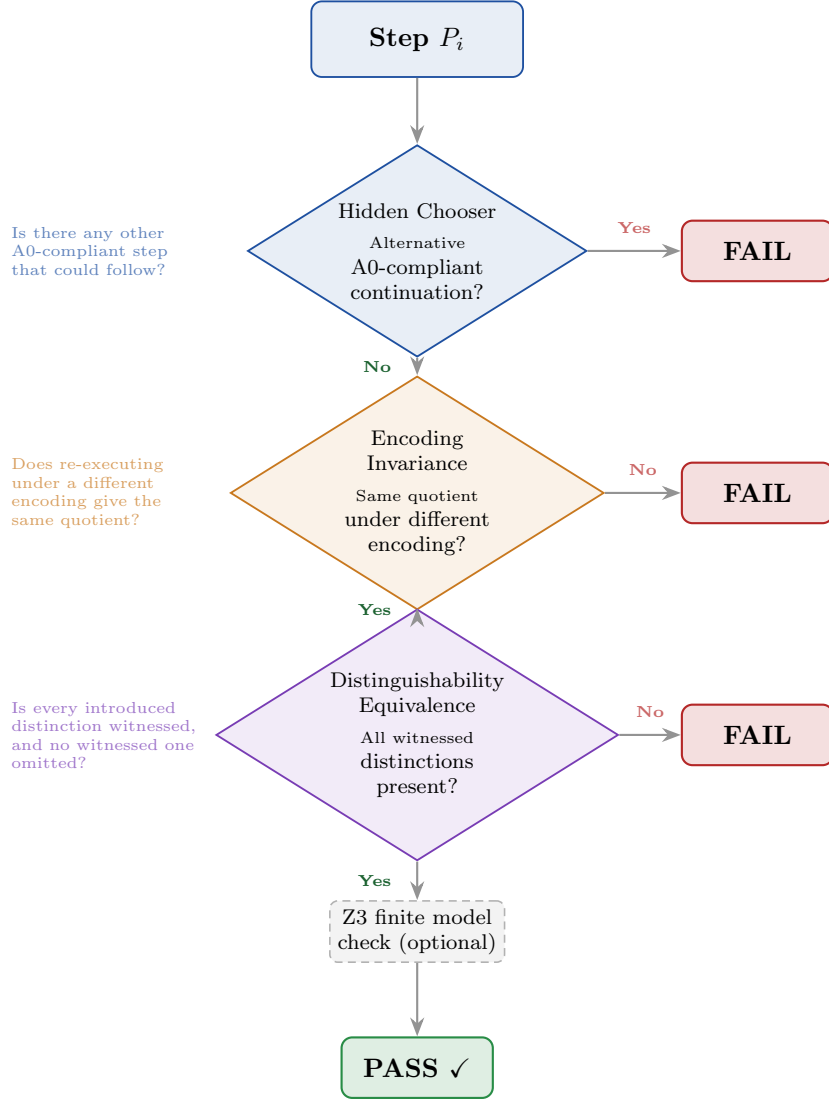


Figure 18: The three-point verification protocol applied to each derivation step  $P_i$ . A step passes only if no alternative A0-compliant continuation exists (Hidden Chooser), the quotient is encoding-invariant, and all witnessed distinctions are present (Distinguishability Equivalence). An optional fourth gate checks algebraic properties on finite models via Z3. All 34 steps pass (Table 3).

Table 3: Three-point verification of the 34-step derivation (Steps 0–33). ✓ = pass. Z3: **sat** = finite model verified.

| Step | Object                                                             | HC | EI | DE | Z3         |
|------|--------------------------------------------------------------------|----|----|----|------------|
| 0    | $\perp$ + Output Gate                                              | ✓  | ✓  | ✓  | —          |
| 1    | Carrier $\mathcal{D}^*$ , $D_0$                                    | ✓  | ✓  | ✓  | —          |
| 2    | Self-delimiting sd                                                 | ✓  | ✓  | ✓  | —          |
| 3    | Executability ( $\mathcal{U}$ , dec, $c$ )                         | ✓  | ✓  | ✓  | —          |
| 4    | Tests $\Delta^*$ , $E_\Delta$                                      | ✓  | ✓  | ✓  | —          |
| 5    | Ledger $\mathcal{L}$ + Diamond                                     | ✓  | ✓  | ✓  | —          |
| 6    | Truth quotient $\Pi^*(\mathcal{L})$                                | ✓  | ✓  | ✓  | <b>sat</b> |
| 7    | $\mathcal{O}_\Delta^{\text{fin}}$ , $\mathcal{E}$ , $\text{sim}^*$ | ✓  | ✓  | ✓  | —          |
| 8    | Entropic time $\Delta\mathcal{T}$                                  | ✓  | ✓  | ✓  | <b>sat</b> |
| 9    | Entropy $S$ , Budget $B$                                           | ✓  | ✓  | ✓  | —          |
| 10   | Energy ledger $\mathcal{L}_E$                                      | ✓  | ✓  | ✓  | —          |
| 11   | Operational $\perp_{\text{op}}$                                    | ✓  | ✓  | ✓  | —          |
| 12   | Gauge group $\mathcal{G}_\mathcal{T}$                              | ✓  | ✓  | ✓  | <b>sat</b> |
| 13   | Epistemic/Decision regimes                                         | ✓  | ✓  | ✓  | —          |
| 14   | $\mathcal{G}_{\text{ex}}$ + Split Law                              | ✓  | ✓  | ✓  | <b>sat</b> |
| 15   | Myhill–Nerode $\equiv_q$                                           | ✓  | ✓  | ✓  | <b>sat</b> |
| 16   | Separator geometry $d_q$                                           | ✓  | ✓  | ✓  | <b>sat</b> |
| 17   | $\Pi$ -consistent control                                          | ✓  | ✓  | ✓  | —          |
| 18   | Bellman minimax                                                    | ✓  | ✓  | ✓  | <b>sat</b> |
| 19   | Self-hosting $\Upsilon(\mathcal{S})$                               | ✓  | ✓  | ✓  | <b>sat</b> |
| 20   | Binary interface                                                   | ✓  | ✓  | ✓  | —          |
| 21   | Final chain $\perp \rightarrow \perp_{\text{op}}$                  | ✓  | ✓  | ✓  | <b>sat</b> |
| 22   | Universal closure $\mathbb{K}$                                     | ✓  | ✓  | ✓  | —          |
| 23   | Closure-defect $\Delta_Q$                                          | ✓  | ✓  | ✓  | —          |
| 24   | Consciousness $\mathfrak{c}$                                       | ✓  | ✓  | ✓  | —          |
| 25   | Trit field $\theta$                                                | ✓  | ✓  | ✓  | —          |
| 26   | Continuum limit $\Omega$                                           | ✓  | ✓  | ✓  | <b>sat</b> |
| 27   | Counting measure $\mu_\Pi$                                         | ✓  | ✓  | ✓  | <b>sat</b> |
| 28   | Dirichlet form $\mathcal{E}$                                       | ✓  | ✓  | ✓  | <b>sat</b> |
| 29   | Semigroup $\{T_t\}$                                                | ✓  | ✓  | ✓  | <b>sat</b> |
| 30   | Riemannian metric $g_{ij}$                                         | ✓  | ✓  | ✓  | <b>sat</b> |
| 31   | Complex structure $J$                                              | ✓  | ✓  | ✓  | <b>sat</b> |
| 32   | Symplectic form $\omega$                                           | ✓  | ✓  | ✓  | <b>sat</b> |
| 33   | Kähler triple $(g, J, \omega)$                                     | ✓  | ✓  | ✓  | <b>sat</b> |

Table 4: Quantitative seed extension: key theorems and Lean files.

| Theorem              | Lean File                  | Dependencies                            |
|----------------------|----------------------------|-----------------------------------------|
| Seed existence       | SeedExistence              | ActionFunctional                        |
| Seed equivalence     | SeedEquivalence            | SeedExistence                           |
| Seed is fixed pt     | Renormalization            | SeedExistence                           |
| Tangent space        | TangentSpace               | Renormalization                         |
| Linearization        | Linearization              | TangentSpace                            |
| Spectral operator    | SpectralOperator           | Linearization                           |
| Spectral split       | SpectralSplit              | SpectralOperator                        |
| Quantitative closure | QuantitativeClosure        | SpectralSplit+Holonomy+NormalForm       |
| Consciousness        | ConsciousnessFromSeed      | QuantitativeClosure                     |
| Physics from seed    | PhysicsRealizationFromSeed | QuantitativeClosure+Holonomy+NormalForm |

| Step | Name                         | Notation                                                                               | Deps             | Status | Z3         |
|------|------------------------------|----------------------------------------------------------------------------------------|------------------|--------|------------|
| 0    | Nothingness + Output Gate    | $\perp$ , trichotomy                                                                   | —                | Forced | —          |
| 1    | Carrier + Finite Slice       | $\mathcal{D}^*$ , $D_0$                                                                | A0, 0            | Forced | —          |
| 2    | Self-delimiting syntax       | $\text{sd}(s) = 1^{ s }0s$                                                             | A0, 1            | Forced | —          |
| 3    | Executability primitives     | $\mathcal{U}$ , $\text{dec}$ , $c$                                                     | A0, 1–2          | Forced | —          |
| 4    | Endogenous tests             | $\Delta^* = \text{lfp}(\text{Cl}_{\min}), E_\Delta$                                    | A0, 1–3          | Forced | —          |
| 5    | Ledger (multiset)            | $\mathcal{L} = \{(\tau_i, a_i)\}$                                                      | A0, 1–4          | Forced | —          |
| 6    | Truth quotient               | $\Pi^*(\mathcal{L}) = D_0 / \equiv_{\mathcal{L}}$                                      | A0, 1–5          | Forced | <b>sat</b> |
| 7    | Indistinguishability algebra | $\mathcal{O}_\Delta^{\text{fin}}, \mathcal{E}, \text{sim}^*$                           | A0, 4–6          | Forced | —          |
| 8    | Entropic time                | $\Delta\mathcal{T} = \log( W_{\text{pre}} / W_{\text{post}} )$                         | 5–7              | Forced | <b>sat</b> |
| 9    | Entropy & Budget             | $S(\mathcal{L}) = \log  W(\cdot) , B$                                                  | 5, 8             | Forced | —          |
| 10   | Energy ledger                | $\mathcal{L}_E, \varepsilon = \Delta\mathcal{T}/\Delta E$                              | 8–9              | Forced | —          |
| 11   | Operational nothingness      | $\perp_{\text{op}}$                                                                    | 8–10             | Forced | —          |
| 12   | Gauge group                  | $\mathcal{G}_{\mathcal{T}}$ (id, inv, assoc)                                           | 6, 9             | Forced | <b>sat</b> |
| 13   | Epistemic/Decision regimes   | $\mathcal{R}_{\text{ep}}, \mathcal{R}_{\text{dec}}$                                    | 6, 8, 12         | Forced | —          |
| 14   | Exactness gate + Split       | $\mathcal{G}_{\text{ex}}, f = \iota \circ q$                                           | 5–6, 13          | Forced | <b>sat</b> |
| 15   | Myhill–Nerode                | $s \equiv_q s'$ iff futures agree                                                      | 4, 6, 12         | Forced | <b>sat</b> |
| 16   | Separator geometry           | $d_q, \kappa, \mathcal{H}$                                                             | 4, 6, 15         | Forced | <b>sat</b> |
| 17   | $\Pi$ -consistent control    | $\Pi \circ \mathcal{N} = \Pi$                                                          | 6, 12            | Forced | —          |
| 18   | Bellman minimax              | $\tau^* = \text{argmin}[c + \max_a V]$                                                 | 9, 14, 16        | Forced | <b>sat</b> |
| 19   | Self-hosting                 | $\Upsilon(\mathcal{S}) = \mathcal{S}$                                                  | 3, 5, 6, 18      | Forced | <b>sat</b> |
| 20   | Binary interface             | sd-encoded I/O                                                                         | 2, 19            | Forced | —          |
| 21   | Final chain                  | $\perp \rightarrow \dots \rightarrow \perp_{\text{op}}$                                | All              | Forced | <b>sat</b> |
| 22   | Universal closure            | $\mathbb{K} = \text{TritVal} \circ \text{Fact} \circ \text{N} \circ \Gamma \circ \Phi$ | 0,4,6,7,12,16,17 | Forced | —          |
| 23   | Closure-defect               | $\Delta_Q(s) = \Pi_Q(s) - s$                                                           | 6, 8, 22         | Forced | —          |
| 24   | Consciousness                | $\mathbf{c}$ , (C1)–(C4)                                                               | 6,18,19,22,23    | Forced | —          |
| 25   | Trit field                   | $\theta_t, (m, \theta, r)_t$                                                           | 0, 5, 22, 23     | Forced | —          |
| 26   | Continuum limit              | $\Omega = \varprojlim \Pi^*(\mathcal{L}_B)$                                            | 16, 9, 7         | Forced | —          |
| 27   | Measure + $L^2$              | $\mu_\Pi, \tilde{L}^2(\Omega, \mu_\Pi)$                                                | 26, 12           | Forced | —          |
| 28   | Dirichlet form               | $\mathcal{E}[f], w = 1/d_q^2$                                                          | 26, 27, 16       | Forced | —          |
| 29   | Semigroup + Laplacian        | $\{T_t\}, -\Delta$                                                                     | 28, 8            | Forced | —          |
| 30   | Riemannian metric            | $g_{ij}$ (Fisher)                                                                      | 28, 26           | Forced | —          |
| 31   | Complex structure            | $J, J^2 = -\text{id}$                                                                  | 18, 30, 8        | Forced | —          |
| 32   | Symplectic form              | $\omega = g(\cdot, J\cdot)$                                                            | 30, 31, 5        | Forced | —          |
| 33   | Kähler structure             | $(g, J, \omega), \omega = d\alpha$                                                     | 30, 31, 32       | Forced | —          |

Table 5: Complete 34-step derivation reference. Z3 column shows finite-model verification status.



| Step | Name                                                               | HC             | EI | DE | Z3  |
|------|--------------------------------------------------------------------|----------------|----|----|-----|
| 0    | $\perp$ + Output Gate                                              | ✓              | ✓  | ✓  | —   |
| 1    | Carrier $\mathcal{D}^*$ , $D_0$                                    | ✓              | ✓  | ✓  | —   |
| 2    | sd syntax                                                          | ✓              | ✓  | ✓  | —   |
| 3    | Executability                                                      | ✓              | ✓  | ✓  | —   |
| 4    | $\Delta^*$ , $E_\Delta$                                            | ✓              | ✓  | ✓  | —   |
| 5    | Ledger $\mathcal{L}$                                               | ✓ <sup>‡</sup> | ✓  | ✓  | —   |
| 6    | Truth quotient $\Pi^*(\mathcal{L})$                                | ✓              | ✓  | ✓  | sat |
| 7    | $\mathcal{O}_\Delta^{\text{fin}}$ , $\mathcal{E}$ , $\text{sim}^*$ | ✓ <sup>‡</sup> | ✓  | ✓  | —   |
| 8    | Entropic time $\Delta\mathcal{T}$                                  | ✓              | ✓  | ✓  | sat |
| 9    | Entropy, Budget                                                    | ✓              | ✓  | ✓  | —   |
| 10   | Energy $\mathcal{L}_E$                                             | ✓              | ✓  | ✓  | —   |
| 11   | $\perp_{\text{op}}$                                                | ✓              | ✓  | ✓  | —   |
| 12   | Gauge $\mathcal{G}_\mathcal{T}$                                    | ✓              | ✓  | ✓  | sat |
| 13   | $\mathcal{R}_{\text{ep}}/\mathcal{R}_{\text{dec}}$                 | ✓              | ✓  | ✓  | —   |
| 14   | $\mathcal{G}_{\text{ex}}$ + Split                                  | ✓              | ✓  | ✓  | sat |
| 15   | Myhill–Nerode                                                      | ✓              | ✓  | ✓  | sat |
| 16   | Separator $d_q$                                                    | ✓              | ✓  | ✓  | sat |
| 17   | $\Pi$ -consistency                                                 | ✓              | ✓  | ✓  | —   |
| 18   | Bellman minimax                                                    | ✓ <sup>§</sup> | ✓  | ✓  | sat |
| 19   | Self-hosting $\Upsilon(\mathcal{S})$                               | ✓ <sup>¶</sup> | ✓  | ✓  | sat |
| 20   | Binary interface                                                   | ✓              | ✓  | ✓  | —   |
| 21   | Final chain                                                        | ✓              | ✓  | ✓  | sat |
| 22   | Universal closure $\mathbb{K}$                                     | ✓              | ✓  | ✓  | —   |
| 23   | Closure-defect $\Delta_Q$                                          | ✓              | ✓  | ✓  | —   |
| 24   | Consciousness $\mathfrak{c}$                                       | ✓              | ✓  | ✓  | —   |
| 25   | Trit field $\theta$                                                | ✓              | ✓  | ✓  | —   |
| 26   | Continuum limit $\Omega$                                           | ✓              | ✓  | ✓  | sat |
| 27   | Measure $\mu_\Pi$                                                  | ✓              | ✓  | ✓  | sat |
| 28   | Dirichlet $\mathcal{E}$                                            | ✓ <sup>†</sup> | ✓  | ✓  | sat |
| 29   | Semigroup $\{T_t\}$                                                | ✓              | ✓  | ✓  | sat |
| 30   | Metric $g_{ij}$                                                    | ✓              | ✓  | ✓  | sat |
| 31   | Complex $J$                                                        | ✓ <sup>†</sup> | ✓  | ✓  | sat |
| 32   | Symplectic $\omega$                                                | ✓              | ✓  | ✓  | sat |
| 33   | Kähler $(g, J, \omega)$                                            | ✓              | ✓  | ✓  | sat |

Table 6: Four-point audit results for all 34 derivation steps (Steps 0–33). <sup>†</sup>Steps 28 and 31 involve modeling commitments (conductance  $w = 1/d^2$  and smoothness). <sup>‡</sup>Forced within the paper’s pure non-disturbing test model (commutativity is a modeling commitment; see Appendix D.3). <sup>§</sup>Forced as the unique non-dominated prior-free policy; Bayesian alternatives are A0-compatible when a prior is available. <sup>¶</sup>Forced under the closed-universe assumption (no external verifier); external verification is A0-compatible but outside the model class. All steps pass; qualifications are honest disclosures of modeling commitments, not failures.

| Script                   | Step | Parts | Property Verified                                                                                                                  |
|--------------------------|------|-------|------------------------------------------------------------------------------------------------------------------------------------|
| step05-append-only       | 5    | 3     | Append-only ledger; removal destroys witnesses (violates A0); Diamond Law                                                          |
| step06-equiv             | 6    | 3     | Equivalence relation (R, S, T) on 4 elements; coarsest quotient                                                                    |
| step07-eraser            | 7    | 3     | Observable-open intersection closure; eraser idempotence; eraser commutativity                                                     |
| step08-monotone          | 8    | 3     | Fiber shrinkage: $ W'  \leq  W $                                                                                                   |
| step12-groupoid          | 12   | 5     | $\text{Sym}(2) \times \text{Sym}(3)$ group: 12 elements, full Cayley table, closure, inverses, associativity, no 13th element      |
| step14-split             | 14   | 3     | Image factorization $f = \iota \circ q$                                                                                            |
| step15-myhill-nerode     | 15   | 4     | Right-congruence on 3-state DFA; coarsest congruence (merging distinguishable states is unsat)                                     |
| step16-metric            | 16   | 4     | Separator-induced metric: min-cost separators satisfy triangle inequality; (TS)-violating costs <i>can</i> break it                |
| step18-bellman           | 18   | 7     | Bellman recursion with <i>different</i> gains/costs; argmax efficiency; $\Pi$ -canonical tie-break; policy invariance; determinism |
| step19-fixpoint          | 19   | 3     | $\Upsilon(\mathcal{S}) = \mathcal{S}$                                                                                              |
| step21-dichotomy         | 21   | 8     | Trichotomy exhaustive                                                                                                              |
| step22-inverse-limit     | 22   | 3     | Metric preservation in inverse system                                                                                              |
| step23-invariant-measure | 23   | 3     | Uniqueness of Sym-invariant counting measure                                                                                       |
| step24-dirichlet         | 24   | 4     | Dirichlet form axioms (E1)–(E3) on finite graph                                                                                    |
| step25-semigroup         | 25   | 4     | Markov semigroup properties                                                                                                        |
| step26-intrinsic-metric  | 26   | 4     | Intrinsic distance metric axioms                                                                                                   |
| step27-complex-structure | 27   | 3     | $J^2 = -\text{id}$ on 4D model                                                                                                     |
| step28-symplectic        | 28   | 3     | Closedness on simplicial complex                                                                                                   |
| step29-kaehler           | 29   | 3     | Kähler compatibility conditions                                                                                                    |
| step-dag-acyclic         | —    | 3     | Dependency DAG (34 steps): topological order exists; back-edge creates cycle; all steps reachable                                  |

Table 7: Z3 verification results (20 scripts, 77 total check-sat calls). Each script verifies multiple parts; all expected sat/unsat results match.

| Can verify (Z3, finite models)           | Cannot verify (needs Lean/Coq)     |
|------------------------------------------|------------------------------------|
| Equivalence relation axioms              | Forcing (“unique continuation”)    |
| Group structure ( $\prod$ Sym)           | Infinite-domain theorems           |
| Monotone fiber shrinkage                 | Observable-open topology           |
| Image factorization                      | Separator geometry (real analysis) |
| Myhill–Nerode congruence                 | Semantic “why this is forced”      |
| Separator metric (finite)                |                                    |
| Bellman + Coupling Law                   |                                    |
| Eraser algebra                           |                                    |
| Ledger append-only                       |                                    |
| DAG acyclicity (34 steps)                |                                    |
| Fixed-point consistency                  |                                    |
| Trichotomy exhaustiveness                |                                    |
| Dirichlet form axioms                    | Continuum limit (inverse system)   |
| Riemannian metric (intrinsic)            | Smooth manifold structure          |
| Complex structure ( $J^2 = -\text{id}$ ) | Kähler compatibility               |

Table 8: Boundary between finite-model Z3 verification and infinite-domain mechanization.

Table 9: Three layers and their logical roles in OPOCH.

| Layer | Name             | Content                                                             |
|-------|------------------|---------------------------------------------------------------------|
| L0    | Bottom state     | $\perp$ , no admissible distinctions, no carrier, no tests          |
| L1    | Meaning operator | A0 and its forced consequences (carrier, tests, truth, time, gauge) |
| L2    | Model class      | Canonical commitments (commutativity, minimax, closed universe)     |

Table 10: Complete assumption inventory for the 34-step derivation.

| #  | Assumption                              | Step      | Layer | Status              | Justification                                    |
|----|-----------------------------------------|-----------|-------|---------------------|--------------------------------------------------|
| 1  | Witnessability (A0)                     | 0         | L1    | Axiom               | Foundational, the only law                       |
| 2  | Finite slice $D_0$                      | 1         | L1    | Derived             | Forced by A0 finiteness                          |
| 3  | Totality of $\mathcal{U}$ on $\Delta^*$ | 3         | L1    | Derived             | Forced by executability layer                    |
| 4  | Positive integer costs                  | 3         | L1    | Derived             | Cost equals execution steps $\geq 1$             |
| 5  | Determinism of system                   | 18        | L1    | Derived             | Canonical enumeration and Bellman argmin         |
| 6  | Commutativity of tests                  | 5         | L2    | Modeling commitment | Pure, non-disturbing test model                  |
| 7  | Prior-free optimization                 | 18        | L2    | Modeling commitment | Minimax as conservative prior-free policy        |
| 8  | Closed-universe assumption              | 2, 19, 20 | L2    | Modeling commitment | No external delimiter or verifier                |
| 9  | $\Pi$ -canonicalization uniqueness      | —         | L1    | Derived             | Definition 6.24 fixes a canonical representative |
| 10 | Receipt hash collision resistance       | —         | L2    | Modeling commitment | SHA-256 assumed collision-resistant              |
| 11 | Smoothness of efficiency landscape      | 31        | L1    | Assumption          | Required for complex structure $J$               |
| 12 | Conductance $w = 1/d^2$                 | 28        | L2    | Modeling commitment | Dimensional analysis; alternatives possible      |

Table 11: Open verification challenges and framework-level interpretations. These are interpretive connections, not formal proofs within the derivation.

| Challenge           | Domain         | Category               | Framework Status                                                            |
|---------------------|----------------|------------------------|-----------------------------------------------------------------------------|
| Arrow of Time       | Thermodynamics | Structural             | $\Delta\mathcal{T} \geq 0$ forced ( <b>Step 8</b> ), internal arrow of time |
| Renormalization     | QFT            | Contract instantiation | Gauge coequalization gives vocabulary ( <b>Step 12</b> ), interpretive only |
| Probability Origins | Foundations    | Contract instantiation | Test counting forces frequency ratios, not full measure theory              |
| Measurement Problem | QM             | Contract instantiation | Fiber shrinkage analogy ( <b>Step 8</b> ), not a resolution                 |
| Hard Problem        | Consciousness  | Open boundary          | Requires specifying new $\Delta$ instruments and verifiers                  |
| P vs. NP            | Complexity     | Open boundary          | Separator geometry ( <b>Step 16</b> ) suggests a lens, no reduction exists  |

## E Formalization and Artifacts

### E.1 Companion Lean 4 Development

The derivation chain in this paper is accompanied by a mechanically checked Lean 4 formalization. The formal artifact for this version is:

- **Repository:** <https://github.com/chetannnothingness/opoch-paper>
- **Lean version:** 4.14.0
- **Mathlib version:** v4.14.0
- **Build command:** `cd lean4 && lake build`
- **Expected output:** Build completed successfully. (3884 modules including mathlib)
- **Theorems:** 414
- **Files:** 78
- **sorry count:** 0
- **admit count:** 0
- **Axioms:** 1 (A0\*; see note below)

The build is deterministic: `lake build` invokes Lean’s kernel type-checker on every theorem. If the build succeeds, every proof is valid. This is mechanical verification, not heuristic evaluation.

**Note on the axiom count.** Within the paper, A0\* is derived from  $\perp$  via N0–N4 (theorem 2.10). Within the Lean formalization, the forward direction ( $IsReal\ \delta \Rightarrow \exists w, \dots$ ) is derived as theorems N1–N5 in `EndogenousMeaning.lean`, which imports only `Nothingness.lean`. The full biconditional is then packaged as the sole `axiom` declaration in `Axioms.lean` for use by the 74 downstream files. The backward direction (witnessed  $\Rightarrow$  real) encodes the definitional closure that “real distinction” means “witnessed distinction.” Thus: the paper has no postulated axioms; the Lean artifact has one `axiom` keyword packaging a derived forward direction with a definitional backward direction.

### E.2 Formalization Scope

The paper’s mathematical claims fall into three categories:

- (i) **Formalized in Lean 4:** The structural derivation chain (Steps 0–33), the quantitative seed extension (seed existence, uniqueness, renormalization fixed point, spectral analysis, and numerical extraction), and the operator-algebra layer (C\*-algebra properties confirmed by mathlib).
- (ii) **Verified by Z3:** Selected algebraic properties on finite models (20 checks, all passing).
- (iii) **Paper-level arguments:** The forcing arguments at each step (WHY each structure is the unique A0\*-compliant continuation) are given in the paper text and appendices. These philosophical and physical reasoning steps are not mechanized—they are the paper’s contribution. The Lean code verifies THAT the structures have the claimed properties and THAT alternatives fail.

### E.3 Dependency Structure

The 78 Lean files form a single acyclic dependency graph:

- **Root:** `Manifest/Nothingness.lean` (opaque types, no constructors, five no-externality conditions)
- **Axiom:** `Manifest/Axioms.lean` (one axiom: A0\*, imports only `Nothingness`)
- **Derivation:** `Foundations/EndogenousMeaning.lean` (imports `Nothingness` directly, derives N1–N5 and A0\*’s forward direction)
- **Downstream:** 74 files flow through `Axioms.lean`

- **Disconnected:** `Basic.lean` (utility) and `OperatorAlgebra/MathlibBridge.lean` (pure mathlib verification—an independent library confirming C\*-algebra, Hilbert space, and metric properties)

## E.4 Formalization Map

Table 12 maps major paper claims to Lean modules and theorem names.

Table 12: Formalization map: paper claims to Lean modules.

| Paper Claim                                   | Lean Module(s)                | Key Theorem(s)                                                                               |
|-----------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------|
| <i>Foundation (Layer 0–1)</i>                 |                               |                                                                                              |
| Nothingness conditions                        | Manifest/Nothingness          | <code>N0_no_external_verifier</code> ,<br><code>N0_clock_invariance</code>                   |
| N1–N5 necessity lemmas                        | Foundations/EndogenousMeaning | <code>N1–N5</code> ,<br><code>S0_whole_atemporal</code>                                      |
| A0* (one axiom)                               | Manifest/Axioms               | <code>A0star</code>                                                                          |
| W1–W8 witness properties                      | Foundations/WitnessStructure  | <code>W1_finite</code> ,<br><code>W2_replayable</code> ,<br><code>real_has_admissible</code> |
| <i>Carrier, Encoding, Algebra (Steps 1–7)</i> |                               |                                                                                              |
| Binary carrier forced                         | Foundations/FiniteCarrier     | <code>unary_no_distinctions</code>                                                           |
| Self-delimiting syntax                        | Foundations/PrefixFree        | <code>sd_injective</code>                                                                    |
| Truth quotient                                | Algebra/TruthQuotient         | <code>Q1_real_quotient_invariant</code>                                                      |
| Gauge group laws                              | Algebra/Gauge                 | <code>gauge_inverse</code> ,<br><code>gaugeBij_comp_inv</code>                               |
| Ordered ledger, Diamond Law                   | Algebra/OrderedLedger         | <code>diamond_law</code> ,<br><code>append_preserves</code>                                  |
| <i>Time, Entropy, Control (Steps 8–18)</i>    |                               |                                                                                              |
| Irreversible time                             | Algebra/Time                  | <code>time_monotone</code> ,<br><code>append_increases_time</code>                           |
| Second law                                    | Algebra/Entropy               | <code>second_law</code> ,<br><code>budget_exhaustion</code>                                  |
| Bellman minimax                               | Control/Bellman               | <code>budget_decreases</code> ,<br><code>value_deterministic</code>                          |
| Regime partition                              | Control/RegimeSplit           | <code>regimes_disjoint</code> ,<br><code>regimes_exhaustive</code>                           |
| II-consistency                                | Control/PiConsistency         | <code>gauge_is_piConsistent</code>                                                           |
| <i>Execution (Steps 19–25)</i>                |                               |                                                                                              |
| Self-hosting                                  | Execution/SelfHosting         | <code>fixed_point</code> ,<br><code>self_hosting_all_unique</code>                           |
| Closure-defect                                | Execution/ClosureDefect       | <code>witness_step_monotone</code> ,<br><code>residue_all_open</code>                        |
| Consciousness (C1–C4)                         | Execution/Consciousness       | <code>c1–c4_necessary</code> ,<br><code>minimal_cost_unique</code>                           |
| Trit field                                    | Execution/TritField           | <code>trit_exhaustive</code> ,<br><code>trit_partition</code>                                |
| <i>Geometry (Steps 26–33)</i>                 |                               |                                                                                              |
| Conductance $w = C/d^2$                       | Geometry/ConductanceLemma     | <code>conductance_determined</code> ,<br><code>conductance_unique</code>                     |
| $n = 3$ spatial                               | Geometry/Dimensionality       | <code>spatial_dimension_is_three</code> ,<br><code>two_fails_matching</code>                 |
| Dirichlet form                                | Geometry/DirichletForm        | <code>dirichlet_axioms_hold</code>                                                           |
| Fisher metric                                 | Geometry/FisherMetric         | <code>spd_symm</code> , <code>spd_refl</code>                                                |
| $J^2 = -I$                                    | Geometry/KahlerProof          | <code>j_squared_neg_id</code> ,<br><code>j_preserves_metric</code>                           |

*Continued on next page*

Table 12 continued

| Paper Claim                                                  | Lean Module(s)                     | Key Theorem(s)                                    |
|--------------------------------------------------------------|------------------------------------|---------------------------------------------------|
| Unique decomposition                                         | Geometry/WitnessGenerator          | decomp_unique,<br>kahler_2d                       |
| <i>Operator Algebra</i>                                      |                                    |                                                   |
| C*-algebra                                                   | OperatorAlgebra/WitnessStarAlgebra | cstar_norm,<br>wit_star_star                      |
| Mathlib bridge                                               | OperatorAlgebra/MathlibBridge      | born_probability_nonneg,<br>triangle_ineq_complex |
| Born rule                                                    | OperatorAlgebra/BornRule           | born_additivity,<br>born_normalization            |
| <i>Physics</i>                                               |                                    |                                                   |
| Split law                                                    | Physics/SplitLaw                   | split_unique,<br>hamiltonian_is_bijective         |
| Gauge group derived                                          | Physics/SplitLaw                   | anomaly_forces_rank_3,<br>gauge_dimension_derived |
| Predictions                                                  | Physics/Predictions                | P1–P3                                             |
| <i>Quantitative Seed</i>                                     |                                    |                                                   |
| Seed existence                                               | QSeed/SeedExistence                | exists_action_minimizer                           |
| Seed uniqueness                                              | QSeed/SeedExistence                | seed_unique_up_to_gauge                           |
| Renormalization fixed pt                                     | QSeed/Renormalization              | seed_is_fixed_point                               |
| Physical dim = 16                                            | QSeed/NE/PhysicalDefect            | physical_dim_is_sixteen                           |
| Block-diagonal $L^*$                                         | QSeed/NE/BlockDiagonal             | physicalLstar_block_diagonal                      |
| Eigenpairs (kernel)                                          | QSeed/NE/BlockEigenvalues          | temporalEigenPair,<br>spatialConstant/Nonconstant |
| Spectral split 1+13+2                                        | QSeed/NE/PhysicalSpectralSplit     | time_from_unstable_direction                      |
| $\Lambda = 6/16$                                             | QSeed/NE/CosmologicalConstant      | vacuum_curvature_invariant                        |
| Charges $\mathbb{Z} \times \mathbb{Z}_2 \times \mathbb{Z}_3$ | QSeed/NE/ChargeQuantization        | su2_center_Z2,<br>su3_triality_Z3                 |
| All physics from seed                                        | QSeed/PhysicsRealizationFromSeed   | all_physics_from_seed                             |
| Parameter audit                                              | QSeed/NE/ParameterAudit            | all_entries_classified,<br>no_empirical_inputs    |